The Alaoglu–Erdős Integer-Exponent Conjecture

Machine-Checked Partial Results, Exact Conditional Structure, Determinant Methods, and the Remaining Barrier

A unified report on the corpus in

https://github.com/VladimirReshetnikov/ProveIt

Bottom line. The conjecture — that $2^x, 3^x \in \mathbb{Z}$ forces $x \in \mathbb{Z}$ — remains open, and the reduction to the four exponentials conjecture explains why no routine combination of present methods can close it. What the corpus does establish is unusually complete. Kernel-verified in Lean: a zero-free region $2^x < 4096$; transcendence of every hypothetical counterexample; multiplicative rank at most two with a canonical primitive odd core; and — conditional on a single counterexample — the *exact* classification

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\}, \qquad \mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\},$$

with unique coordinates, exact dyadic block counts, a sharp quadratic cumulative asymptotic, and vanishing block Fourier modes. These reduce the problem to one narrow statement: no primitive odd $w > 1$ has a positive power of w^ϑ in $\mathbb{Z}[1/3]$.

This report adds, at paper level, two determinant methods — an arbitrary-node Vandermonde repulsion inequality and a generalized-Vandermonde hierarchy over the mixed exponent semigroup $\mathbb{N}_0 + \mathbb{N}_0\vartheta$, the latter giving an independent explicit $O((\log X)^2)$ dyadic bound — together with a diagnosis of exactly why they stop short: archimedean interpolation needs $\asymp (\log X)^2$ nodes at scale X, while a counterexample supplies only $\asymp \log X$. It also records a new sufficient condition of a different kind: if the products $\log p \log q$ of two prime logarithms are $\mathbb{Q}$-linearly independent, the conjecture follows.

This document merges and extends two earlier AI-assisted reports on the same corpus.

August 20, 2026

Abstract

Let $\vartheta = \log 3/\log 2$. The Alaoglu–Erdős conjecture asserts that a real x with $2^x, 3^x \in \mathbb{Z}$ must be an integer; equivalently, the only positive integers m with $m^\vartheta \in \mathbb{Z}$ are the powers of two. It is the first unresolved 2×2 instance of the four exponentials problem, while its three-base analogue follows from six exponentials and is formalized here by an interpolation determinant.

This unified report has four purposes. First, it catalogues the machine-checked corpus with exact Lean identifiers: zero-free regions, the Gelfond–Schneider dichotomy, multiplicative rank two, the canonical primitive odd core, the rational-power index and radical degree, localized-radical equivalences, arbitrary-order arithmetic-progression obstructions, shrinking-target Diophantine bounds, fractional-part gap equivalences, and the exact conditional structure theory — the free monoid $\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta$ with a unique least generator, exact block counts, and the sharp quadratic asymptotic. Second, it develops two new determinant methods. Arbitrary candidate tuples on the lattice curve $y = x^\vartheta$ satisfy

$$\prod_{0\le i<j\le r} (m_j - m_i) \ \ge\ \frac{r!}{|\,(\vartheta)_{\underline{r}}\,|}\, m_0^{r-\vartheta},$$

with no arithmetic-progression hypothesis; and the mixed exponent semigroup supplies positive integer generalized Vandermonde determinants whose repulsion exponent tends to one, yielding the explicit unconditional bound $\#(\mathcal{C} \cap [X, 2X]) \le 10000 \max\{1, \log X\}^2$ by a route independent of the prime-valuation rank theorem. Third, it quantifies the remaining gap: the determinant hierarchy misses the conditional population by exactly one factor of $\log X$. Fourth, it records new equivalent and sufficient conditions — block-growth dichotomies (including the observation that the conjecture is equivalent to the dyadic block count being 1 for *infinitely many* exponents), a compact synchronized S-integral orbit formulation, and the prime-log-product independence criterion — and closes with a prioritized research program and a formalization blueprint.

Contents

1 Executive summary

1.1 What is proved, and at what level of trust

The full conjecture remains open. Several routes were pushed to their exact failure points: the three-base determinant method does not shrink to two bases without proving a case of the four exponentials conjecture; equidistribution of multiples does not defeat exponentially shrinking integrality gaps; local curvature obstructions cannot reach the sparse global structure; rational approximation to a hypothetical least generator stalls at the same transcendence input; and every successful reduction terminates at the same transcendence-theoretic wall, isolated in Section 6. Two lines of attack opened after that survey – interpolation determinants on arbitrary candidate nodes, and generalized Vandermonde determinants over the full semigroup of integral mixed monomials – do produce new unconditional partial results, and a third, the prime-log-product criterion of Section 12, isolates a clean sufficient condition of a different flavor. None of them closes the final gap.

To keep unlike kinds of evidence separate, every important statement in this report carries one of five labels.

[kernel-verified Lean]
: A proof source is present in the repository, lies on the audited `ExponentialIdentities.lean` import surface, and has been accepted by the Lean kernel in direct or full builds of the cited modules (toolchain 4.31/4.32 series, Mathlib pinned by the repository manifest). No `sorry`, no extra axioms, no `native_decide`.

[paper proof in this report]
: A conventional mathematical proof is supplied in this document. It is intended to be suitable for formalization, but no claim of kernel checking is made.

[Lean draft, not yet accepted]
: A Lean source for the statement exists in the repository tree, but it has *not* yet been accepted by the kernel at the pinned state. A draft is weaker evidence than **[paper proof in this report]** prose backed by a written proof, not stronger: it records intent to formalize, nothing more.

[interval computation]
: A finite statement was checked by outward-rounded interval arithmetic. The script, parameters, worst margins, and manifest hashes are recorded in Section 19 and Appendix F. The trusted base includes CPython and `mpmath`; this is not interchangeable with a Lean certificate.

[classical/open background]
: A classical theorem quoted from the literature.

An earlier audit, performed against commit `423ac24`, correctly observed that three modules cited by the survey — `FiniteCheck4096`, `OddCore` and `FractionalPartDensity` — were absent then. That discrepancy is resolved: the three modules were subsequently committed in `98646b455` and are kernel-verified; the survey's temporary pending-verification remark was removed in `716617d72`. Statements from these modules are therefore labeled [kernel-verified Lean] below. The kernel-checked feature wave also includes `PrimitiveOddCore`, `LeastSolution`, `ThreeDenominatorNormalization`, `RationalPowerIndex`, `PrimitiveGenerator`, `ExactCounting`, `QuadraticAsymptotic`, `BlockWeyl`, `BlockEquidistribution`, `OutputNormalization`, `RationalThirdBase`, `RadicalReformulation`, `PrimitiveRadical`, `RadicalDegree`, `SharperProgressions`, `ThirdDifference`, and `HigherDifference`. Their public headline theorems were audited with `##print axioms`; only the standard Mathlib axioms `propext`, `Classical.choice`, and `Quot.sound` occur. The new determinant and reformulation modules are still drafts; the per-statement ledger is Appendix A and the module inventory is Appendix B.

1.2 Headline results

1. **Verified zero-free region.** If $2^x, 3^x \in \mathbb{Z}$ and $2^x < 4096$, then $x \in \mathbb{Z}$; equivalently every nonintegral solution has $x \geq 12$. [kernel-verified Lean] An independent interval scan extends the finite barrier to $2^x < 2^{20}$, i.e. $x > 20$. [interval computation]
2. **Transcendence.** Every nonintegral solution is transcendental; for x with $2^x \in \mathbb{Z}$, algebraicity of x is *equivalent* to integrality. $\vartheta = \log_2 3$ is transcendental. [kernel-verified Lean]
3. **Exact conditional structure.** Conditional on a counterexample there is a unique gcd-normalized odd core $w > 1$, every candidate has unique coordinates $2^i w^j$, and the solution set has the exact free-monoid classification $\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta$ with a unique least nonintegral generator β, which is affinely primitive. Candidate counts below N are at most $c\log_2 N \cdot c\log_3 N$, each dyadic block $[2^T, 2^{T+1})$ holds at most $c\log_3(2^{T+1})$ candidates, the exact block count is $\lfloor (N+1)/\beta \rfloor$, the cumulative count satisfies $C(T)/T^2 \to 1/(2\beta)$, and the conjecture is equivalent to subquadratic growth along 2^T. Exact block averages of every finite trigonometric polynomial converge to its constant Fourier coefficient. [kernel-verified Lean] The extension to all continuous tests and interval indicators remains paper-level. [paper proof in this report]
4. **Unconditional gap equivalences.** The conjecture is equivalent to the existence of any nonempty open subinterval of $(0,1)$ free of fractional parts $\{x\}$ of nonintegral solutions, and to those fractional parts being bounded away from 0, or from 1; the convergent denominators of ϑ obey $q_{n+1} < 2m^{q_n}$. [kernel-verified Lean]
5. **Sharper local combinatorics.** For every $r \geq 2$, the sharp criterion $|\,(\vartheta)_{\underline{r}}\,|\, h^r < n^{r-\vartheta}$ forbids $r+1$ candidates in arithmetic progression, uniformly in the progression length. The underlying arbitrary-order forward-difference mean-value identity is formalized as well, and for the three-term case the rational effective criterion $h^{400} \leq n^{83}$ (step exponent $83/400 = 0.2075$) is verified independently. [kernel-verified Lean]
6. **New: arbitrary-node repulsion and the semigroup hierarchy.** Any $r+1$ distinct natural candidates – with no arithmetic-progression hypothesis – obey a quantitative product-of-gaps lower bound, giving explicit local packing bounds and a best ordinary-power exponent $0.241503\ldots$ at determinant order 4 (Section 14). Enlarging the exponent set to the mixed semigroup $\mathbb{N}_0 + \mathbb{N}_0\vartheta$ yields arbitrarily large positive integer generalized Vandermonde determinants, a near-linear repulsion exponent $1 - O(r^{-1/2})$, and the unconditional dyadic bound $\#(\mathcal{C} \cap [X, 2X]) \leq 10000 \max\{1, \log X\}^2$ (Section 15). This route is independent of the repository's prime-valuation rank theorem. [paper proof in this report] Lean sources `ArbitraryNodeDeterminant`, `MixedExponentSemigroup`, `SemigroupDeterminant`,

`FallingFactorialBound`, and `SemigroupWeightBound` exist but are not yet accepted. **[Lean draft, not yet accepted]**

7. **New: a sufficient condition from products of prime logarithms.** Suppose $2^x = A \in \mathbb{Z}$ and $3^x = M \in \mathbb{Z}$. Eliminating x gives the exact identity

$$\log A \, \log 3 = \log M \, \log 2.$$

Expanding A and M into prime factorizations turns this into an integer-coefficient linear form in the pairwise products $\log p \, \log q$ of prime logarithms, and the form is nontrivial precisely when $x \notin \mathbb{Z}$. Hence: *if the products* $\log p \, \log q$*, over unordered pairs of primes, are linearly independent over* $\mathbb{Q}$*, the conjecture holds* (Section 12). That independence is a consequence of Schanuel's conjecture. It is different in kind from the four exponentials conjecture – it constrains a bilinear lattice of logarithm products rather than a 2×2 logarithm determinant – so it supplies a genuinely separate route to the same conclusion, not a reformulation of the same wall. **[paper proof in this report]** A Lean module `LogProductIndependence` is in preparation and has not yet been accepted by the kernel. **[Lean draft, not yet accepted]**

8. **The irreducible remainder.** The conjecture is equivalent to: no odd integer $u > 1$ satisfies $u^{\vartheta} \in \mathbb{Z}[1/3]$, with a symmetric base-swapped formulation and an equivalent reduction to canonical power-primitive odd bases. The least radical index gives an irreducible binomial and the exact algebraic degree. **[kernel-verified Lean]** The exclusion itself remains open even for $u = 3$.

1.3 The strongest new conclusions

Determinant hierarchy (Sections 14 and 15). For every $r \geq 2$, any $r+1$ distinct natural candidates obey a quantitative product-of-gaps lower bound, with no arithmetic-progression hypothesis. Enlarging the exponent set from ordinary powers to the mixed semigroup $\mathbb{N}_0 + \mathbb{N}_0\vartheta$ supplies arbitrarily large positive integer generalized Vandermonde determinants; their archimedean upper bounds give a near-linear repulsion exponent $1 - O(r^{-1/2})$ and, unconditionally,

$$\#(\mathcal{C} \cap [X, 2X]) \ll (\log X)^2,$$

with the explicit constant 10000 proved below. Paper proof in this report; a Lean formalization is drafted but not yet accepted. **[paper proof in this report]** **[Lean draft, not yet accepted]**

The one-logarithm deficit (Section 16). To make the semigroup determinant nearly force a gap comparable with X, one needs $r \asymp (\log X)^2$ candidate nodes. Conditional on a counterexample, the exact ProveIt classification supplies only $\asymp \log X$ candidates in $[X, 2X]$. Purely archimedean interpolation, at its present strength, therefore misses the conditional population by exactly one power of $\log X$. A successful refinement must either reduce the determinant's node requirement to $O(\log X)$ or extract additional nonarchimedean size from the same nodes. **[paper proof in this report]**

1.4 Why this is not a proof

The conjecture is already a special integral instance of the real four exponentials conjecture. In the natural 2×2 matrix

$$\begin{pmatrix} 1 \cdot \log 2 & 1 \cdot \log 3 \\ x \log 2 & x \log 3 \end{pmatrix},$$

the four exponentials are $2, 3, 2^x, 3^x$. For a nonintegral solution, both the row pair and the column pair are rationally linearly independent, yet all four exponentials are algebraic. The conclusion needed here is therefore exactly what four exponentials predicts and what the known six exponentials theorem does not supply in a 2×2 configuration.

The additional integrality hypotheses are stronger than mere algebraicity, and they do yield the new determinant inequalities of Sections 14 and 15 and the bilinear criterion of Section 12. They do not, by themselves, produce enough nodes, enough lower-bound divisibility, or the required independence of logarithm products to cross the four-exponentials threshold. Conjecture 2.1 therefore remains marked open throughout this report.

2 The problem and its equivalent forms

2.1 Original statement

Conjecture 2.1 (Alaoglu–Erdős, two-base integer form)**.** *For every $x \in \mathbb{R}$,*

$$2^x \in \mathbb{Z} \quad \textit{and} \quad 3^x \in \mathbb{Z} \qquad \Longrightarrow \qquad x \in \mathbb{Z}.$$

Status: recorded in Lean as the proposition `AlaogluErdosConjecture`, without being asserted.

Since 2^x and 3^x are positive, the hypotheses actually put them in $\mathbb{N}$; and $2^x \in \mathbb{N}$ forces $x \geq 0$, since a negative exponent gives $0 < 2^x < 1$. All solutions therefore live in $[0, \infty)$ (`nonneg_of_two_rpow_integer`), and "integer" may be replaced throughout by "positive natural number."

Historical placement. Questions of this type go back to the 1944 study of colossally abundant numbers by Alaoglu and Erdős [1], where the arithmetic nature of simultaneous rational powers p^x, q^x of distinct primes controls the quotients of consecutive colossally abundant numbers — a theme reaching back to Ramanujan's superior highly composite numbers [2]. Alaoglu and Erdős isolated the two-prime assertion explicitly, regarded it as very likely, could not prove it, and reported a communicated three-prime result of Siegel. In modern language Siegel's case is supplied by the six exponentials theorem, whereas the two-base statement sits at the unresolved 2×2 threshold of the four exponentials conjecture (Schneider's first problem [9]; see Waldschmidt [10, 13]). The elementary-looking integrality question is thus one of the canonical concrete tests of the gap between six and four exponentials, not an isolated recreational puzzle. As of the dated literature check for this report, the four exponentials conjecture remains open.

2.2 Candidate-base normalization

Put

$$\vartheta := \frac{\log 3}{\log 2} = \log_2 3 = 1.584962500721156\ldots$$

and define the *candidate set*

$$\mathcal{C} := \{m \in \mathbb{N} : m > 0,\ m^\vartheta \in \mathbb{N}\}.$$

Lemma 2.2 (Exponent–candidate bijection)**.** *The maps $x \mapsto 2^x$ and $m \mapsto \log_2 m$ are mutually inverse bijections between $\mathcal{S} := \{x \in \mathbb{R} : 2^x, 3^x \in \mathbb{Z}\}$ and $\mathcal{C}$. Under the bijection, $3^x = m^\vartheta$.*

Proof. If $x \in \mathcal{S}$, set $m = 2^x \in \mathbb{N}$. Then $m^\vartheta = \exp(\log m \cdot \log 3/\log 2) = \exp(x \log 3) = 3^x \in \mathbb{N}$. Conversely, if $m \in \mathcal{C}$ and $x = \log_2 m$, then $2^x = m$ and the same identity gives $3^x = m^\vartheta \in \mathbb{N}$. $\square$

Status: [kernel-verified Lean] —
`exists_twoBaseNaturalCandidate_of_two_three_rpow_integer`,
`TwoBaseNaturalCandidate.integer_powers`.

Integral solutions correspond exactly to powers of two, by the identifier

`twoBaseCandidateExponent_isInteger_iff_isPowerOfTwo`,

so

$$\boxed{\text{Conjecture 2.1} \iff \mathcal{C} = \{2^n : n \in \mathbb{N}_0\}.} \tag{1}$$

This equivalence is itself formalized, as

`alaogluErdosConjecture_iff_candidates_are_powers_of_two`.

The candidate formulation is well suited to unique factorization, finite certification, finite differences of t^ϑ, multiplicative-rank arguments, and the determinant constructions of this report, and it is the formulation in which most of the corpus is stated.

Two closure properties of $\mathcal{S}$ are elementary but essential: $\mathcal{S}$ is closed under addition and under multiplication by positive naturals, so it is an additive submonoid of $[0, \infty)$ containing $\mathbb{N}$ (`rpow_integer_add`, `rpow_integer_nat_mul`).

2.3 The logarithmic determinant

Writing

$$M = 2^x, \qquad A = 3^x,$$

a solution is equivalently a positive integer pair satisfying

$$\log M\ \log 3 - \log A\ \log 2 = 0, \tag{2}$$

and the conjecture says that all positive integer solutions of (2) are

$$(M, A) = (2^n, 3^n), \qquad n \in \mathbb{N}_0.$$

Thus the problem is the vanishing of a 2×2 determinant of logarithms of algebraic numbers,

$$\det \begin{pmatrix} \log M & \log 2 \\ \log A & \log 3 \end{pmatrix} = 0,$$

with the additional information that two of the four entries are logarithms of *unknown integers*. This is not merely an analogy with the four exponentials conjecture; it is its exact local geometry.

2.4 One-parameter subgroup form

Let

$$\Gamma = \{(2^t, 3^t) : t \in \mathbb{R}\} \subset \mathbb{R}^2_{>0}.$$

Then Conjecture 2.1 is the statement

$$\Gamma \cap \mathbb{N}^2 = \{(2^n, 3^n) : n \in \mathbb{N}_0\}.$$

The curve Γ is a real one-parameter subgroup of the multiplicative torus $\mathbb{G}_m^2(\mathbb{R})$, and the assertion is that it meets the integer lattice points of the torus only at the obvious places. The difficulty is that its direction vector $(\log 2, \log 3)$ is transcendental rather than algebraic, so standard algebraic-subgroup intersection theorems do not apply directly.

2.5 The four-exponentials reduction

Suppose x is nonintegral and $2^x, 3^x$ are algebraic. The rational-exponent argument (Lemma 2.3) shows that x cannot be rational, so $1, x$ are $\mathbb{Q}$-linearly independent; and $\log 2, \log 3$ are $\mathbb{Q}$-linearly independent since $2^a 3^b = 1$ forces $a = b = 0$. The four exponentials built from these two pairs are

$$e^{1\cdot\log 2} = 2, \quad e^{1\cdot\log 3} = 3, \quad e^{x\log 2} = 2^x, \quad e^{x\log 3} = 3^x,$$

all four algebraic — contrary to the four exponentials conjecture. This reduction is formalized: `alaogluErdosConjecture_of_realFourExponentialsConjecture`. **[kernel-verified Lean]**

The reduction is a diagnostic, not just context: any proposed elementary argument must exploit integrality, positivity, or arithmetic structure beyond bare algebraicity of those four values, or else it would prove a substantial case of the four exponentials conjecture by another name.

2.6 Rational and algebraic exponents

Lemma 2.3 (Rational exponent)**.** *If $x \in \mathbb{Q}$ and $2^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$. Consequently every nonintegral solution is irrational.*

Proof. Write $x = a/b$ in lowest terms with $b > 0$. If $2^{a/b} = m \in \mathbb{N}$, then $2^a = m^b$; comparing the exponent of 2 gives $a = b\,v_2(m)$, so $b \mid a$ and $b = 1$. □

Status: **[kernel-verified Lean]** —
`IntegerExponent.integer_of_rational_of_two_rpow_integer`,
`irrational_of_not_integer_of_two_rpow_integer`.

Gelfond–Schneider upgrades "irrational" to "transcendental" and removes the algebraic irrational case entirely.

Lemma 2.4 (Transcendence dichotomy). *If $2^x \in \mathbb{Z}$, then either $x \in \mathbb{Z}$ or x is transcendental. In particular every counterexample to Conjecture 2.1 is a transcendental number, and $\vartheta = \log 3/\log 2$ is itself transcendental.*

Status: **[kernel-verified Lean]** — `transcendental_of_not_integer_of_two_three_rpow_integer`, `transcendental_logThreeDivLogTwo`; the supporting Gelfond–Schneider material is discussed in Section 4.

Together with the finite verification these lemmas already remove several whole regimes. In summary form, the pinned corpus proves the following unconditional restrictions on a hypothetical counterexample x.

- x is irrational, and in fact transcendental. **[kernel-verified Lean]**
- ϑ is transcendental. **[kernel-verified Lean]**
- $x \geq 12$, by the kernel-checked zero-free region $2^x < 4096$. **[kernel-verified Lean]**
- $2^x \geq 2^{20}$, hence $x \geq 20$, by the independent interval scan; this raises the barrier only under the stated software trust boundary and is not a kernel certificate. **[interval computation]**
- The real four exponentials conjecture implies Conjecture 2.1. **[kernel-verified Lean]**

These facts eliminate the rational, algebraic-irrational, small, and routine three-exponentials cases. They also explain why an elementary manipulation of logarithms is unlikely to finish the proof: the unknown lies exactly where present transcendence theory changes from theorem to conjecture.

2.7 The three-base theorem and its determinant architecture

By contrast with the two-base problem, three multiplicatively independent bases fall to the six exponentials mechanism, and the special case needed here is fully formalized.

Theorem 2.5 (Three-base theorem). *If $2^x, 3^x, 5^x \in \mathbb{Z}$ for a real x, then $x \in \mathbb{Z}$.*

Status: **[kernel-verified Lean]** — modules `IntegerExponent`, `IntegerExponent.DeterminantBound`, `IntegerExponent.ExponentialKernel`, `IntegerExponent.Grid`, following Waldschmidt's square-determinant proof [12].

In outline, for irrational x one forms a large square matrix $\mathcal{A}_{ij} = \exp(a_i b_j)$, where the row arguments a_i encode monomials in $2, 3, 5$ and the column arguments b_j encode affine forms in $1, x$. Irrationality makes the column parameters pairwise distinct and multiplicative independence makes the row parameters pairwise distinct; a total-positivity argument gives $\det \mathcal{A} \neq 0$; integrality of all entries forces $|\det \mathcal{A}| \geq 1$; and an interpolation estimate with explicit size choices gives $|\det \mathcal{A}| < 1$, a contradiction.

The dimension count is exactly where the argument stops short of Conjecture 2.1. With three independent row directions and two column directions the analytic upper bound decays faster than the arithmetic lower bound 1 permits, and the contradiction closes. Shrinking to two bases removes one row direction, and the same balance then lands precisely on the classical boundary where the decay below one fails. Nothing in the proof is wasteful enough to be recovered by sharper constants: a successful two-base determinant of this shape would be a new four-exponentials argument, which is why Section 14 and Section 15 enlarge the *node set* rather than tune the estimate.

A denominator-controlled rational third output. The rational-output extension now formalized is deliberately conditional. Let $a > 1$ and assume the monomial map

$$(i, j, k) \longmapsto 2^i 3^j a^k$$

is injective. If $2^x = m_2$ and $3^x = m_3$ are natural numbers, $a^x = q \in \mathbb{Q}$, and for some $u, v \in \mathbb{N}_0$ the reduced denominator satisfies

$$\operatorname{den}(q) \mid m_2^u m_3^v,$$

then $x \in \mathbb{Q}$, and in fact $x \in \mathbb{Z}$ because 2^x is integral. A concrete corollary obtains the monomial injectivity when a has a prime divisor different from 2 and 3.

On the determinant side, a nonzero $m \times m$ real determinant whose entries become integers after multiplication by a common positive denominator Q satisfies $|\det A| \geq Q^{-m}$. The explicit analytic estimate also retains a denominator margin: for $m = 2r$, $r > 2$, $192C \leq r$, and $bs + 2r < r^2$, the verified upper-bound expression remains strictly below 1 even after multiplication by b^s.

Status: [kernel-verified Lean] — `one_div_pow_le_abs_det_of_common_denominator`, `exponential_det_numeric_bound_with_denominator`, `rational_of_two_three_a_rpow_rational_of_den_dvd_outputs`, and `integer_of_two_three_a_rpow_rational_of_prime_factor_of_den_dvd_outputs` in `RationalThirdBase`. These are denominator-divisibility and explicit-margin results; the unrestricted determinant grid for an arbitrary rational third-base output has *not* been completed.

3 Repository timeline and verification status

The formal development advanced in three waves of Lean work, interleaved with four documentation milestones. All commits are in the repository [16]; the mathematical role of each is summarized below.

Commit	Role
`f492c18f1`	Three-base theorem via interpolation determinants (`IntegerExponent` and submodules).
`5ed4e6185`	Merge of the three-base branch.

Commit	Role
`a37d97f9d`	Basic two-base module: the conjecture as a proposition, closure under natural multiples, irrationality of nonintegral solutions, endpoint gaps, numerical bounds for ϑ, the $2^x < 10$ exclusion, second-difference obstruction, aligned-block 2/3 counting.
`4a9b03d12`	Major corpus: finite checks through $2^x < 256$; arithmetic descent; multiplicative rank; candidate structure; zero density; shrinking targets; the 3-smooth theorem; the four-exponentials reduction; transcendence consequences; and a complete Gelfond–Schneider port.
`18a85e6b8`	Merge of the two-base branch.
`423ac24f0`	First survey document (`two-base-partial-results.tex`) with rendered PDF.
`98646b455`	Three further modules, kernel-verified: `FractionalPartDensity` (density dichotomy and gap equivalences), `OddCore` (common odd core, improved counting, dyadic blocks), `FiniteCheck4096` (zero-free region to $2^x < 4096$).
`716617d72`	Survey finalized after kernel acceptance of the 4096 region.
`11ac3093f`	First independent research report (audit at `423ac24f0`, exact conditional structure, interval scan to 2^{20}); one of the sources merged into the present document.
`83ef82ab1`	Merge of the two earlier documents into a single expanded report.
`46f5b67e3`	Localization, sharp arithmetic progressions and an exact generator: `Localization`, `SharpProgression`, and the first form of `PrimitiveGenerator`.
`5d3982ceb`	Merge of the parallel conditional-structure and asymptotic formalizations: canonical primitive generator and exact solution monoid, least solution, exact dyadic counting, the quadratic cumulative asymptotic, block Weyl sums and block equidistribution, simultaneous output normalization and power-of-three denominator normalization, the rational-power index, the radical modules (radical reformulation, primitive radical, exact radical degree), the higher-difference obstruction, and denominator-controlled rational third-base outputs.
`8c9c59224`	Denominator-growth constraint $q_{n+1} < 2m^{q_n}$ on the continued fraction of a hypothetical solution (`DenominatorGrowth`), together with the report analysis naming the obstruction as a vanishing integer-coefficient linear form in products of two prime logarithms.
`2ece309cc`	Second research report: arbitrary-node Vandermonde repulsion, the semigroup generalized-Vandermonde hierarchy and its dyadic $O((\log X)^2)$ bound, the one-logarithm deficit analysis, and the compact-orbit and block-growth reformulations; the other source merged into the present document.

The audit performed in `11ac3093f` treated results of the then-uncommitted modules as article-level claims; with `98646b455` in the tree and kernel-verified, those results are now cited as committed Lean. The interval scan to 2^{20} remains at the **[interval computation]** level by design. The feature modules merged in `5d3982ceb` prove the cumulative quadratic asymptotic and the finite-trigonometric-polynomial block averages; they do *not* yet formalize the stronger continuous-test-function or interval-counting equidistribution theorem, and that distinction is maintained throughout this report.

3.1 Evidence convention

The evidence convention of this report is source-tree plus build-level, not rebuild-on-demand. Every claim marked **[kernel-verified Lean]** was checked against the cited Lean source and theorem statement at the repository state above, and the corresponding module is on the aggregate import surface `ExponentialIdentities.lean`, where it is recorded as accepted by the Lean kernel. The label does not assert that a separate fresh build was run while preparing this document; it asserts that the theorem exists, in the stated form, on an import surface the kernel has accepted.

Three weaker labels are used deliberately. **[interval computation]** marks a finite interval computation with an explicit software trust boundary and no kernel certificate; **[paper proof in this report]** marks a conventional proof given in this report, written with formalization in mind but not yet translated; and **[Lean draft, not yet accepted]** marks a Lean module that exists in the tree as a draft but has *not* been accepted by the kernel, and therefore carries no more weight than **[paper proof in this report]**. The complete label-by-label inventory is in Appendix A, and the per-module inventory in Appendix B.

4 The kernel-verified corpus

Throughout this section x denotes a hypothetical nonintegral solution, $m = 2^x \in \mathbb{N}$, $n = 3^x \in \mathbb{N}$. Every statement here is **[kernel-verified Lean]** unless labeled otherwise.

4.1 Fractional-part gaps and shrinking targets

An integral power caught strictly between consecutive integral powers of its base must clear both by at least one full unit; dividing through normalizes this to the fractional part.

Theorem 4.1 (Fractional-part gaps)**.** *Let $x \in (N, N+1)$ with $N \in \mathbb{N}$, $t = x - N$, and suppose $2^x, 3^x \in \mathbb{Z}$. Then*

$$1 + 2^{-N} \le 2^t \le 2 - 2^{-N}, \qquad 1 + 3^{-N} \le 3^t \le 3 - 3^{-N}.$$

Status: **[kernel-verified Lean]** — `two_three_fractional_part_gaps`, `two_three_rpow_gaps_near_nat`, from `rpow_integer_gap_between_consecutive_powers`.

Near the endpoints the excluded zones have width $\asymp 2^{-N}$: they *shrink* as the integer part grows. Applied to all multiples kx (which are again solutions), this yields explicit Diophantine lower bounds.

Theorem 4.2 (Effective irrationality bound)**.** *Let $x \notin \mathbb{Z}$ satisfy $2^x \in \mathbb{Z}$ and write $m = 2^x$. Then for all integers p and all $k \ge 1$,*

$$\left| x - \frac{p}{k} \right| \ \ge\ \frac{1}{2\, k\, m^k}.$$

Exact (slightly stronger) radii, and two-base versions taking the better of the base-2 and base-3 radii, are formalized as well.

Status: [kernel-verified Lean] — `rational_approximation_exponential_lower_bound`, `two_three_multiple_distance_to_integer_lower_bound`, `exists_natural_base_rational_approximation_lower_bound` in `ShrinkingTarget`.

In shorthand, $\|kx\|_{\mathbb{Z}} \gtrsim 2^{-kx}$. This bound is compatible with ordinary Diophantine approximation: Dirichlet guarantees infinitely many k with $\|kx\|_{\mathbb{Z}} < 1/k$, and the shrinking radius 2^{-kx} is incomparably smaller. The interplay between this exponential gap and equidistribution is discussed in Section 20.

The bound dualizes into a growth constraint on the continued fraction of a solution: if $|x - p/q| < 1/(qQ)$ with $q, Q \geq 1$, then $Q < 2m^q$. Applied to consecutive convergent denominators, which satisfy $|x - p_n/q_n| < 1/(q_n q_{n+1})$, this gives

$$q_{n+1} \; < \; 2\, m^{\, q_n} \; :$$

the partial quotients of a hypothetical nonintegral solution are capped by a single exponential in the preceding denominator, with base its own output $m = 2^x$.

Status: [kernel-verified Lean] — `approximation_quality_lt_of_noninteger_solution` in `DenominatorGrowth`.

4.2 Verified zero-free regions and certificate architecture

Theorem 4.3 (Zero-free region)**.** *If $2^x, 3^x \in \mathbb{Z}$ and $2^x < 4096$, then $x \in \mathbb{Z}$. Equivalently, every nonintegral solution satisfies $x \geq 12$, and every candidate that is not a power of two is at least 4096.*

Status: [kernel-verified Lean] — `integer_of_two_three_rpow_integer_of_two_rpow_lt_4096`, `twelve_le_of_not_integer_of_two_three_rpow_integer`, `le_of_twoBaseNonintegerCandidate_4096` in `FiniteCheck4096`, extending `FiniteCheck` ($2^x < 100$) and `FiniteCheck256` ($2^x < 256$, hence $x \geq 8$).

The certificate architecture is worth recording, both because it is the repository's model of a trustworthy large computation and because scaling it is a named research direction. For each non-power-of-two m the certificate is a pair of exact integer power comparisons against rational enclosures $L_p/L_q < \vartheta < U_p/U_q$ drawn from continued-fraction convergents of ϑ: from

$$a^{L_q} < m^{L_p} \quad \text{and} \quad m^{U_p} < (a+1)^{U_q}$$

one traps $a < m^{\vartheta} < a+1$, so $m^{\vartheta} \notin \mathbb{Z}$. The $2^x < 256$ stage verifies per-value certificates by `norm_num` with three enclosure tiers (exponent sizes ∼500, ∼1500, ∼25000). The extension to 4096 verifies 3836 certificates through a single kernel-replayed `decide` over a table generated by PARI/GP and revalidated independently with exact big-integer arithmetic in Python. Five tiers suffice, with tier populations 78, 488, 3241, 28, 1:

tier	lower convergent L_p/L_q	upper convergent U_p/U_q
0	569/359	485/306
1	1054/665	1539/971
2	25781/16266	24727/15601
3	50508/31867	125743/79335
4	176251/111202	301994/190537

Only the value $m = 2762$, whose ϑ-power is within $3 \cdot 10^{-9}$ of an integer in logarithmic scale, needs tier 4; its certificate compares integers of roughly 90,000 digits, which the kernel's exact arithmetic replays without difficulty. No axioms beyond the Lean kernel are involved; in particular no `native_decide`.

4.3 Arithmetic descent

Theorem 4.4 (Affine descent). *If the two integral outputs factor simultaneously as $2^x = 2^r u^k$ and $3^x = 3^r v^k$ with $r \geq 0$, $k \geq 1$, then $(x - r)/k$ is again a solution. Combined with the $x \geq 8$ region this gives the unconditional constraint*

$$r + 8k \leq x$$

for nonintegral x; in particular common perfect-power shapes of the two outputs are strongly restricted.

Status: [kernel-verified Lean] — `affine_descent_integral_powers`, `base_depth_add_eight_mul_degree_le`, `min_output_base_factorization_add_eight_le` in `ArithmeticRigidity`. With the 4096 region the constant 8 improves to 12 by bookkeeping; with the interval scan of Section 19 it would improve to 20 at the [interval computation] level.

For the *least* counterexample the same descent later becomes an exact primitivity statement in Section 5: no nontrivial matched decomposition exists at all.

4.4 Local progression obstructions and zero density

The function $t \mapsto t^{\vartheta}$ is strictly convex with second differences at moderate steps lying strictly between 0 and 1 — hence never integers.

Theorem 4.5 (Committed AP obstruction). *Let $n, h \geq 1$ with $h^5 \leq n$. Then $n, n+h, n+2h$ are not all candidates. In particular no three consecutive naturals are candidates, and among the positive integers up to $3N$ at most $2N$ are candidates.*

Status: [kernel-verified Lean] — `not_three_arithmetic_progression_twoBaseNaturalCandidates`, `not_three_consecutive_logThreeDivLogTwo_rpow_integers`, `twoBaseNaturalCandidate_Icc_card_le`. The sharper criterion $\vartheta(\vartheta-1)h^2 < n^{2-\vartheta}$ is kernel-verified in `SharperProgressions`, and the analogous four-term criterion is kernel-verified in `ThirdDifference`. The arbitrary-order statement for every $r \geq 2$ is `not_arithmetic_progression_twoBaseNaturalCandidates_of_curvature` in `HigherDifference`, which formalizes a generic iterated forward-difference mean-value identity and applies uniformly in the order r.

For finite verification the same curvature mechanism is also available in a purely rational effective form: if $h^{400} \leq n^{83}$ — step exponent $83/400 = 0.2075$, against the $1/5$ of Theorem 4.5 — then $n, n+h, n+2h$ are not all candidates.

Status: [kernel-verified Lean] — `not_three_arithmetic_progression_twoBaseNaturalCandidates_sharp` in `SharpProgression`, with the kernel-replayed enclosure $\vartheta < 317/200$ obtained from the exact integer comparison $3^{200} < 2^{317}$.

The hypothesis bounding h is essential: for steps comparable to n the second difference is large and the obstruction vanishes. This locality is a genuine barrier — a *global* progression obstruction would, for example, forbid candidates with odd part $2^s \pm 1$ via progressions such as $(2^i,\ 2^{i+t-1},\ 2^i(2^t-1))$, but no such global statement is available; under a counterexample the exact monoid of Section 5 shows sparse candidates whose additive relations the curvature argument cannot see.

Feeding the local obstruction into the theory of Roth numbers gives quantitative sparsity.

Theorem 4.6 (Zero density)**.** *For every $H \geq 1$ the number of candidates below N is at most $H^5 + \lceil (N-H^5)/H \rceil\, r_3(H)$, where $r_3(H)$ is the maximal size of a 3-AP-free subset of $\{0, \ldots, H-1\}$. Consequently the candidate counting function is $o(N)$: the candidates, and a fortiori the values 2^x of nonintegral solutions, have asymptotic density zero.*

Status: [kernel-verified Lean] — `twoBaseNaturalCandidatesBelow_card_le`, `twoBaseNaturalCandidatesBelow_isLittleO_id`, `tendsto_twoBaseNonintegerCandidatesBelow_density_zero` in `ZeroDensity` and `NonintegerDensity`.

4.5 Multiplicative rank two and the common odd core

Theorem 4.7 (Rank two)**.** *The prime-exponent vectors $\mathbf{v}(a) = (v_p(a))_p$ of all candidates span a rational space of dimension at most two. Consequently at most $(c\log_2 N)^2$ candidates lie below N.*

Status: [kernel-verified Lean] — `finrank_span_primeExponentVectors_le_two`, `twoBaseNaturalCandidatesBelow_card_le_clog_sq` in `MultiplicativeRank`. The mechanism abstracts the determinant method: three candidates with independent exponent vectors would give three multiplicatively independent bases, contradicting the three-base theorem of Section 2 through the irrationality of ϑ (`irrational_logThreeDivLogTwo`).

Since 2 is itself a candidate, all odd parts lie on at most one further rational direction. Pairwise this yields the cross-valuation identity, and globally a common odd core.

Lemma 4.8 (Cross-valuation identity)**.** *Let $a, b \in \mathcal{C}$ and let p be an odd prime with $v_p(a) > 0$. Then for every odd prime q,*

$$v_p(a)\ v_q(b) = v_p(b)\ v_q(a)\,.$$

Status: [kernel-verified Lean] — `odd_factorization_cross_mul` in `CandidateStructure`; the fixed-power relations `fixed_power_relation_of_odd_prime_factor` follow.

Theorem 4.9 (Common odd core)**.** *There is a single odd $w > 1$ such that every candidate equals $2^i w^j$ for some $i, j \in \mathbb{N}_0$; candidates that are not powers of two have $j \geq 1$. In exponent coordinates: every solution has the form $x = i + j\log_2 w$ with $i, j \in \mathbb{N}_0$.*

Status: [kernel-verified Lean] — `exists_common_odd_core`, `exists_common_odd_core_split`, `exists_common_odd_core_exponent` in `OddCore`. The canonical (gcd-normalized) version with uniqueness of w and of each coordinate pair is `existsUnique_primitive_common_odd_core` in `PrimitiveOddCore`.

Corollary 4.10 (Improved counting)**.** *The number of candidates below N is at most $\mathrm{clog}_2 N \cdot \mathrm{clog}_3 N$, and each dyadic block $[2^T, 2^{T+1})$ contains at most $\mathrm{clog}_3(2^{T+1}) \approx 0.631\,(T+1)$ candidates. In exponent coordinates, the number of solutions with $x \in [T, T+1)$ grows at most linearly in T, against the quadratic global bound.*

Status: [kernel-verified Lean] — `twoBaseNaturalCandidatesBelow_card_le_clog_mul_clog`, `twoBaseNonintegerCandidatesBelow_card_le_clog_mul_clog`, `twoBaseNaturalCandidatesInDyadicBlock_card_le` in `OddCore`.

The logarithmic-square order cannot be improved by counting alone: if a candidate with an odd prime factor exists, multiplicative closure generates injective boxes $2^i a^j$, so the count below $2^k a^l$ is at least kl (`twoBaseNaturalCandidatesBelow_card_ge_box`, `twoBaseNonintegerCandidatesBelow_card_ge_box`) — a conditional lower bound of the same quadratic-in-logarithm shape, made exact in Section 5.

4.6 Transcendence

The corpus contains a complete Lean proof of the Gelfond–Schneider theorem (Hilbert's seventh problem), following Hua's exposition [4], ported from Michail Karatarakis's Mathlib formalization (Mathlib PR #42911, Apache 2.0).

Theorem 4.11 (Gelfond–Schneider)**.** *If α, β are algebraic, $\alpha \neq 0, 1$, and β is irrational, then α^β is transcendental.*

Status: [kernel-verified Lean] —
`GelfondSchneider.transcendental_cpow_of_isAlgebraic_of_irrational`.

Corollary 4.12 (Algebraic–integral dichotomy)**.** *If $2^x \in \mathbb{Z}$ then x is either an integer or transcendental; algebraicity of such x is equivalent to integrality. Every nonintegral two-base solution is transcendental, and $\vartheta = \log_2 3$ is transcendental.*

Status: [kernel-verified Lean] —
`transcendental_of_not_integer_of_two_rpow_integer`,
`isAlgebraic_iff_integer_of_two_rpow_integer`,
`transcendental_of_not_integer_of_two_three_rpow_integer`,
`transcendental_logThreeDivLogTwo`.

The dichotomy does not resolve the conjecture — the unknown exponent is allowed to be transcendental — but it constrains every construction below: the primitive generator β of Section 5 is transcendental, as is $\log_2 w$ for the odd core.

4.7 Classification of auxiliary bases: the 3-smooth theorem

Theorem 4.13 (3-smooth classification)**.** *Suppose $x \notin \mathbb{Z}$ and $2^x, 3^x \in \mathbb{Z}$. Then for a positive natural a, the power a^x is an integer if and only if $a = 2^u 3^v$ is 3-smooth. In particular a third base with any prime factor other than 2 and 3 forces $x \in \mathbb{Z}$.*

Status: [kernel-verified Lean] —
`rpow_integer_iff_eq_two_pow_mul_three_pow_of_not_integer`,
`integer_of_two_three_a_rpow_integer_of_prime_factor` in `ThreeSmooth`.

This is an exact classification of bases for a *fixed* hypothetical counterexample. It should not be confused with the classification of candidate values 2^x as x ranges over all solutions, which is the subject of the next section. Note also the scope limitation explained in Section 20: the theorem controls the quantity b^x, not divisibility of the integer 2^x by b.

4.8 Unconditional gap equivalences

Because natural multiples of a solution are solutions, one nonintegral solution generates infinitely many, with irrational generator; Dirichlet approximation and a rotation-stepping argument spread their fractional parts densely.

Theorem 4.14 (Density dichotomy)**.** *If some $x \notin \mathbb{Z}$ satisfies $2^x, 3^x \in \mathbb{Z}$, then for every $t \in [0,1]$ and $\varepsilon > 0$ there is a nonintegral solution y with $|\{y\} - t| < \varepsilon$; and the set of nonintegral solutions is infinite.*

Theorem 4.15 (Gap equivalences)**.** *Each of the following is equivalent to Conjecture 2.1:*

(i) *some nonempty open interval $(u, v) \subseteq (0, 1)$ contains no fractional part of a nonintegral solution;*

(ii) *the fractional parts of nonintegral solutions are bounded away from 1;*

(iii) *the fractional parts of nonintegral solutions are bounded away from 0.*

Status: **[kernel-verified Lean]** — `dense_fract_of_noninteger_solution`, `infinite_noninteger_solutions_of_exists`, `alaogluErdosConjecture_iff_fract_avoids_interval`, `alaogluErdosConjecture_iff_fract_bounded_away_one`, `alaogluErdosConjecture_iff_fract_bounded_away_zero` in `FractionalPartDensity`, built on the reusable `exists_pos_nat_fract_close_of_irrational`.

Remark 4.16 (Consistency pincer)*.* Theorem 4.1 bounds $\{x\}$ away from 0 and 1 by margins $\asymp 2^{-\lfloor x \rfloor}$ that decay with the height of the solution, while Theorem 4.14 produces fractional parts arbitrarily close to 0 and 1 along multiples of unbounded height. The two results meet edge-to-edge without contradiction, and Theorem 4.15 shows the decay is essential: *any* height-independent gap, however small and wherever located, is exactly as strong as the full conjecture. The elementary gap bounds are therefore of optimal shape, and the fractional-part axis measures the entire remaining distance to Conjecture 2.1.

These equivalences are *unconditional* biconditionals. Conditional on failure, the exact structure of the next section gives kernel-verified convergence for every finite trigonometric polynomial and the paper-level strengthening to full blockwise Weyl equidistribution (Section 5).

5 Exact conditional structure of all solutions

The most consequential fact in the current corpus is not another exclusion range. It is the complete description of *all* solutions under the hypothesis that even one nonintegral solution exists. Conditional on failure of Conjecture 2.1, the solution set is not merely sparse or contained in some rank-two set: it is a free rank-two additive monoid with an explicit canonical generator, exactly countable in every dyadic block, and equidistributed blockwise on the circle.

Throughout this section we write

$$\mathcal{S} := \{x \in \mathbb{R} : 2^x \in \mathbb{N} \text{ and } 3^x \in \mathbb{N}\}, \qquad \mathcal{C} := \{2^x : x \in \mathcal{S}\}, \qquad \mathcal{P} := \{(2^x, 3^x) : x \in \mathcal{S}\},$$

and we say a counterexample exists when $\mathcal{S} \not\subseteq \mathbb{N}_0$. The canonical primitive odd core, the rational-power index, the power-of-three denominator normalization, the unique least solution, the simultaneous primitive output normalization, the exact free-monoid classification, the exact unit-block and dyadic counts, the cumulative quadratic asymptotic, and the finite-Fourier block averages are all **[kernel-verified Lean]**. Only the extension of the block averages from finite trigonometric polynomials to arbitrary continuous tests or interval indicators (Theorem 5.10) remains **[paper proof in this report]**.

5.1 The primitive common odd core

Theorem 5.1 (Primitive common odd core)**.** *Assume $\mathcal{C}$ contains a non-power of two. Then there is a unique odd integer $w > 1$ that is power-free in the sense*

$$\gcd\{v_p(w) : p \mid w\} = 1,$$

and every candidate $b \in \mathcal{C}$ has a unique representation

$$b = 2^i w^j, \qquad i, j \in \mathbb{N}_0.$$

Proof. Choose a non-power-of-two candidate a. Its odd valuation vector $A = (A_q)_{q \neq 2}$, $A_q = v_q(a)$, is nonzero and finitely supported. Let

$$g = \gcd\{A_q : A_q > 0\}, \qquad c_q = A_q/g, \qquad w = \prod_{q \neq 2} q^{c_q}.$$

Then w is odd, greater than one, and the positive c_q have gcd one.

Let $b \in \mathcal{C}$ and set $B_q = v_q(b)$. Fix an odd prime p with $A_p > 0$. By the cross-valuation identity of Section 4, $A_p B_q = B_p A_q$ for all odd q. If $A_q = 0$, this forces $B_q = 0$. On the support of A it says

$$B_q = \frac{B_p}{A_p} A_q = \Big(\frac{gB_p}{A_p}\Big) c_q.$$

Put $t = gB_p/A_p \in \mathbb{Q}_{\geq 0}$. Every tc_q is an integer. Since the c_q have gcd one, there are integers r_q, finitely many nonzero, with $\sum r_q c_q = 1$; therefore $t = \sum_q r_q(tc_q) \in \mathbb{Z}$. Writing $j = t \in \mathbb{N}_0$, the odd part of b is exactly w^j, so $b = 2^i w^j$ with $i = v_2(b)$.

Uniqueness of (i, j) for fixed w follows from the odd valuations and the 2-adic valuation. If w' is another primitive common core, the odd valuation vectors of w and w' generate the same rational ray and are both primitive integer vectors, hence equal. □

Status: [kernel-verified Lean]

`existsUnique_primitive_common_odd_core` in `PrimitiveOddCore`. The module defines `oddPrimeValuationGCD`, proves its equivalence with power-theoretic primitivity for $w > 1$, and kernel-checks both uniqueness of the normalized core and uniqueness of every pair (i, j). The earlier non-normalized existence statement `exists_common_odd_core` recorded in Section 4 is subsumed by it.

The gcd normalization is exactly what makes w canonical: it prevents a hidden perfect power from being absorbed into the exponent j, so that both the core and the coordinates are determined by $\mathcal{C}$ alone.

Set

$$\alpha := \log_2 w, \qquad E := 3^\alpha = w^\vartheta.$$

Every candidate exponent then has the form $x = i + j\alpha$. The number α is irrational (otherwise the rational-exponent lemma applied to $2^\alpha = w$ would make it integral, forcing the odd $w > 1$ to be a power of two), and in fact transcendental by the algebraic–integral dichotomy of Section 4. The candidate condition becomes

$$3^{i+j\alpha} = 3^i E^j \in \mathbb{N},$$

and at this stage the admissible pairs (i, j) still form an unknown subset of $\mathbb{N}_0^2$. The next subsection shows that this subset is completely rigid.

5.2 The rational-power index and the denominator normalization

Because a non-power-of-two candidate exists, some $j > 0$ has $E^j \in \mathbb{Q}$. Let

$$d := \min\{j \in \mathbb{N}_{>0} : E^j \in \mathbb{Q}\}.$$

Lemma 5.2 (All rational powers are multiples of the least one). *For every $j \in \mathbb{Z}$: $E^j \in \mathbb{Q} \iff d \mid j$.*

Proof. The set $J := \{j \in \mathbb{Z} : E^j \in \mathbb{Q}\}$ is an additive subgroup of $\mathbb{Z}$ (it contains 0, products add exponents, and reciprocals negate them), and every nonzero subgroup of $\mathbb{Z}$ equals $d\mathbb{Z}$ for its least positive element d. □

Status: [kernel-verified Lean]

`rationalPowerIndices`, `AddSubgroup.Int.exists_least_positive_generator`, and `exists_rationalPowerIndex` in `RationalPowerIndex`.

Write the positive rational E^d in lowest terms. Its denominator has no prime other than 3: taking a candidate with odd-core exponent $j = dk > 0$ and 2-exponent i, integrality of $3^i(E^d)^k$ shows that any other prime in the reduced denominator would survive in the k-th power. Hence there are unique integers $a \geq 0$ and $A \geq 1$ with

$$E^d = \frac{A}{3^a}, \qquad \text{and} \qquad 3 \nmid A \ \ \text{when } a > 0, \tag{3}$$

so that $a = 0$ or $3 \nmid A$ in every case. Define

$$\beta := a + d\alpha = a + d\log_2 w, \qquad M := 2^\beta = 2^a w^d, \qquad A = 3^\beta = 3^a E^d. \tag{4}$$

Thus β is itself a solution, and it is irrational because α is. (Where the denominator normalization is written with a named integer numerator c, the identity $E^d = c/3^a$ with $a = 0$ or $3 \nmid c$ is exactly (3) with $c = A = 3^\beta$.)

Status: [kernel-verified Lean]

The denominator-support step is `exists_three_pow_denominator_of_rational_power` in `ThreeDenominatorNormalization`; the normalized integrality criterion is `three_pow_mul_normalized_pow_integer_iff` in `RationalPowerIndex`.

Equation (3) also has a field-theoretic reading: E is a root of the binomial $X^d - A/3^a$, which is the minimal polynomial of E over $\mathbb{Q}$, so $[\mathbb{Q}(E) : \mathbb{Q}] = d$. That computation, together with the equivalent one-number and localized-radical formulations of the conjecture that follow from it, is developed in Section 6; we do not repeat it here.

5.3 The exact primitive-generator theorem

Theorem 5.3 (Exact primitive generator and solution monoid). *Assume Conjecture 2.1 is false. With w, d, a, β, M, A as above:*

(i) *every solution has a unique representation* $x = n + k\beta$ *with* $n, k \in \mathbb{N}_0$*, that is,*

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\};$$

(ii) *every candidate has a unique representation* $m = 2^n M^k$ *with* $n, k \in \mathbb{N}_0$*, that is,*

$$\mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\};$$

(iii) *every output pair has a unique representation* $(2^x, 3^x) = (2^n M^k, 3^n A^k)$*;*

(iv) β *is the unique least nonintegral solution.*

Proof. Let $x \in \mathcal{S}$. By Theorem 5.1, $2^x = 2^i w^j$ for unique $i, j \in \mathbb{N}_0$, so $x = i + j\alpha$. From $3^x = 3^i E^j \in \mathbb{N}$ we get $E^j \in \mathbb{Q}$, so Lemma 5.2 gives $j = dk$ for some $k \in \mathbb{N}_0$. Using (3),

$$3^x = 3^i (E^d)^k = 3^{i-ak} A^k.$$

Because $A/3^a$ is reduced, this is an integer exactly when $i \geq ak$. Put $n = i - ak \in \mathbb{N}_0$; then

$$x = i + dk\alpha = n + k(a + d\alpha) = n + k\beta.$$

Conversely, for any $n, k \in \mathbb{N}_0$,

$$2^{n+k\beta} = 2^n M^k \in \mathbb{N}, \qquad 3^{n+k\beta} = 3^n A^k \in \mathbb{N},$$

so $n + k\beta \in \mathcal{S}$. If $n + k\beta = n' + k'\beta$ with $k \neq k'$, then $\beta = (n' - n)/(k - k') \in \mathbb{Q}$, a contradiction; hence the representation is unique. This proves (i), and (ii)–(iii) follow by exponentiation. A nonintegral solution has $k \geq 1$; the least such value is attained at $(n, k) = (0, 1)$ and equals β, uniquely so by uniqueness of the representation. □

Status: [kernel-verified Lean]

`exists_canonical_primitiveGenerator_of_not_alaogluErdosConjecture` in `PrimitiveGenerator`, built from `exists_exact_solution_monoid_of_fixed_common_oddCore` in `RationalPowerIndex`. The Lean statement supplies the canonical odd core w, the rational-power index d, the depth a, the numerator A, and the unique-coordinate clauses in one package. Conditional on failure, $\mathcal{S}$ is not merely contained in a rank-two additive set: every solution has a unique pair of coordinates in the free additive monoid on 1 and the irrational least generator β. The set of occurring odd-core exponents is exactly $d\,\mathbb{N}_0$, with the power of two governed by one linear threshold.

Packaged algebraically, Theorem 5.3 says that

$$(\mathbb{N}_0^2, +) \xrightarrow{\sim} (\mathcal{S}, +), \quad (n, k) \mapsto n + k\beta, \qquad (\mathbb{N}_0^2, +) \xrightarrow{\sim} (\mathcal{C}, \cdot), \quad (n, k) \mapsto 2^n M^k,$$

are monoid isomorphisms, and that the output-pair monoid $\mathcal{P}$ is freely generated by $(2, 3)$ and (M, A). Every multiplicative relation among candidates is the obvious coordinate relation; there is no room for an unexpected coincidence. This exact conditional model is the benchmark against which any future density, progression, or descent argument must be tested.

The phrase "least nonintegral" is materially stronger than choosing an arbitrary counterexample. It forces affine primitivity: no nontrivial common root can be extracted from the two output integers while preserving the same base-adic offset. It is the arithmetic analogue of choosing a primitive lattice direction, and the next subsection turns it into exact exclusions.

5.4 Primitivity of the least output pair

Corollary 5.4 (No matched base depth)**.** *For the primitive pair* $(M, A) = (2^\beta, 3^\beta)$ *we have* $\min\{v_2(M), v_3(A)\} = 0$*; that is,* M *is odd or* $3 \nmid A$ *(possibly both).*

Proof. If both valuations were positive, then $2^{\beta-1} = M/2$ and $3^{\beta-1} = A/3$ would be integers, so $\beta - 1$ would be a smaller nonintegral solution. □

Corollary 5.5 (No common perfect-power degree)**.** *There is no* $k \geq 2$ *for which both* M *and* A *are perfect* k*-th powers.*

Proof. If $M = U^k$ and $A = V^k$, then $2^{\beta/k} = U$ and $3^{\beta/k} = V$, so β/k is a smaller nonintegral solution. □

Corollary 5.6 (Exact affine primitivity)**.** *If* $M = 2^r U^k$ *and* $A = 3^r V^k$ *with* $r \in \mathbb{N}_0$*,* $k \in \mathbb{N}_{>0}$*, and* $U, V \in \mathbb{N}$*, then* $r = 0$ *and* $k = 1$*.*

Proof. The affine descent theorem of Section 4 produces the solution $(\beta - r)/k$, which is nonintegral since β is irrational; if $r > 0$ or $k > 1$ it is strictly smaller than β, contradicting leastness. □

Status: [kernel-verified Lean]

`exists_least_twoBaseNonintegerSolution`,
`IsLeastTwoBaseNonintegerSolution.unique`,
`IsLeastTwoBaseNonintegerSolution.not_both_base_dvd_outputs`,
`IsLeastTwoBaseNonintegerSolution.no_common_perfect_power`, and
`IsLeastTwoBaseNonintegerSolution.affine_primitive` in `LeastSolution`.

The odd-core normalization of M has a symmetric counterpart on the other output, and the two can be imposed simultaneously.

Theorem 5.7 (Simultaneous primitive output normalization)**.** *If the conjecture fails, its least nonintegral solution* β *admits simultaneous factorizations*

$$M = 2^\beta = 2^a w^d, \qquad A = 3^\beta = 3^b v^e,$$

where $w > 1$ *is odd and power-primitive,* $v > 1$ *is power-primitive with* $3 \nmid v$*, and* $d, e > 0$*. Moreover* $a = 0$ *or* $b = 0$*; the exponents* a *and* b *are the exact base-adic depths;* d *and* e *are the gcds of the prime valuations of the corresponding base-free parts; and*

$$\gcd(\gcd(d, e), |b - a|) = 1.$$

Since every counterexample satisfies $\beta \geq 12$*, these normalizations also obey the size dichotomy*

$$w^d \geq 4096 \qquad \textit{or} \qquad v^e \geq 3^{12}.$$

Status: [kernel-verified Lean]

`exists_simultaneous_primitive_output_normalization`, `IsLeastTwoBaseNonintegerSolution.simultaneousCoordinateGCD_eq_one`, and `exists_large_primitiveCore_of_not_alaogluErdosConjecture` in `OutputNormalization`. The gcd-one statement simultaneously controls the two outer power degrees and the difference of the two base-adic depths; it is strictly stronger than merely excluding a common perfect-power degree (Corollary 5.5).

These restrictions are strong, but they still admit infinitely many formal integer parameter patterns. The unresolved analytic content is untouched: the pair (M, A) must satisfy

$$\frac{\log M}{\log 2} = \frac{\log A}{\log 3},$$

and no known local valuation argument turns the depth dichotomy $a = 0$ or $b = 0$, or the gcd-one condition, into a contradiction. Every constraint proved so far is compatible with the existence of such a pair; what fails to exist must fail for a transcendence reason, not a valuation-theoretic one.

5.5 Exact counting

The free-monoid description makes the population of a hypothetical counterexample completely explicit, block by block.

Theorem 5.8 (Exact unit-block and dyadic counts)**.** *Assume a counterexample exists and let β be the primitive generator. For every $N \in \mathbb{N}_0$, the nonintegral solutions in $[N, N+1)$ are precisely*

$$x_{N,k} = \lceil N - k\beta \rceil + k\beta, \qquad 1 \le k \le \Big\lfloor \frac{N+1}{\beta} \Big\rfloor,$$

so their number is exactly $\lfloor (N+1)/\beta \rfloor$. Consequently

$$\#(\mathcal{S} \cap [N, N+1)) = \#(\mathcal{C} \cap [2^N, 2^{N+1})) = 1 + \Big\lfloor \frac{N+1}{\beta} \Big\rfloor.$$

Proof. For fixed $k \ge 1$ there is at most one integer n with $N \le n + k\beta < N+1$; such an $n \ge 0$ exists if and only if $k\beta < N+1$, and then $n = \lceil N - k\beta \rceil$. Irrationality of β makes $(N+1)/\beta$ nonintegral, which gives the stated range. The unique integral solution in the block is N itself, whence the additional 1; passing through $m = 2^x$ turns the exponent block $[N, N+1)$ into the magnitude block $[2^N, 2^{N+1})$ bijectively. □

Status: [kernel-verified Lean]

`ExactCounting` proves the explicit ceiling-image parametrization, `twoBaseNonintegerSolutionsInUnitBlock_ncard`, `twoBaseNaturalCandidatesInDyadicBlock_card`, and the exact finite-sum identity `cumulativeTwoBaseCandidateCount_eq`. `QuadraticAsymptotic` identifies that cumulative count with the candidate count below 2^T and proves the sharp bounds and the limit in (5).

In particular a failed conjecture would produce only linearly many candidates in the logarithmic block index,

$$\#(\mathcal{C} \cap [X, 2X]) \asymp \log X,$$

and the kernel-verified zero-free region gives $\beta \geq 12$, hence $\beta > 12$ because β is irrational. The block count is therefore at most $\lfloor N/12 \rfloor$ (and at most $\lfloor N/20 \rfloor$ under the interval scan, which is **[interval computation]**). Summing over blocks, the cumulative candidate count below 2^T is

$$C(T) = T + \sum_{r=1}^{T} \left\lfloor \frac{r}{\beta} \right\rfloor = \frac{T^2}{2\beta} + O(T). \tag{5}$$

More precisely, Lean proves the two-sided bound and the limit

$$\frac{T(T+1)}{2\beta} \leq C(T) \leq \frac{T(T+1)}{2\beta} + T, \qquad \frac{C(T)}{T^2} \longrightarrow \frac{1}{2\beta}.$$

This is conditionally sharper than the unconditional verified bounds ($(\operatorname{clog}_2 N)^2$ and $\operatorname{clog}_2 N \cdot \operatorname{clog}_3 N$ of Section 4) and matches them in leading order: with $\beta > 12$ the leading coefficient $1/(2\beta)$ is below $1/24$.

There is also an unconditional growth reformulation, obtained by comparing the two cases. Under the conjecture the count below 2^T is exactly T; under failure the preceding limit is the strictly positive constant $1/(2\beta)$. Hence

$$\text{Conjecture 2.1 holds} \iff \frac{\#(\mathcal{C} \cap [1, 2^T))}{T^2} \longrightarrow 0, \tag{6}$$

which is `alaogluErdosConjecture_iff_candidate_count_below_two_pow_subquadratic`. **[kernel-verified Lean]**

This exact population is the correct benchmark for every proposed density or determinant argument. A method that proves only $O((\log X)^2)$ candidates in $[X, 2X]$, for instance, is a valid unconditional theorem but cannot contradict Theorem 5.8: it is weaker than the conditional truth it would need to refute.

5.6 Blockwise irrational rotation and equidistribution

By Theorem 5.8 the fractional part of $x_{N,k}$ is $\{k\beta\}$, so a high unit block contains exactly an initial segment

$$\{\beta\},\ \{2\beta\},\ \ldots,\ \{K_N\beta\}, \qquad K_N = \left\lfloor \frac{N+1}{\beta} \right\rfloor,$$

of one irrational rotation orbit, and $K_N \to \infty$.

Theorem 5.9 (Finite Fourier tests on exact blocks)**.** *Assume a counterexample exists and let β be its primitive generator. For every finitely supported coefficient function $c : \mathbb{Z} \to \mathbb{C}$, put*

$$P_c(t) = \sum_{h \in \mathbb{Z}} c(h) e^{2\pi i h t}.$$

Then, on the exact ceiling-parametrized points $x_{N,k} = \lceil N - k\beta \rceil + k\beta$,

$$\frac{1}{K_N} \sum_{k=1}^{K_N} P_c(x_{N,k}) \longrightarrow c(0) \qquad (N \to \infty).$$

Status: [kernel-verified Lean]

`BlockWeyl` proves vanishing of every nonzero normalized Fourier mode on the exact block points, and `BlockEquidistribution` proves `tendsto_primitiveUnitBlockTrigonometricAverage` for every finitely supported coefficient function, with limit `c 0`. This is convergence against finite trigonometric polynomials; it is not yet a theorem for arbitrary continuous tests.

Theorem 5.10 (Blockwise Weyl equidistribution)**.** *Assume a counterexample exists. For every interval $I \subseteq [0,1]$ whose boundary has measure zero,*

$$\frac{\#\{x \in (\mathcal{S} \setminus \mathbb{N}_0) \cap [N, N+1) : \{x\} \in I\}}{\#\big((\mathcal{S} \setminus \mathbb{N}_0) \cap [N, N+1)\big)} \longrightarrow |I| \qquad (N \to \infty).$$

Proof. By Theorem 5.8 the relevant fractional parts are $\{\beta\}, \{2\beta\}, \dots, \{K_N\beta\}$ with $K_N = \lfloor (N+1)/\beta \rfloor \to \infty$. For every nonzero integer h the Weyl sum is a finite geometric series,

$$\frac{1}{K}\sum_{k=1}^{K} e^{2\pi i h k\beta} = \frac{e^{2\pi i h\beta}\big(1 - e^{2\pi i h K\beta}\big)}{K\big(1 - e^{2\pi i h\beta}\big)} \longrightarrow 0,$$

since β is irrational and hence $e^{2\pi i h\beta} \neq 1$. Weyl's criterion concludes. □

Status: [paper proof in this report]

The finite-trigonometric-polynomial input is kernel-verified by Theorem 5.9, but the Stone–Weierstrass and Weyl-criterion extension to all continuous tests, and then to interval indicators, has not been formalized. No effective discrepancy rate follows without further Diophantine input on β: the geometric-sum proof controls each fixed Fourier mode but not uniformly over many modes.

This regularity is attractive, and it is worth stating plainly why it does not by itself decide anything. The integrality condition being tested is not a fixed positive-measure target in the circle; it is an exponentially shrinking target, since the relevant approximation quality must improve like the reciprocal of the integer sizes involved. Equidistribution at a fixed scale is simply the wrong instrument for a target whose measure tends to zero geometrically, and that mismatch recurs in the analytic approaches examined later in this report.

6 Equivalent one-number and radical formulations

The exact conditional structure of Section 5 compresses a statement quantified over all real x into a statement about a single real number at a time. This section records that compression in its two kernel-verified forms — a localization at the prime 3 and a radical-algebraicity normalization — and then names the exact arithmetic obstruction that remains.

6.1 Localization at the prime three

Theorem 6.1 (Localization reformulation)**.** *Conjecture 2.1 is false if and only if there exists an odd integer $u > 1$ with*

$$u^{\vartheta} \in \mathbb{Z}[1/3],$$

equivalently, an odd $u > 1$ and $a \in \mathbb{N}_0$ with $3^a u^\vartheta \in \mathbb{N}$.

Proof. If x is a counterexample, write $2^x = 2^a u$ with u odd; since 2^x is not a power of two, $u > 1$. Using $(2^a)^\vartheta = 3^a$,

$$u^\vartheta = \frac{(2^x)^\vartheta}{(2^a)^\vartheta} = \frac{3^x}{3^a} \in \mathbb{Z}[1/3].$$

Conversely, if $u^\vartheta = B/3^a$ with $B \in \mathbb{N}$, set $x = a + \log_2 u$; then $2^x = 2^a u \in \mathbb{N}$ and $3^x = 3^a u^\vartheta = B \in \mathbb{N}$, and $x \notin \mathbb{Z}$ since otherwise the odd $u > 1$ would be a power of two. ☐

Status: [kernel-verified Lean] — `alaogluErdosConjecture_iff_odd_rpow_not_localized` in `Localization` and `alaogluErdosConjecture_iff_oddLocalizedRadicalExclusion` in `RadicalReformulation`; the candidate-level variant `alaogluErdosConjecture_iff_no_odd_multiple_candidate`, also in `Localization`, is proved independently and phrases the same content in terms of admissible candidate bases. The equivalence is kernel-verified; the localized-radical exclusion on its right-hand side remains the open conjecture. The allowance of 3-power denominators is exactly the freedom created by shifting the exponent by an integer.

The denominator restriction to powers of 3 is not cosmetic. Only the prime 3 can occur in the denominator, because the exponent shift $x \mapsto x + 1$ multiplies 2^x by 2 and 3^x by 3; the odd part u absorbs the 2-adic freedom completely and $\mathbb{Z}[1/3]$ absorbs the remaining 3-adic freedom. Everything else has been eliminated.

6.2 Reduction to power-primitive odd bases

An odd $u > 1$ that is a proper power, $u = w^d$ with w odd and power-primitive, satisfies $u^\vartheta = (w^\vartheta)^d$. Unique primitive-power decomposition therefore lets the quantifier in Theorem 6.1 range over power-primitive bases only, at the cost of an extra exponent $d \geq 1$.

Status: [kernel-verified Lean] — `PrimitiveRadical` proves unique primitive-power decomposition of every integer $u > 1$ together with the exact equivalence `alaogluErdosConjecture_iff_primitiveOddLocalizedRadicalExclusion`. Proper-power choices of u introduce no genuinely new cases.

6.3 Canonical radical algebraicity

In the notation of the least generator of Section 5, a counterexample produces natural numbers w, d, a, A with $w > 1$ odd and power-primitive, $d > 0$, and

$$\beta = a + d \log_2 w, \qquad 2^\beta = 2^a w^d, \qquad 3^\beta = A,$$

normalized so that $a = 0$ or $3 \nmid A$. Setting

$$E := w^\vartheta = 3^{\log_2 w},$$

one has $E^d = A/3^a \in \mathbb{Q}$, and d is the least positive exponent for which E^d is rational. Minimality of d has an exact algebraic consequence.

Proposition 6.2 (Exact radical degree). *The binomial $X^d - A/3^a$ is irreducible over $\mathbb{Q}$, it is the minimal polynomial of E, and E is algebraic of exact degree d.*

Proof. For a positive rational q, the binomial $X^d - q$ can be reducible only if q is a p-th power in $\mathbb{Q}$ for some prime $p \mid d$ (the additional $4 \mid d$ exceptional case concerns -4 times a fourth power and is irrelevant for $q > 0$). If $q = r^p$ with $p \mid d$, the positive real $E^{d/p}$ is the positive p-th root of q, hence rational, contradicting minimality of d. □

Status: [kernel-verified Lean] — `irreducible_X_pow_sub_C_of_least_rational_power` in `RadicalDegree` proves the general norm argument, and `oddCoreRpow_exact_radical_degree` specializes it to the least normalized radical index. It identifies the minimal polynomial with $X^d - A/3^a$ and proves that the degree is exactly d.

A counterexample therefore produces the very special algebraic number $E = 3^{\log_2 w}$, reached from a transcendental exponent. Taking logarithms of $E^d = A/3^a$:

$$d \log w \, \log 3 = (\log A - a \log 3) \log 2. \tag{7}$$

This is a *bilinear* relation among logarithms of integers: each side is a product of two logarithms, not a linear form in logarithms with algebraic coefficients. Baker's theorem [3] bounds linear forms $\sum \alpha_i \log \gamma_i$ with algebraic α_i away from zero, and no substitution turns (7) into such a form, because the coefficient of $\log w$ on the left is $d \log 3$, itself a logarithm rather than an algebraic number. The four exponentials conjecture does forbid the relation. Equation (7) is thus a precise description of the transcendence-theoretic wall, not a disguised elementary factorization problem.

6.4 Where the wall stands in the smallest case

The smallest structural question already meets the open core. A 3-smooth candidate $m = 2^a 3^b$ with $b \geq 1$ would require $3^a E_3^b \in \mathbb{N}$ for

$$E_3 = 3^{\log_2 3} = e^{(\log 3)^2/\log 2} = 5.7045224946911176\ldots,$$

and even the irrationality of E_3 is open: it is a "three-logarithms" quantity outside the reach of Gelfond–Schneider, Baker's method, and the six exponentials theorem. No currently known technique excludes the odd core $w = 3$; this is why the corpus emphasizes counting, structure, and verified finite regions rather than infinite families of exclusions, and why Conjecture 6.3 below is the honest name of the obstruction.

Conjecture 6.3 (Radical four-exponentials target). *For every odd integer $u > 1$, $u^{\log_2 3} \notin \mathbb{Z}[1/3]$. Equivalently, for every power-primitive odd $w > 1$ and every $d \geq 1$, $(3^{\log_2 w})^d \notin \mathbb{Z}[1/3]$.*

By Theorem 6.1 and the primitive reduction above, both formulations are exactly equivalent to Conjecture 2.1; neither is proved. A proof of the special case $u = 3$ — even of the weaker assertion that $3^{\log_2 3}$ is irrational — by any genuinely new mechanism would already be a meaningful breakthrough.

7 The exact four-exponentials barrier

7.1 The two-by-two exponential matrix

Suppose x is a nonintegral solution. Consider the row vector of parameters $(1, x)$ and the column vector of parameters $(\log 2, \log 3)$. The four exponentials formed from the products are

$$\exp(1 \cdot \log 2) = 2, \quad \exp(1 \cdot \log 3) = 3, \quad \exp(x \log 2) = 2^x, \quad \exp(x \log 3) = 3^x,$$

all four algebraic — indeed positive integers. The two row parameters $1, x$ are $\mathbb{Q}$-linearly independent because x is irrational: a rational nonintegral exponent is excluded by the rational-exponent argument of Section 2. The two column parameters $\log 2, \log 3$ are $\mathbb{Q}$-linearly independent because $2^a 3^b = 1$ forces $a = b = 0$, that is, no nonzero powers of 2 and 3 agree. The real four exponentials conjecture predicts that at least one of the four displayed values must be transcendental, a contradiction [10].

Status: **[kernel-verified Lean]** —
`alaogluErdosConjecture_of_realFourExponentialsConjecture` in
`FourExponentialsReduction` formalizes exactly this implication. It does not assume four exponentials as an axiom: the conjectural statement is defined as a proposition and the implication is proved conditionally on it. Four exponentials itself is open, so this is context and diagnosis, not a proof of Conjecture 2.1.

The reduction is a diagnostic and not merely background. Any proposed elementary argument must exploit integrality, positivity, or arithmetic structure beyond bare algebraicity of those four values, or else it would prove a substantial case of the four exponentials conjecture by another name.

7.2 Why six exponentials does not specialize

The six exponentials theorem applies to two independent row parameters together with three independent column parameters, or to three rows and two columns. Here only two canonical logarithmic directions are available. Adding a third base 5 supplies the missing direction and yields the kernel-verified three-base theorem

$$2^x, 3^x, 5^x \in \mathbb{Z} \quad \Longrightarrow \quad x \in \mathbb{Z},$$

proved by an interpolation determinant. In the two-base problem, any additional base built from powers of 2 and 3 has logarithm in the rational span of $\log 2$ and $\log 3$ and therefore does not create an independent column.

The dimension count is what decides the outcome. With three independent row directions and two column directions the analytic size estimate beats the arithmetic lower bound $|\det| \geq 1$; in the 2×2 configuration the same balance lands exactly on the classical boundary, where the required decay below one fails. A successful two-base determinant argument would, by construction, be a new four-exponentials argument.

The classification of auxiliary bases makes the obstruction precise: `ThreeSmooth` and the denominator-controlled results in `RationalThirdBase` show that an auxiliary base with a prime factor outside $\{2, 3\}$ would supply a genuinely new column, but its corresponding exponential value is not forced to be an integer — or even algebraic — by the original hypotheses. Bases supported only on 2 and 3 do not increase the transcendence rank.

7.3 Integrality is stronger than algebraicity, but not by enough

Four exponentials asks only for algebraicity. The values here are integers, so there is extra discrete structure available, and the determinant machinery of Section 14 and Section 15 extracts some of it: the relevant determinants have integer entries, so a nonzero determinant is a nonzero integer and therefore satisfies $|\det| \geq 1$.

That inequality is the whole of the extra archimedean information. A nonzero integer has absolute value at least one, and nothing in the integrality hypotheses improves the constant 1 by itself. To cross the four-exponentials threshold one would need a strictly stronger arithmetic input, for instance:

- a proof that the determinants are divisible by a large integer, replacing $|\det| \geq 1$ by $|\det| \geq D$ with D growing in the size parameters; or
- a proof that a suitable common denominator is anomalously small, improving the arithmetic side rather than the analytic side.

Both routes ask for divisibility rather than size, which is precisely why the p-adic direction is given priority in the research program of Section 21.

Status: [paper proof in this report] for the determinant hierarchy cited here: the arbitrary-node repulsion theorem and the semigroup generalized-Vandermonde bounds are paper proofs in this report. **[Lean draft, not yet accepted]** — `ArbitraryNodeDeterminant`, `SemigroupDeterminant`, and `MixedExponentSemigroup` contain Lean drafts that the kernel has not yet accepted. The reduction of Conjecture 2.1 to four exponentials is **[kernel-verified Lean]**, but four exponentials is open and the conjecture remains open with it.

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8 Rational-function rigidity and closure properties

The exact primitive-generator theorem of Section 5 determines the additive structure of $\mathcal{S}$ completely under failure of Conjecture 2.1. It has a consequence that deserves to be isolated: a hypothetical solution monoid is not merely additively free of rank two, it is *rigid* against every nonlinear rational operation over the algebraic numbers. Addition and multiplication by naturals are the only operations that stay inside $\mathcal{S}$, and every apparent exception collapses to an identity in one indeterminate. The results below are exact equivalences, several of them unconditional, and they translate directly into closure reformulations of the conjecture itself.

8.1 Evaluation injectivity at the generator

Throughout this section, β denotes the canonical least nonintegral solution supplied by the generator theorem of Section 5, so that

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\}$$

with unique coordinates n, k. The transcendence dichotomy of Section 4 makes β transcendental over $\mathbb{Q}$. The rational-function statements need the slightly stronger form over the algebraic closure.

Lemma 8.1 (Evaluation is injective over $\overline{\mathbb{Q}}$)**.** *Let $\gamma \in \mathbb{C}$ be transcendental over $\mathbb{Q}$. Then no nonzero $P \in \overline{\mathbb{Q}}[T]$ satisfies $P(\gamma) = 0$; equivalently, the evaluation homomorphism $\overline{\mathbb{Q}}[T] \to \mathbb{C}$, $P \mapsto P(\gamma)$, is injective.*

Proof. If $P(\gamma) = 0$ for some nonzero $P \in \overline{\mathbb{Q}}[T]$, then γ is algebraic over $\overline{\mathbb{Q}}$. Since $\overline{\mathbb{Q}}/\mathbb{Q}$ is an algebraic extension, the tower property makes γ algebraic over $\mathbb{Q}$, contradicting the hypothesis. Injectivity follows by applying this to differences. □

Status: [paper proof in this report] — the transcendence of β itself is **[kernel-verified Lean]** (Section 4, `Transcendence`); the only new content here is the standard tower argument that upgrades transcendence over $\mathbb{Q}$ to transcendence over $\overline{\mathbb{Q}}$.

8.2 The rational-function intersection theorem

Theorem 8.2 (Rational-function rigidity at the primitive generator)**.** *Assume Conjecture 2.1 is false and let β be the canonical generator. Let $R(T) \in \overline{\mathbb{Q}}(T)$ have no pole at β. Then*

$$\boxed{R(\beta) \in \mathcal{S} \quad \Longleftrightarrow \quad R(T) = n + kT \text{ in } \overline{\mathbb{Q}}(T) \text{ for some } n, k \in \mathbb{N}_0.}$$

Proof. The reverse implication is additive closure of $\mathcal{S}$ together with closure under multiplication by naturals: $R(\beta) = n + k\beta \in \mathcal{S}$.

For the forward implication write $R = P/Q$ with $P, Q \in \overline{\mathbb{Q}}[T]$ and $Q(\beta) \neq 0$. If $R(\beta) \in \mathcal{S}$, the exact coordinates give $R(\beta) = n + k\beta$ for unique $n, k \in \mathbb{N}_0$, hence

$$P(\beta) - (n + k\beta)\, Q(\beta) = 0.$$

The left-hand side is the value at β of the polynomial $P(T) - (n + kT)Q(T) \in \overline{\mathbb{Q}}[T]$. By Lemma 8.1 that polynomial is identically zero, so $P(T) = (n + kT)Q(T)$ and $R(T) = n + kT$ in $\overline{\mathbb{Q}}(T)$. □

Status: [paper proof in this report]

The theorem allows poles away from β, permits arbitrary cancellation, and takes the full algebraic closure as coefficient field. It can be read as the exact intersection

$$\mathcal{S} \cap \overline{\mathbb{Q}}(\beta) = \mathbb{N}_0 + \mathbb{N}_0\beta, \tag{8}$$

with the extra information that every rational expression realizing an element of that intersection is *identically* the corresponding affine expression — there is no accidental representation.

Corollary 8.3 (Rational self-maps of the solution set)**.** *Under failure, the rational functions over $\overline{\mathbb{Q}}$ that are regular on $\mathcal{S}$ and map $\mathcal{S}$ into itself are exactly the maps*

$$T \longmapsto n + kT, \qquad n, k \in \mathbb{N}_0.$$

Proof. Such a function in particular sends β into $\mathcal{S}$, so Theorem 8.2 identifies it as $n + kT$ with $n, k \in \mathbb{N}_0$. Conversely, translation by a natural and dilation by a natural preserve $\mathbb{N}_0 + \mathbb{N}_0\beta$. □

Status: [paper proof in this report]

Proposition 8.4 (Multivariate reduction to one indeterminate)**.** *Assume failure, let $x_i = n_i + k_i\beta \in \mathcal{S}$ for $i = 1, \dots, r$, and let $F \in \overline{\mathbb{Q}}(X_1, \dots, X_r)$ be regular at $(x_1, \dots, x_r)$. Then*

$$F(x_1, \dots, x_r) \in \mathcal{S}$$

if and only if the one-variable specialization

$$G(T) := F(n_1 + k_1 T, \dots, n_r + k_r T)$$

is identically $p + qT$ in $\overline{\mathbb{Q}}(T)$ for some $p, q \in \mathbb{N}_0$.

Proof. Write $F = P/Q$ with $P, Q \in \overline{\mathbb{Q}}[X_1, \dots, X_r]$ and $Q(x_1, \dots, x_r) \neq 0$. The specialized denominator $Q(n_1 + k_1 T, \dots, n_r + k_r T)$ is a polynomial in T that does not vanish at $T = \beta$, hence is not the zero polynomial. Therefore G is a well-defined element of $\overline{\mathbb{Q}}(T)$ with no pole at β, and $G(\beta) = F(x_1, \dots, x_r)$. Now apply Theorem 8.2. □

Status: [paper proof in this report]

Every rational operation on finitely many hypothetical solutions is thus reduced to an elementary identity between polynomials in a single indeterminate, with coordinates in $\mathbb{N}_0$. No analytic input is required beyond the transcendence already verified by the kernel.

8.3 Exact multiplication criterion

The next statement needs no assumption at all: the branch in which the conjecture holds is trivial, and the branch in which it fails is governed by the generator.

Theorem 8.5 (Products of two solutions)**.** *For all $x, y \in \mathcal{S}$, unconditionally,*

$$\boxed{xy \in \mathcal{S} \quad \Longleftrightarrow \quad x \in \mathbb{N}_0 \text{ or } y \in \mathbb{N}_0.}$$

Proof. If one factor, say x, lies in $\mathbb{N}_0$, then xy is a natural multiple of a solution and therefore a solution; this direction is unconditional.

For the converse, split on Conjecture 2.1. If the conjecture holds then $\mathcal{S} = \mathbb{N}_0$ and the right-hand side is automatically true, so there is nothing to prove. Under failure, write the unique coordinates

$$x = n + k\beta, \qquad y = m + \ell\beta, \qquad n, k, m, \ell \in \mathbb{N}_0.$$

If neither factor is integral then $k, \ell > 0$ and

$$xy = nm + (n\ell + mk)\beta + k\ell\,\beta^2,$$

with $k\ell > 0$. Membership $xy \in \mathcal{S}$ would force $xy = r + s\beta$ for some $r, s \in \mathbb{N}_0$, producing a nonzero polynomial of degree exactly two over $\mathbb{Q}$ vanishing at the transcendental number β. Equivalently, apply Theorem 8.2 to $R(T) = (n + kT)(m + \ell T)$, which is not affine. Both routes are impossible. □

Status: [paper proof in this report] — note that only transcendence over $\mathbb{Q}$ is used, which is [kernel-verified Lean]; the appeal to $\overline{\mathbb{Q}}$ in Theorem 8.2 is a convenience, not a necessity.

Corollary 8.6 (Finite products)**.** *Let $x_1, \ldots, x_r \in \mathcal{S}$ be positive. Then*

$$\prod_{i=1}^{r} x_i \in \mathcal{S} \quad \Longleftrightarrow \quad \text{at most one } x_i \text{ is nonintegral.}$$

If a zero factor is allowed, the product is $0 \in \mathcal{S}$ and the criterion is vacuous.

Proof. If the conjecture holds, every x_i is integral and both sides are true. Under failure, the positive integral factors multiply to a positive natural, so after removing them the product is a natural multiple of a product of nonintegral solutions. Writing each remaining factor as $n_i + k_i\beta$ with $k_i > 0$ and expanding, the result is a polynomial in β whose degree equals the number of nonintegral factors and whose leading coefficient is positive. Degree at least two is excluded by Theorem 8.2, while degree zero or one is preserved by natural dilation and translation. □

Status: [paper proof in this report]

Corollary 8.7 (Internal power test)**.** *For every $x \in \mathcal{S}$ and every integer $r \geq 2$,*

$$x^r \in \mathcal{S} \quad \Longleftrightarrow \quad x \in \mathbb{N}_0.$$

In particular $x \in \mathbb{N}_0$ holds exactly when $x^2 \in \mathcal{S}$: a single squaring test detects integrality.

Proof. Apply Theorem 8.5 with $y = x$ and induct, or apply Corollary 8.6 to r equal factors, noting that $x = 0$ satisfies both sides. □

Status: [paper proof in this report]

8.4 Exact quotient criterion

The same evaluation argument classifies division. Division is more delicate than multiplication because even the true solution set $\mathbb{N}_0$ is not closed under arbitrary quotients, so the criterion cannot be a clean dichotomy; it is instead an exact divisibility condition on coordinates.

Theorem 8.8 (Quotients of solutions)**.** *Assume Conjecture 2.1 is false and write the unique coordinates*

$$x = n + k\beta, \qquad y = m + \ell\beta > 0.$$

Exactly one of the two cases $\ell = 0$ and $\ell > 0$ applies, and in that case $x/y \in \mathcal{S}$ if and only if the corresponding condition holds:

(a) *$\ell = 0$, so that $y = m$ is a positive natural, and $m \mid n$ and $m \mid k$;*

(b) *$\ell > 0$, and $x = qy$ for some $q \in \mathbb{N}_0$, equivalently $n = qm$ and $k = q\ell$.*

In particular, if x and y are both nonintegral, then

$$\frac{x}{y} \in \mathcal{S} \quad \Longleftrightarrow \quad \frac{x}{y} \in \mathbb{N}_0.$$

Proof. Suppose $x/y = r + s\beta$ with $r, s \in \mathbb{N}_0$. Clearing the denominator gives

$$n + k\beta = mr + (ms + \ell r)\beta + \ell s\,\beta^2.$$

The difference of the two sides is a polynomial in β with rational coefficients, so transcendence of β forces every coefficient to vanish:

$$\ell s = 0, \qquad k = ms + \ell r, \qquad n = mr.$$

If $\ell = 0$ then $y = m$ and $y > 0$ gives $m \geq 1$; the equations read $n = mr$ and $k = ms$, which is precisely $m \mid n$ and $m \mid k$. If $\ell > 0$ then $s = 0$, so $n = mr$ and $k = \ell r$, that is $x = ry$ with $r \in \mathbb{N}_0$. The converses follow by direct substitution.

For the final claim, both x and y nonintegral means $k, \ell > 0$, so case (b) applies and $x/y = r \in \mathbb{N}_0$. ☐

Status: **[paper proof in this report]**

Corollary 8.9 (Reciprocals)**.** *Unconditionally, for $x \in \mathcal{S}$ with $x > 0$,*

$$\frac{1}{x} \in \mathcal{S} \quad \Longleftrightarrow \quad x = 1.$$

Proof. If the conjecture holds then $\mathcal{S} = \mathbb{N}_0$ and the claim is the elementary fact that a positive natural with natural reciprocal equals 1. Under failure apply Theorem 8.8 with numerator $1 = 1+0\cdot\beta$ and denominator $y = x$. If $x = m$ is integral, case (a) gives $m \mid 1$, so $m = 1$. If x is nonintegral, case (b) gives $1 = qx$ for some $q \in \mathbb{N}_0$; here $q = 0$ is impossible, and $q \geq 1$ would make $x = 1/q$ rational, contradicting the irrationality of every nonintegral solution. ☐

Status: **[paper proof in this report]**

A hypothetical solution monoid is therefore additively rich — a free additive monoid of rank two — and almost maximally hostile to every multiplicative operation: products, powers, reciprocals and quotients all escape $\mathcal{S}$ except in the degenerate cases forced by the additive structure itself.

8.5 Closure formulations of the conjecture

Because the multiplication criterion is unconditional, it converts directly into equivalent statements of the conjecture in terms of closure.

Theorem 8.10 (Multiplicative closure equivalences)**.** *The following are equivalent.*

(i) *Conjecture 2.1 is true.*

(ii) *$\mathcal{S}$ is closed under multiplication.*

(iii) *$\mathcal{S}$ is closed under squaring.*

(iv) *For every real x,*

$$2^x, 3^x \in \mathbb{Z} \quad \Longrightarrow \quad 2^{x^2}, 3^{x^2} \in \mathbb{Z}.$$

Proof. If the conjecture holds then $\mathcal{S} = \mathbb{N}_0$, which is closed under multiplication, so (i) implies (ii); and (ii) trivially implies (iii). Statements (iii) and (iv) are the same assertion, written once for $\mathcal{S}$ and once for the defining integrality conditions. Finally, assume (iii). For any $x \in \mathcal{S}$ we get $x^2 \in \mathcal{S}$, so Corollary 8.7 forces $x \in \mathbb{N}_0$. Hence $\mathcal{S} = \mathbb{N}_0$, which is Conjecture 2.1. □

Status: [paper proof in this report]

The candidate formulation of Section 2 carries the same content in multiplicative language. For positive reals define the commutative operation

$$m \star n := 2^{(\log_2 m)(\log_2 n)} = m^{\log_2 n} = n^{\log_2 m}. \tag{9}$$

If $m = 2^x$ and $n = 2^y$, then $m \star n = 2^{xy}$, so $\star$ is exactly the transport of multiplication of exponents through the exponent–candidate bijection of Section 2.

Corollary 8.11 (Candidate star-product)**.** *For $m, n \in \mathcal{C}$,*

$$m \star n \in \mathcal{C} \quad \Longleftrightarrow \quad m \text{ is a power of } 2 \text{ or } n \text{ is a power of } 2.$$

Consequently Conjecture 2.1 is equivalent both to closure of $\mathcal{C}$ under $\star$ and to the apparently weaker diagonal condition

$$m \in \mathcal{C} \quad \Longrightarrow \quad m \star m \in \mathcal{C}.$$

Proof. Put $x = \log_2 m$ and $y = \log_2 n$, so that $x, y \in \mathcal{S}$ and $m \star n = 2^{xy}$. Membership $m \star n \in \mathcal{C}$ is equivalent to $xy \in \mathcal{S}$, and m is a power of two exactly when $x \in \mathbb{N}_0$; now apply Theorem 8.5. The two closure equivalences are the image of Theorem 8.10(ii) and (iii) under the same bijection. □

Status: [paper proof in this report]

These are genuine equivalences and not a proof. There is at present no independent mechanism forcing nonlinear closure: nothing in the archimedean, valuation-theoretic, or determinant toolkits of Sections 4, 14 and 15 supplies a reason for $\mathcal{C}$ to be closed under $\star$. Their value is diagnostic — they identify the missing property as a single crisp failure of semiring-like structure, rather than as a quantitative deficit.

8.6 Formalization cost

The multiplication, power and quotient criteria are inexpensive Lean targets, and a route exists that avoids building a general rational-function library. For a new module — `RationalRigidity` would be a natural name — the plan is:

1. case split on `AlaogluErdosConjecture`, discharging the true branch by $\mathcal{S} = \mathbb{N}_0$;

2. in the failure branch obtain the canonical generator and the unique coordinates from the packaged statement

   `exists_canonical_primitiveGenerator_of_not_alaogluErdosConjecture`

   in the module `PrimitiveGenerator`;

3. expand the relevant equality as a polynomial of degree at most two in β with rational coefficients;

4. invoke the kernel-verified transcendence in `Transcendence` to force every coefficient to vanish;

5. discharge the remaining natural-number divisibility and coordinate conditions with `omega`.

Only the full rational-function theorem, Theorem 8.2, needs evaluation injectivity over $\overline{\mathbb{Q}}$ and hence Lemma 8.1; the multiplication theorem, the power test, the quotient criterion and the closure equivalences need nothing beyond transcendence over $\mathbb{Q}$, which is already accepted by the kernel.

Status: Planned; no draft yet. The mathematics is **[paper proof in this report]** throughout this section, and the prerequisites are **[kernel-verified Lean]**.

8.7 What the rigidity statements buy

The practical payoff is that they sharply restrict how a counterexample could be manufactured from a smaller one. A natural strategy for producing new solutions from old ones is to apply some algebraic operation — square a solution, multiply two of them, take a ratio, invert, or substitute solutions into a rational function — and hope to land back in $\mathcal{S}$, either to run a descent or to build an infinite family with better arithmetic properties than the canonical generator. Equation (8) closes that door completely: the only rational functions over $\overline{\mathbb{Q}}$ that map solutions to solutions are the affine maps $T \mapsto n + kT$ with natural coefficients, which reproduce nothing that the additive monoid $\mathbb{N}_0 + \mathbb{N}_0\beta$ did not already contain. Any construction of a new counterexample from an old one must therefore be transcendental in an essential way, or must leave the solution set and return by a route that is not rational over the algebraic numbers. In the other direction, the closure equivalences of Theorem 8.10 and Corollary 8.11 convert the conjecture into a purely structural question — is $\mathcal{C}$ closed under $\star$? — which is attractive precisely because it has no quantitative parameter to optimize, and which is a warning that no counting or equidistribution bound alone will produce the answer. Finally, all of these statements are short, elementary consequences of results already accepted by the kernel, which makes them the cheapest genuine additions to the verified corpus that this report proposes.

9 Three-smooth outputs and transcendental product exponents

Call a positive integer *three-smooth* when it has no prime factor outside $\{2, 3\}$, that is, when it equals $2^u 3^v$ for some $u, v \in \mathbb{N}_0$. The kernel-verified classification of auxiliary bases recorded in Section 4 settles which bases B satisfy $B^x \in \mathbb{Z}$ for a *fixed* nonintegral solution x: exactly the three-smooth ones. This section turns that classification onto the outputs $2^x, 3^x$ themselves, and then onto second-level exponents xy formed from two solutions.

Two distinct things come out, and they have different provenance. The first is an unconditional integrality test: a solution is integral precisely when its own two outputs are three-smooth. It uses only the kernel-verified transcendence of ϑ and unique factorization, so it is cheap to add to the Lean development. The second is an upgrade of "not an integer" to "transcendental", obtained from the six exponentials theorem. That upgrade converts several apparently weaker *algebraicity* statements into exact reformulations of Conjecture 2.1, at the cost of a classical input that the repository has not formalized in general form.

9.1 The two outputs cannot both be three-smooth

Theorem 9.1 (Three-smooth output test)**.** *For every $x \in \mathcal{S}$ the following are equivalent:*

(i) $x \in \mathbb{N}_0$*;*

(ii) *both* 2^x *and* 3^x *are three-smooth;*

(iii) 6^x *is three-smooth.*

Proof. Condition (i) gives (ii) immediately, since 2^x and 3^x are then powers of 2 and 3 respectively. Conditions (ii) and (iii) are equivalent because 2^x and 3^x are positive integers with product 6^x, and a product of positive integers is three-smooth exactly when both factors are.

For (ii)⇒(i), write

$$2^x = 2^p 3^q, \qquad 3^x = 2^r 3^s$$

with $p, q, r, s \in \mathbb{N}_0$. Taking logarithms and dividing by $\log 2$ and $\log 3$ respectively gives

$$x = p + q\vartheta, \qquad x = s + \frac{r}{\vartheta},$$

where $\vartheta = \log_2 3$. Eliminating x and clearing the denominator yields the quadratic relation

$$q\,\vartheta^2 + (p - s)\,\vartheta - r = 0 \tag{10}$$

with rational — indeed integer — coefficients. The repository proves ϑ transcendental, so every coefficient in (10) vanishes: $q = 0$, $p = s$, and $r = 0$. Hence $2^x = 2^p$ and $x = p \in \mathbb{N}_0$. □

Status: [paper proof in this report]

The statement is unconditional and does not use the conditional generator, the exact solution monoid, or any six-exponentials input. Its only nonelementary ingredient is the kernel-verified transcendence of ϑ,

`transcendental_logThreeDivLogTwo`

in `Transcendence`, together with unique factorization. A Lean proof is expected to be short; see Section 22 and the skeleton at the end of this section.

The test is sharp in a useful direction: it rules out one of the four a priori smoothness patterns of the least output pair.

Corollary 9.2 (Primitive-core trichotomy). *Assume Conjecture 2.1 fails, and use the simultaneous least-output normalization of Section 5,*

$$M = 2^{\beta} = 2^{a} w^{d}, \qquad A = 3^{\beta} = 3^{b} v^{e},$$

with $w > 1$ odd and power-primitive, $v > 1$ power-primitive and coprime to 3, and $d, e > 0$. Then at most one of $w = 3$ and $v = 2$ holds, and exactly one of the following three patterns occurs:

M	A	*primitive cores*
three-smooth	*not three-smooth*	$w = 3,\ v \neq 2$
not three-smooth	*three-smooth*	$w \neq 3,\ v = 2$
not three-smooth	*not three-smooth*	$w \neq 3,\ v \neq 2$

The missing fourth pattern, both outputs three-smooth, is impossible.

Proof. An odd, three-smooth, power-primitive integer greater than one is a power of 3, and primitivity forces the exponent to be 1; hence M is three-smooth exactly when $w = 3$. Symmetrically a three-smooth power-primitive integer greater than one that is coprime to 3 is exactly 2, so A is three-smooth exactly when $v = 2$. The least nonintegral solution β lies in $\mathcal{S} \setminus \mathbb{N}_0$, so Theorem 9.1 forbids both outputs from being three-smooth, which is the exclusion $\neg(w = 3 \wedge v = 2)$. □

Status: [paper proof in this report]

The normalization itself, including power-primitivity of w and v, the depth dichotomy $a = 0$ or $b = 0$, and the gcd-one condition, is **[kernel-verified Lean]** in `OutputNormalization` (Section 5). Only the exclusion of the fourth pattern is new here.

The two exceptional rows are precisely the cases the corpus already identifies as hardest: $w = 3$ makes 3^{ϑ} the sharp radical target of Section 6, and $v = 2$ is its base-swapped mirror with $2^{1/\vartheta} = 2^{\log_3 2}$. The third row is generic and carries two independent outside-prime supports.

9.2 Third-base transcendence from six exponentials

The remaining statements of this section rest on one published transcendence theorem, which we state in the form we use.

Theorem 9.3 (Six exponentials theorem; Siegel, Lang, Ramachandra). *Let u_1, u_2 be complex numbers linearly independent over $\mathbb{Q}$, and let v_1, v_2, v_3 be complex numbers linearly independent over $\mathbb{Q}$. Then at least one of the six numbers $e^{u_i v_j}$, $1 \le i \le 2$, $1 \le j \le 3$, is transcendental.*

Status: [classical/open background] — classical, not formalized in this repository in general form.

See Waldschmidt [10, 11]. Every consequence below is tagged **[classical/open background]** for the same reason: the repository formalizes only the integer-entry specialization of the underlying interpolation determinant, which yields the 3-smooth classification of Section 4 but not the transcendence conclusion.

Lemma 9.4 (Third-base transcendence)**.** *Let $y \in \mathcal{S} \setminus \mathbb{N}_0$ and let $B \in \mathbb{N}_{>0}$ be not three-smooth. Then B^y is transcendental.*

Proof. The numbers $\log 2, \log 3, \log B$ are linearly independent over $\mathbb{Q}$: a nontrivial rational relation among them would give a nontrivial multiplicative relation among 2, 3, and B, which unique factorization forbids because B has a prime factor outside $\{2, 3\}$. Also 1 and y are linearly independent over $\mathbb{Q}$, since y is irrational — a rational solution is integral by the kernel-verified rational-exponent lemma of Section 4, and $y \notin \mathbb{N}_0$.

Apply Theorem 9.3 with $u_1 = 1$, $u_2 = y$ and $v_1 = \log 2$, $v_2 = \log 3$, $v_3 = \log B$. The six values are

$$2, \quad 3, \quad B, \quad 2^y, \quad 3^y, \quad B^y,$$

and the first five are integers, hence algebraic: $2, 3, B$ by hypothesis and $2^y, 3^y$ because $y \in \mathcal{S}$. Therefore the sixth, B^y, is transcendental. □

Status: [**classical/open background**] — consequence of the six exponentials theorem (Theorem 9.3).

Combining Lemma 9.4 with the kernel-verified 3-smooth classification of Section 4 gives an exact algebraicity trichotomy for integer bases: for every $y \in \mathcal{S} \setminus \mathbb{N}_0$ and every $B \in \mathbb{N}_{>0}$,

$$B^y \in \overline{\mathbb{Q}} \iff B^y \in \mathbb{Z} \iff B = 2^u 3^v \text{ for some } u, v \in \mathbb{N}_0. \tag{11}$$

There is no algebraic-irrational middle case. Any base that is not three-smooth produces a transcendental value outright.

9.3 Products of noninteger solutions

The set $\mathcal{S}$ is closed under addition and under multiplication by natural numbers, but it is not known to be closed under multiplication of two of its own nonintegral elements. The next theorem says exactly what happens at the product exponent xy.

Theorem 9.5 (Product-exponent dichotomy)**.** *Let $x, y \in \mathcal{S}$.*

(a) *If $x \in \mathbb{N}_0$ or $y \in \mathbb{N}_0$, then $2^{xy} \in \mathbb{N}$ and $3^{xy} \in \mathbb{N}$.*

(b) *If $x \notin \mathbb{N}_0$ and $y \notin \mathbb{N}_0$, then each of 2^{xy} and 3^{xy} is either an integer or transcendental, at least one of the two is transcendental, and*

$$6^{xy} \text{ is transcendental.}$$

Proof. Part (a) is natural-number dilation: if $y = n \in \mathbb{N}_0$ then $xy = nx$ is a natural multiple of a solution and therefore a solution, and symmetrically in x.

For part (b) put $M_x = 2^x$ and $A_x = 3^x$, both positive integers, and note

$$2^{xy} = M_x^y, \qquad 3^{xy} = A_x^y.$$

If M_x is three-smooth, the 3-smooth classification of Section 4 makes M_x^y an integer; if M_x is not three-smooth, Lemma 9.4 makes it transcendental. The same alternative holds for A_x^y. By Theorem 9.1 applied to the nonintegral solution x, the two outputs M_x and A_x are not both three-smooth, so at least one of $2^{xy}, 3^{xy}$ is transcendental.

Finally $6^x = M_x A_x$ is a positive integer, and it is not three-smooth, since a prime outside $\{2, 3\}$ dividing one factor divides the product. Applying Lemma 9.4 with base 6^x and exponent y gives $(6^x)^y = 6^{xy}$ transcendental. ☐

Status: [**classical/open background**] for part (b) — via Theorem 9.3.

Part (a) is elementary and its underlying closure property, that natural multiples of a solution are solutions, is [**kernel-verified Lean**].

Corollary 9.6 (Diagonal transcendence)**.** *Let $x \in \mathcal{S} \setminus \mathbb{N}_0$. Then 6^{x^2} is transcendental and at least one of 2^{x^2}, 3^{x^2} is transcendental. More precisely,*

$$2^{x^2} \in \mathbb{N} \iff 2^x \text{ is three-smooth,}$$
$$3^{x^2} \in \mathbb{N} \iff 3^x \text{ is three-smooth,}$$

and failure of either equivalence's right-hand side makes the corresponding value transcendental.

Proof. Take $y = x$ in Theorem 9.5(b). The two displayed equivalences are the alternative used in that proof, applied to the bases $M_x = 2^x$ and $A_x = 3^x$: a three-smooth base gives an integer by the 3-smooth classification of Section 4, and a base that is not three-smooth gives a transcendental value by Lemma 9.4. ☐

Corollary 9.7 (Mixed three-smooth bases)**.** *Let $x, y \in \mathcal{S} \setminus \mathbb{N}_0$ and let $u, v \geq 1$ be integers. Then $(2^u 3^v)^{xy}$ is transcendental.*

Proof. The positive integer

$$(2^u 3^v)^x = (2^x)^u (3^x)^v$$

is divisible by every prime occurring in either output, because $u, v \geq 1$. By Theorem 9.1 at least one output has a prime factor outside $\{2, 3\}$, so this base is not three-smooth. Apply Lemma 9.4 with exponent y. ☐

Status: [**classical/open background**] for both corollaries — via Theorem 9.3.

9.4 A global second-level trichotomy

The exact generator theorem of Section 5 makes the preceding dichotomy uniform: which of the three patterns occurs is a property of the counterexample as a whole, not of the particular pair (x, y).

Theorem 9.8 (Uniform second-level pattern)**.** *Assume Conjecture 2.1 fails and let $M = 2^\beta$, $A = 3^\beta$ for the primitive generator β. For every $x, y \in \mathcal{S} \setminus \mathbb{N}_0$, exactly one of the following three patterns holds, and the pattern does not depend on x or y:*

condition on (M, A)	2^{xy}	3^{xy}
M three-smooth, A not	*integer*	*transcendental*
A three-smooth, M not	*transcendental*	*integer*
neither three-smooth	*transcendental*	*transcendental*

Proof. By the exact primitive-generator theorem of Section 5, write $x = n + k\beta$ with $n, k \in \mathbb{N}_0$ and $k \geq 1$, so that

$$2^x = 2^n M^k, \qquad 3^x = 3^n A^k.$$

Multiplying by a power of the distinguished smooth prime cannot remove a prime factor outside $\{2, 3\}$, and $k \geq 1$ keeps every such prime of M present in M^k; hence 2^x is three-smooth exactly when M is, and likewise 3^x is three-smooth exactly when A is. Now apply Theorem 9.5(b) together with the alternative used in its proof. The fourth pattern, both M and A three-smooth, is excluded by Theorem 9.1 applied to β. □

Status: [classical/open background] — via Theorem 9.3.

The structural input, namely $\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\}$ with unique coordinates, is **[kernel-verified Lean]** in `PrimitiveGenerator` (Section 5).

This suggests a clean three-case research split, matching Corollary 9.2. The two one-smooth-output cases reduce to the exceptional primitive cores $w = 3$ and $v = 2$, where the target radicals are 3^{ϑ} and $2^{1/\vartheta}$ respectively and where Section 6 already locates the wall. In the generic case both second iterates are transcendental and two independent outside-prime supports are available, which is the configuration a mixed archimedean and non-archimedean determinant would exploit; see Section 21.

9.5 Algebraicity formulations equivalent to the conjecture

Theorem 9.9 (Algebraicity tests)**.** *Each of the following is equivalent to Conjecture 2.1.*

(i) *For every $x \in \mathcal{S}$, $6^{x^2} \in \overline{\mathbb{Q}}$.*

(ii) *For every $x \in \mathcal{S}$, both $2^{x^2} \in \overline{\mathbb{Q}}$ and $3^{x^2} \in \overline{\mathbb{Q}}$.*

(iii) *For every $x, y \in \mathcal{S}$, $6^{xy} \in \overline{\mathbb{Q}}$.*

(iv) *For one fixed pair of integers $u, v \geq 1$, every $x \in \mathcal{S}$ satisfies $(2^u 3^v)^{x^2} \in \overline{\mathbb{Q}}$.*

(v) *For every $x \in \mathcal{S}$, the integer 6^x is three-smooth.*

Proof. If Conjecture 2.1 holds, then $\mathcal{S} = \mathbb{N}_0$, every exponent appearing in (i)–(iv) is a nonnegative integer, and all the listed values are positive integers, hence algebraic; condition (v) is immediate because $6^x = 2^x 3^x$ is then a power of 6.

Conversely, suppose the conjecture fails and fix $x \in \mathcal{S} \setminus \mathbb{N}_0$. Then (i) and (ii) fail by Corollary 9.6, (iii) fails by taking $y = x$ in Theorem 9.5(b), (iv) fails by Corollary 9.7 with $y = x$, and (v) fails by Theorem 9.1. □

Status: [classical/open background] for (i)–(iv) — via Theorem 9.3. **[paper proof in this report]** for (v).

Equivalence (v) needs only Theorem 9.1, hence only the kernel-verified transcendence of ϑ; it is therefore the one item in this list that is within immediate reach of the Lean development.

Items (i)–(iv) replace integrality by the apparently much weaker target of algebraicity, and the six exponentials theorem makes the replacement lossless. That sharpens the statement of what is missing: one would need an independent reason why some nonlinear iterate such as 6^{x^2} is algebraic, derived from the two original integer values 2^x and 3^x alone. No mechanism in the present corpus produces algebraicity at the second level, which is consistent with the four-exponentials diagnosis of Section 7: the two-base configuration supplies only two independent logarithmic directions, and squaring the exponent does not create a third.

9.6 Formalization boundary

The division of labour in this section is unusually clean, and it marks exactly where the repository can advance without new transcendence theory.

Theorem 9.1, Corollary 9.2, part (a) of Theorem 9.5, and equivalence (v) of Theorem 9.9 use only kernel-verified ingredients: transcendence of ϑ, unique factorization, the closure of $\mathcal{S}$ under natural multiples, the simultaneous output normalization, and the 3-smooth classification in `ThreeSmooth`. A proposed module `OutputSmoothness` would host them. With the predicate

$$\mathrm{ThreeSmooth}(m) :\Longleftrightarrow \exists\, u, v \in \mathbb{N}_0,\ m = 2^u 3^v,$$

the headline statement is a biconditional between integrality of the exponent and joint smoothness of the two outputs:

```
theorem integer_iff_outputs_threeSmooth
    {x : R} (hx : TwoBaseIntegralSolution x) :
    x in Set.range ((^) : Z -> R) <->
      ThreeSmooth (natOutputTwo hx) /\
      ThreeSmooth (natOutputThree hx)
```

Listing 1: Proposed output-smoothness signature.

The reverse direction is the interesting one, and it follows the proof of Theorem 9.1: take logarithms of the two smooth factorizations, assemble (10) as a polynomial of degree at most two in ϑ with integer coefficients, apply

`transcendental_logThreeDivLogTwo`

to force every coefficient to vanish, and discharge the remaining natural-number arithmetic with `omega`. Mathlib's polynomial evaluation interface is the natural way to build $qX^2 + (p - s)X - r$ and invoke the definition of transcendence.

Everything else in this section — Lemma 9.4, part (b) of Theorem 9.5, Corollaries 9.6 and 9.7, Theorem 9.8, and items (i)–(iv) of Theorem 9.9 — needs the six exponentials theorem in general form. The repository's interpolation determinant is currently specialized to integer entries: a candidate matrix has integer entries, a nonzero integer determinant satisfies $|D| \geq 1$, and that inequality is what drives the 3-smooth classification. The natural formal route to Lemma 9.4 is to replace the bound $|D| \geq 1$ by a number-field norm bound for a nonzero *algebraic* determinant, retaining the same analytic upper bound from the interpolation estimate. Packaging that as an "algebraic entries of bounded height" interface would recover the transcendence conclusion while reusing the existing determinant infrastructure, and would place the general six exponentials theorem within the same framework. This is a substantial addition, not a refactor, and it is listed among the transcendence-theoretic priorities of Section 21.

10 Perfect-power content, affine descent, and primitive-output statistics

Section 5 closes the conditional structure question: if Conjecture 2.1 fails, then $\mathcal{S}$ is the free rank-two additive monoid generated by 1 and the canonical least nonintegral solution β, and the least output pair $(M, A) = (2^\beta, 3^\beta)$ is affinely primitive. Both facts are **[kernel-verified Lean]**. What that corpus does not yet record is the *exact* perfect-power content of the output pair at an arbitrary point of the monoid: the kernel-verified corollaries exclude matched decompositions at β, but say nothing about which matched decompositions actually occur elsewhere.

This section supplies that classification. It determines exactly which simultaneous factorizations

$$2^x = 2^r U^d, \qquad 3^x = 3^r V^d$$

can occur for $x \in \mathcal{S}$, identifies the maximal common perfect-power degree of the output pair with a gcd of coordinates, and then converts the classification into exact limiting statistics for primitive output pairs. Everything here is conditional on failure and rests on the kernel-verified monoid classification; the classification and the statistics themselves are **[paper proof in this report]**.

Throughout, assume failure and fix the canonical generator β of Section 5, with

$$M = 2^\beta, \qquad A = 3^\beta.$$

For $x = n + k\beta$ with $n, k \in \mathbb{N}_0$, exact coordinates give

$$2^x = 2^n M^k, \qquad 3^x = 3^n A^k, \tag{12}$$

and the pair (n, k) is uniquely determined by x. The coordinate gcd $\gcd(n, k)$ turns out to be an intrinsic arithmetic invariant of the output pair, not merely of its coordinates.

10.1 Simultaneous perfect powers

Theorem 10.1 (Exact common-power criterion)**.** *Let $x = n + k\beta \in \mathcal{S}$ and let $r \geq 1$. The following are equivalent:*

(i) *both 2^x and 3^x are perfect r-th powers in $\mathbb{N}$;*

(ii) *$x/r \in \mathcal{S}$;*

(iii) *$r \mid n$ and $r \mid k$.*

Proof. If $2^x = U^r$ and $3^x = V^r$ with $U, V > 0$, uniqueness of positive real r-th roots gives

$$2^{x/r} = U, \qquad 3^{x/r} = V,$$

so $x/r \in \mathcal{S}$. The converse follows by raising the two integer values at x/r to the r-th power.

For the coordinate form, write $x/r = p + q\beta$ in the unique coordinates supplied by the exact primitive-generator theorem of Section 5. Then $n + k\beta = rp + rq\beta$, and uniqueness of coordinates forces $n = rp$ and $k = rq$, which is exactly the pair of divisibility conditions. Conversely, if $r \mid n$ and $r \mid k$, then $(n/r) + (k/r)\beta \in \mathcal{S}$ and equals x/r. ☐

For $x > 0$ define the *common output power degree*

$$g(x) := \max\{r \geq 1 : 2^x \text{ and } 3^x \text{ are both perfect } r\text{-th powers}\}.$$

Corollary 10.2 (Coordinate gcd)**.** *If $x = n + k\beta > 0$, then*

$$g(x) = \gcd(n, k),$$

with the convention $\gcd(n, 0) = n$. *The output pair is simultaneously power-primitive exactly when* $\gcd(n, k) = 1$.

Proof. By Theorem 10.1 the admissible degrees r are precisely the common divisors of n and k, and the set of common divisors of a pair that is not $(0, 0)$ has a largest element, namely $\gcd(n, k)$. ☐

Consequently every nonzero output pair has a unique decomposition

$$(2^x, 3^x) = (U^g, V^g), \qquad g = \gcd(n, k),$$

where $(U, V) = (2^{x/g}, 3^{x/g})$ is a simultaneously power-primitive output pair belonging to the solution x/g. The repository proves power-primitivity for the least pair only; this corollary computes the intrinsic power degree at every point of the solution monoid. For integral $x = n$ it returns $g(n) = n$, which is the familiar statement that 2^n and 3^n are perfect n-th powers and no better; for $x = \beta$ it returns $g(\beta) = \gcd(0, 1) = 1$, which is exactly the kernel-verified exclusion of a common perfect-power degree at the least pair.

Status: **[paper proof in this report]** — Theorem 10.1 and Corollary 10.2 are paper results of this report. Their inputs are kernel-verified: the unique-coordinate clause of the exact primitive-generator theorem (`PrimitiveGenerator`) and the least-solution corollaries of `LeastSolution`. The implication (i)⇒(ii) is the special case $r = 0$ of the Lean affine descent theorem `affine_descent_integral_powers` in `ArithmeticRigidity`; the converse and the coordinate criterion (iii) are new here.

10.2 Matched affine decompositions

A descent that strips the same distinguished-base depth and the same outer power from both outputs is the natural common generalization of the two operations above. The repository's affine descent theorem shows that such a stripping produces a solution; the following criterion says exactly when one exists.

Theorem 10.3 (Matched affine decomposition criterion)**.** *Let $x = n + k\beta \in \mathcal{S}$, let $d \geq 1$, and let $r \in \mathbb{N}_0$. The following are equivalent:*

(i) *there exist $U, V \in \mathbb{N}_{>0}$ with*
$$2^x = 2^r U^d, \qquad 3^x = 3^r V^d;$$

(ii) $(x - r)/d \in \mathcal{S}$*;*

(iii) $0 \leq r \leq n$, $d \mid k$, *and* $d \mid (n - r)$.

When these conditions hold,

$$\frac{x-r}{d} = \frac{n-r}{d} + \frac{k}{d}\,\beta, \qquad U = 2^{(x-r)/d}, \qquad V = 3^{(x-r)/d}.$$

Proof. Conditions (i) and (ii) are equivalent by uniqueness of positive real roots, applied after removing the common factor 2^r from the first output and 3^r from the second. Expanding (ii) in unique coordinates gives
$$n + k\beta = r + d(p + q\beta)$$
for some $p, q \in \mathbb{N}_0$, so $n = r + dp$ and $k = dq$; this is exactly (iii), and it also yields the displayed formulas. Conversely, (iii) lets one define $p = (n-r)/d \in \mathbb{N}_0$ and $q = k/d \in \mathbb{N}_0$, and then $p + q\beta \in \mathcal{S}$ equals $(x - r)/d$. □

At the least pair the criterion collapses. For $x = \beta$ we have $n = 0$ and $k = 1$, so (iii) forces $r = 0$ and $d = 1$: no nontrivial matched decomposition of (M, A) exists. That is the kernel-verified exact affine primitivity statement of Section 5, recovered here as one instance of a criterion valid at every point.

For a nonintegral solution $x = n + k\beta$ with $k > 0$, Euclidean division $n = qk + r$ with $0 \leq r < k$ gives a canonical reduced affine presentation

$$x = r + k(q + \beta), \tag{13}$$

and hence
$$2^x = 2^r\bigl(2^q M\bigr)^k, \qquad 3^x = 3^r\bigl(3^q A\bigr)^k.$$

The normalization $0 \leq r < k$ makes the presentation unique. It separates the outer affine degree k — an invariant of the point, being its β-coordinate — from the reduced common base depth r.

Status: [paper proof in this report] — Theorem 10.3 and the reduced presentation (13) are paper results. The implication (i)⇒(ii) is the kernel-verified affine descent theorem; the equivalence with the coordinate condition (iii), and hence the classification of *all* matched decompositions, is new.

10.3 What is already formal, and what is new here

The table below separates the two layers. The middle column is the kernel-verified statement about the least pair (M, A); the right column is what this section adds for a general point $x = n + k\beta$ of the monoid.

Statement	Kernel-verified at β	Extension in this section
Matched base depth	$\min\{v_2(M)\,, v_3(A)\} = 0$, in `LeastSolution`	Admissible depths at x are exactly $0 \le r \le n$ with $d \mid (n-r)$
Common perfect-power degree	no $k \ge 2$ with M, A both k-th powers, in `LeastSolution`	the exact degree is $g(x) = \gcd(n,k)$ (Corollary 10.2)
Affine primitivity	$M = 2^r U^k$, $A = 3^r V^k$ force $r = 0$, $k = 1$, in `LeastSolution`	full criterion for every x, with the least pair as the case $(n,k) = (0,1)$ (Theorem 10.3)
Affine descent	$(x-r)/k \in \mathcal{S}$ from a matched factorization, in `ArithmeticRigidity`	the descent condition is an equivalence, not merely an implication
Output normalization	$M = 2^a w^d$, $A = 3^b v^e$ with $a = 0$ or $b = 0$ and $\gcd(\gcd(d,e), \lvert b-a\rvert) = 1$, in `OutputNormalization`	unchanged; that invariant constrains the two *separate* degrees, whereas g measures only the common one

The kernel-verified identifiers behind the middle column are

```
IsLeastTwoBaseNonintegerSolution.not_both_base_dvd_outputs
```

```
IsLeastTwoBaseNonintegerSolution.no_common_perfect_power
```

```
IsLeastTwoBaseNonintegerSolution.affine_primitive
```

in `LeastSolution`, together with `exists_simultaneous_primitive_output_normalization` in `OutputNormalization` and `affine_descent_integral_powers` in `ArithmeticRigidity`. None of the right-hand column is machine-checked. A formalization would be light: the coordinate criteria follow from the already accepted unique-representation clause, so the missing work is bookkeeping over $\mathbb{N}_0^2$ rather than new arithmetic.

10.4 Why this does not yet descend below β

If n and k have a common divisor, Theorem 10.1 descends to $x/\gcd(n,k)$. If a matched decomposition exists, Theorem 10.3 descends to $(x-r)/d$. But exact coordinates guarantee that the descended point is again $p + q\beta$ with $p, q \in \mathbb{N}_0$, and it lies below β only when $q = 0$, in which case it is integral and carries no information. The least nonintegral point β is already primitive for every descent of this shape, which is precisely why the kernel-verified corollaries at β are exclusions rather than contradictions.

A successful descent argument therefore needs a transformation not captured by common perfect powers and common distinguished-base depths. Both operations act inside the monoid $\mathbb{N}_0^2$; anything that could close the problem must leave it.

10.5 Total and primitive counts

Corollary 10.2 converts ordinary coprimality statistics of lattice points into exact statistics of hypothetical output pairs. For $T > 0$ put

$$N_\beta(T) = \#\{(n,k) \in \mathbb{N}_0^2 \setminus \{(0,0)\} : n + k\beta < T\}$$

and

$$P_\beta(T) = \#\{(n,k) \in \mathbb{N}_0^2 : n + k\beta < T,\ \gcd(n,k) = 1\}.$$

The first counts nonzero solutions below T; by Corollary 10.2 the second counts exactly those output pairs with no nontrivial common perfect-power degree.

Counting each column separately, for every real $U > 0$,

$$N_\beta(U) = \sum_{0 \le k < U/\beta} \lceil U - k\beta \rceil - 1 = \frac{U^2}{2\beta} + O_\beta(U+1), \tag{14}$$

uniformly in U. For integer T this is the kernel-verified cumulative count $C(T)$ of Section 5 minus the point $x = 0$, so (14) agrees in leading order with `QuadraticAsymptotic`; the uniform real-variable form used below is elementary.

Theorem 10.4 (Primitive output pairs). *As $T \to \infty$,*

$$P_\beta(T) = \frac{T^2}{2\beta\zeta(2)} + O_\beta(T \log T) = \frac{3T^2}{\pi^2\beta} + O_\beta(T \log T).$$

Consequently

$$\frac{P_\beta(T)}{N_\beta(T)} \longrightarrow \frac{1}{\zeta(2)} = \frac{6}{\pi^2}.$$

Proof. Every nonzero coordinate pair has the unique factorization

$$(n,k) = d\,(n_0,k_0), \qquad d = \gcd(n,k), \quad \gcd(n_0,k_0) = 1,$$

and $n + k\beta < T$ holds if and only if $n_0 + k_0\beta < T/d$. Hence

$$N_\beta(T) = \sum_{d \ge 1} P_\beta(T/d),$$

a finite sum for each T, and Möbius inversion gives

$$P_\beta(T) = \sum_{d \le T} \mu(d)\, N_\beta(T/d).$$

Inserting the uniform estimate (14),

$$\begin{aligned} P_\beta(T) &= \frac{T^2}{2\beta} \sum_{d \le T} \frac{\mu(d)}{d^2} + O_\beta\left(\sum_{d \le T} \Big(\frac{T}{d} + 1 \Big) \right) \\ &= \frac{T^2}{2\beta\zeta(2)} + O_\beta(T \log T), \end{aligned}$$

where the tail $\sum_{d>T} \mu(d)/d^2 = O(1/T)$ contributes $O_\beta(T)$. Dividing by (14) gives the stated limit. □

The constant $6/\pi^2$ is universal: it does not depend on the hypothetical generator at all. The geometry of the monoid contributes the factor $1/(2\beta)$, and the primitive-lattice density contributes $1/\zeta(2)$. In particular no choice of counterexample can make the primitive proportion small.

10.6 Distribution of the maximal common power degree

For fixed $g \geq 1$ let

$$P_{\beta,g}(T) := \#\{x = n + k\beta : 0 < x < T,\ \gcd(n,k) = g\}$$

be the number of solutions below T whose output pair has common power degree exactly g.

Corollary 10.5 (Exact-degree distribution)**.** *For fixed $g \geq 1$,*

$$P_{\beta,g}(T) = P_\beta(T/g) = \frac{3T^2}{\pi^2 \beta g^2} + O_\beta\!\left(\frac{T}{g}\log\frac{2T}{g}\right).$$

The limiting probability that an output pair below T has maximal common power degree exactly g is

$$\frac{6}{\pi^2 g^2}.$$

Proof. The map $(n_0, k_0) \mapsto g\,(n_0, k_0)$ is a bijection from the primitive pairs below T/g onto the pairs of gcd exactly g below T, so $P_{\beta,g}(T) = P_\beta(T/g)$; now apply Theorem 10.4 and divide by (14). ☐

The probabilities sum to one, since $\sum_{g\geq 1} 6/(\pi^2 g^2) = 1$. Likewise, by Theorem 10.1 both outputs are perfect r-th powers exactly when $r \mid n$ and $r \mid k$, so

$$\#\{0 < x < T : 2^x \text{ and } 3^x \text{ are both perfect } r\text{-th powers}\} = N_\beta(T/r), \tag{15}$$

and the limiting proportion is $1/r^2$.

10.7 Exact generating series

The coordinate monoid has an equally explicit Laplace series. For $s > 0$,

$$\begin{aligned} Z_\beta(s) &:= \sum_{x\in\mathcal{S}\setminus\{0\}} e^{-sx} = \sum_{n,k\geq 0} e^{-s(n+k\beta)} - 1 \\ &= \frac{1}{(1-e^{-s})(1-e^{-\beta s})} - 1, \end{aligned} \tag{16}$$

the two factors being the two free generators 1 and β. Möbius inversion in the degree gives the primitive-output series

$$Z_\beta^{\mathrm{prim}}(s) = \sum_{d\geq 1} \mu(d)\, Z_\beta(ds).$$

At the candidate level the same factorization appears multiplicatively: since $\mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\}$ with unique coordinates,

$$\sum_{m\in\mathcal{C}} m^{-s} = \frac{1}{(1-2^{-s})(1-M^{-s})}, \qquad \Re s > 0. \tag{17}$$

These identities make the two free generators visible in analytic form and give clean starting points for weighted Tauberian refinements of Theorem 10.4.

Status: **[paper proof in this report]** — Theorem 10.4, Corollary 10.5, and the series (16)–(17) are paper results of this report. Their only nonelementary input is the kernel-verified exact monoid classification of Section 5, whose leading-order count is itself **[kernel-verified Lean]** through `ExactCounting` and `QuadraticAsymptotic`. The finite scans of Section 19 play no role here; the exponent bound in force remains the kernel-verified $x \geq 12$.

10.8 Interpretation

Roughly 60.79% of hypothetical output pairs would be simultaneously power-primitive, and the remaining ones split across degrees $g \geq 2$ with the exact frequencies $6/(\pi^2 g^2)$. This is not pressure toward a contradiction. It is precisely the primitive proportion expected of a free rank-two monoid, and it would be the same for any generator: the statistic is diagnostic, not restrictive.

Its diagnostic value is a warning about method. The kernel-verified primitivity corollaries at β, and the classification of this section at a general point, both say that perfect-power and matched-depth structure is the exception among hypothetical solutions. Any descent argument built on extracting a common root or a common base depth therefore acts on a minority of the conditional population — asymptotically the proportion $1 - 6/\pi^2 \approx 0.3921$ of pairs that admit any nontrivial common degree at all — and leaves the primitive majority untouched. A method that closes the problem must engage the power-primitive pairs directly, which is exactly where the transcendence content of Section 7 sits.

11 Candidate divisibility and ambient root saturation

The exact classification of Section 5 describes the conditional candidate set by its *generators*: under failure, $\mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\}$ with unique coordinates. Generation alone says nothing about how candidates sit inside $\mathbb{N}$ under the two operations that descent arguments and searches actually use — divisibility, and extraction of roots. This section answers both exactly. The candidate monoid has a simple valuation geometry: divisibility is coordinatewise order on a two-dimensional lattice cone, $\mathcal{C}$ is closed under gcd and lcm, and the set of integers some power of which is a candidate is again an explicit cone — one that contains $\mathcal{C}$ with an exact finite-index defect measured by the rational-power index d of Section 5, which is also the exact radical degree of Section 6.

Every statement below is **[paper proof in this report]**. Its inputs — the canonical primitive odd core, the rational-power index, and the exact primitive-generator theorem — are **[kernel-verified Lean]**; what is added here is elementary lattice combinatorics on top of them. None of it is a step toward a contradiction: as the closing discussion makes explicit, the closure properties proved here are compatible with failure by construction, and the kernel-verified exponent bound remains $x \geq 12$.

11.1 A two-dimensional valuation cone

Lemma 11.1 (Valuation coordinates and divisibility)**.** *Assume Conjecture 2.1 is false and let w, d, a, β, M be the canonical data of Section 5, so that $w > 1$ is odd and power-primitive, $d > 0$, and*

$$M = 2^\beta = 2^a w^d.$$

Put $U := w^d$, the odd part of M, and for $n, k \in \mathbb{N}_0$ write

$$m(n,k) := 2^n M^k = 2^{n+ak} U^k, \qquad s := v_2(m(n,k)) = n + ak.$$

Then:

(i) $U > 1$;

(ii) *the map $m(n,k) \mapsto (s,k)$ is a bijection from $\mathcal{C}$ onto the lattice cone*

$$\Lambda_a := \{(s,k) \in \mathbb{N}_0^2 : s \geq ak\};$$

(iii) *writing $m_i = m(n_i, k_i)$ and $s_i = n_i + ak_i$,*

$$m_1 \mid m_2 \quad \Longleftrightarrow \quad s_1 \leq s_2 \;\; \textit{and} \;\; k_1 \leq k_2. \tag{18}$$

Proof. (i) If $U = 1$ then $M = 2^a$ and $\beta = a \in \mathbb{Z}$, contradicting the irrationality of the least generator.

(ii) By the exact primitive-generator theorem of Section 5, every candidate is $m(n,k)$ for a unique pair $(n,k) \in \mathbb{N}_0^2$. The reparameterization $(n,k) \mapsto (n+ak, k)$ is affine with inverse $(s,k) \mapsto (s-ak, k)$, and $n \geq 0$ holds exactly when $s \geq ak$.

(iii) Since U is odd, $v_2(m(n,k)) = s$, and for an odd prime p we have $v_p(m(n,k)) = k\, v_p(U)$, which vanishes unless $p \mid U$. Divisibility in $\mathbb{N}$ is comparison of all prime valuations. If $m_1 \mid m_2$ then comparison at 2 gives $s_1 \leq s_2$, and comparison at any prime $p \mid U$ — one exists by (i) — gives $k_1 v_p(U) \leq k_2 v_p(U)$ with $v_p(U) > 0$, hence $k_1 \leq k_2$. Conversely those two inequalities force every prime valuation of m_1 to be at most the corresponding valuation of m_2. □

Thus candidate divisibility is nothing but the coordinatewise order on Λ_a. The inequality $s \geq ak$ is the only constraint, and it records the base-adic depth a of the least generator: a candidate cannot carry the odd part U^k without carrying at least ak factors of two along with it.

11.2 The candidate gcd/lcm lattice

Theorem 11.2 (Candidate gcd/lcm lattice)**.** *Unconditionally, $\mathcal{C}$ is closed under* gcd *and* lcm*. If Conjecture 2.1 fails, then in the notation of Lemma 11.1,*

$$\begin{aligned}\gcd(m_1,m_2) &= m(\min(s_1,s_2) - a\min(k_1,k_2),\ \min(k_1,k_2)),\\ \operatorname{lcm}(m_1,m_2) &= m(\max(s_1,s_2) - a\max(k_1,k_2),\ \max(k_1,k_2));\end{aligned}$$

equivalently, in cone coordinates gcd *and* lcm *are coordinatewise minimum and maximum on Λ_a. Candidate divisibility is therefore a distributive lattice, and Λ_a is a sublattice of $\mathbb{N}_0^2$.*

Proof. If the conjecture holds then $\mathcal{C} = \{2^n : n \in \mathbb{N}_0\}$, whose gcds and lcms are again powers of two; this gives the unconditional half.

Assume failure. By Lemma 11.1 the candidate $m(n,k)$ has valuation vector s at the prime 2 and $k\, v_p(U)$ at each odd p. Gcd and lcm take coordinatewise minima and maxima of valuation vectors, so

$$\gcd(m_1,m_2) = 2^{\min(s_1,s_2)} U^{\min(k_1,k_2)}, \qquad \operatorname{lcm}(m_1,m_2) = 2^{\max(s_1,s_2)} U^{\max(k_1,k_2)}.$$

It remains to check that both exponent pairs lie in Λ_a. Suppose $s_1 \le s_2$; then $\min(s_1, s_2) = s_1 \ge ak_1 \ge a\min(k_1, k_2)$ because $a \ge 0$. For the maximum, suppose $k_1 \le k_2$; then $\max(s_1, s_2) \ge s_2 \ge ak_2 = a\max(k_1, k_2)$. Hence Λ_a is closed under coordinatewise min and max, that is, it is a sublattice of $\mathbb{N}_0^2$, and the translation back to (n, k) coordinates is the stated substitution $n = s - ak$. A sublattice of a product of chains is distributive. □

Status: [paper proof in this report] — the proof uses only unique factorization in $\mathbb{N}$ together with the **[kernel-verified Lean]** exact primitive-generator theorem of Section 5. No transcendence input and no analytic estimate enters.

Corollary 11.3 (Counting candidate divisors)**.** *Under failure, the number of candidates dividing $m(n, k)$, including 1 and $m(n, k)$ itself, is*

$$(k + 1)(n + 1) + \frac{ak(k + 1)}{2}.$$

Proof. By (18) a candidate divisor corresponds to a point $(s', j) \in \Lambda_a$ with $0 \le j \le k$ and $aj \le s' \le n + ak$. For fixed j there are $n + ak - aj + 1$ admissible values of s'. Summing the arithmetic progression over $j = 0, 1, \dots, k$ gives $(k + 1)(n + 1) + a\sum_{i=0}^{k} i$, which is the stated value. □

Remark 11.4 (Candidates are not divisor-closed). The full divisor count of $m(n, k) = 2^s U^k$ is $(s + 1)\prod_{p|U}(k\, v_p(U) + 1)$, which for $k \ge 1$ and U with more than one prime factor grows much faster than the count of Corollary 11.3. So $\mathcal{C}$ is emphatically not closed under taking divisors; it is closed only under the two lattice operations. The gap between the two counts is a concrete measure of how thin the candidate sublattice is inside the divisor lattice of a single candidate.

11.3 Fixed prime support

The lattice picture has an immediate consequence for the shape of a hypothetical counterexample branch: the supports do not vary at all.

Corollary 11.5 (Prime-support rigidity)**.** *Assume failure and use the simultaneous primitive output normalization of Section 5,*

$$M = 2^\beta = 2^a w^d, \qquad A = 3^\beta = 3^b v^e.$$

Then for $x = n + k\beta$,

$$2^x = 2^{\,n+ak} w^{dk}, \qquad 3^x = 3^{\,n+bk} v^{ek},$$

so every nonintegral solution — that is, every solution with $k \ge 1$ — has exactly the same odd prime support $\operatorname{supp}(w)$ *in its base-2 output, and exactly the same prime support away from 3, namely* $\operatorname{supp}(v)$*, in its base-3 output. The only variation across the entire nonintegral branch is whether the distinguished primes 2 and 3 occur at all.*

Proof. Substituting M and A into $(2^x, 3^x) = (2^n M^k, 3^n A^k)$ gives the displayed factorizations. For $k \ge 1$ the odd part of 2^x is w^{dk}, and $\operatorname{supp}(w^{dk}) = \operatorname{supp}(w)$ because $dk \ge 1$; similarly for v^{ek}. The exponents $n + ak$ and $n + bk$ may vanish, which is exactly the stated freedom at 2 and 3. □

A counterexample therefore locks the whole nonintegral branch onto a single fixed pair of finite prime sets. Computational reconnaissance should exploit this directly, enumerating primitive cores and supports rather than raw integers scanned independently; the finite verifications reported in Section 19 are **[interval computation]** and this observation does not change their status, only the shape of the search that would extend them.

11.4 Ambient root saturation and the index d

Reparameterizing Lemma 11.1 by the exponent of the primitive core w rather than of $U = w^d$ gives the description that governs roots. With $s = v_2(m)$ and t the exponent of w in the odd part of m,

$$\mathcal{C} = \{2^s w^t : s, t \in \mathbb{N}_0,\ d \mid t,\ ds \geq at\}. \tag{19}$$

Indeed $m = 2^n M^k$ has $s = n + ak$ and $t = dk$, so $d \mid t$, and $n \geq 0$ is equivalent to $s \geq at/d$, that is, to $ds \geq at$.

Definition 11.6 (Ambient root saturation)**.** For a multiplicative submonoid $H \subseteq \mathbb{N}_{>0}$ put

$$\mathrm{Sat}(H) := \{u \in \mathbb{N}_{>0} : u^r \in H \text{ for some } r \geq 1\},$$

the set of positive integers admitting some positive power inside H.

Theorem 11.7 (Exact ambient root saturation)**.** *Assume Conjecture 2.1 is false. Then*

$$\mathrm{Sat}(\mathcal{C}) = \{2^s w^t : s, t \in \mathbb{N}_0,\ ds \geq at\},$$

and for $u = 2^s w^t$ in this set the least positive r with $u^r \in \mathcal{C}$ is

$$r_{\min}(u) = \frac{d}{\gcd(d,t)}.$$

Proof. First let $u = 2^s w^t$ with $s, t \in \mathbb{N}_0$ and $ds \geq at$, and put $r = d/\gcd(d,t)$. Then $rt = d \cdot t/\gcd(d,t)$ is divisible by d, and multiplying $ds \geq at$ by $r > 0$ gives $d(rs) \geq a(rt)$. By (19), $u^r = 2^{rs} w^{rt} \in \mathcal{C}$, so $u \in \mathrm{Sat}(\mathcal{C})$. Conversely, if $u^{r'} \in \mathcal{C}$ for some $r' \geq 1$ then (19) forces $d \mid r't$, since w is power-primitive and hence the exponent of w in the odd part of $u^{r'}$ is exactly $r't$. Now $d \mid r't$ is equivalent to $(d/\gcd(d,t)) \mid r'$, so $r' \geq r$ and r is least.

Now suppose merely that $u \in \mathbb{N}_{>0}$ satisfies $u^r \in \mathcal{C}$ for some $r \geq 1$, and write $u^r = 2^S w^{dk}$ with $dS \geq a\,dk$. A prime divides u if and only if it divides u^r, so no prime outside $\{2\} \cup \mathrm{supp}(w)$ divides u. Put $c_p := v_p(w)$ for $p \mid w$; comparing valuations in $u^r = 2^S w^{dk}$ gives

$$r\,v_p(u) = dk\,c_p \qquad \text{for every } p \mid w.$$

Because w is power-primitive, $\gcd\{c_p : p \mid w\} = 1$, so there are integers r_p, almost all zero, with $\sum_p r_p c_p = 1$; then

$$\frac{dk}{r} = \sum_p r_p \cdot \frac{dk\,c_p}{r} = \sum_p r_p\,v_p(u) \in \mathbb{Z}.$$

Call this integer $t \geq 0$. Then $v_p(u) = tc_p$ for every $p \mid w$, so the odd part of u is exactly w^t, while $S = rs$ for $s = v_2(u)$. Substituting $S = rs$ and $dk = rt$ into $dS \geq a\,dk$ and dividing by $r > 0$ yields $ds \geq at$. Hence $u = 2^s w^t$ lies in the stated cone. $\square$

Status: [paper proof in this report] — the only nonformal ingredient is the Bézout step, which is the same primitivity mechanism already kernel-verified in `PrimitiveOddCore` through `oddPrimeValuationGCD`. Everything else is unique factorization and integer arithmetic.

Corollary 11.8 (Ambient saturation index)**.** *Under failure, $\mathcal{C} = \mathrm{Sat}(\mathcal{C})$ if and only if $d = 1$, and in general*

$$d = \min\{r \geq 1 : \exists\, s \in \mathbb{N}_0,\ (2^s w)^r \in \mathcal{C}\}.$$

Proof. Take $t = 1$ and any $s \geq a$, so that $ds \geq a = at$; by Theorem 11.7 the integer $2^s w$ lies in $\mathrm{Sat}(\mathcal{C})$ with $r_{\min} = d/\gcd(d, 1) = d$. This proves the displayed formula, and it shows that $2^s w \in \mathcal{C}$ forces $d = 1$. Conversely, if $d = 1$ then the divisibility condition $d \mid t$ in (19) is vacuous and the two sets coincide. ☐

Corollary 11.9 (Which integer roots of a candidate are again candidates)**.** *Assume failure and let $m = 2^n M^k \in \mathcal{C}$ and $r \geq 1$.*

(a) *There is an integer u with $u^r = m$ if and only if $r \mid n + ak$ and $r \mid dk$; such a u always lies in $\mathrm{Sat}(\mathcal{C})$.*

(b) *That integer root is itself a candidate if and only if $r \mid n$ and $r \mid k$, in which case $u = 2^{n/r} M^{k/r}$.*

Proof. (a) Write $m = 2^s w^t$ with $s = n + ak$ and $t = dk$. If $u^r = m$ then, exactly as in the proof of Theorem 11.7, u is supported in $\{2\} \cup \mathrm{supp}(w)$, $r \mid s$, and $t/r \in \mathbb{Z}$ by the Bézout step; conversely $u = 2^{s/r} w^{t/r}$ is an integer r-th root when $r \mid s$ and $r \mid t$. Membership in $\mathrm{Sat}(\mathcal{C})$ is immediate from Definition 11.6.

(b) If $u \in \mathcal{C}$ then $u = 2^{n'} M^{k'}$ for unique $n', k' \in \mathbb{N}_0$, and $u^r = 2^{rn'} M^{rk'}$ together with uniqueness of coordinates gives $n = rn'$ and $k = rk'$. The converse is the direct computation $\left(2^{n/r} M^{k/r}\right)^r = m$. ☐

Observation 11.10 (Normal intrinsically, saturated only up to index d)**.** Let $L := \{2^s w^t : s, t \in \mathbb{Z}\}$ be the ambient multiplicative group generated by 2 and w inside $\mathbb{Q}^\times_{>0}$, and let $L_\mathcal{C}$ be the group generated by $\mathcal{C}$. Then

$$L_\mathcal{C} = \{2^s w^t : s \in \mathbb{Z},\ d \mid t\}, \qquad [L : L_\mathcal{C}] = d, \qquad \mathcal{C} = \mathrm{Sat}(\mathcal{C}) \cap L_\mathcal{C}.$$

In particular $\mathrm{Sat}(\mathcal{C})$ is itself a submonoid of $\mathbb{N}_{>0}$, namely the full lattice points of the rational cone $ds \geq at$, $s, t \geq 0$.

Proof. $L_\mathcal{C}$ is generated by 2 and $M = 2^a w^d$, so it consists of the elements $2^{i+aj} w^{dj}$ with $i, j \in \mathbb{Z}$, which is the stated set; the quotient $L/L_\mathcal{C}$ is isomorphic to $\mathbb{Z}/d\mathbb{Z}$ through the exponent of w. Intersecting the cone description of Theorem 11.7 with the congruence $d \mid t$ returns (19). Closure of the cone under multiplication is additivity of the inequality $ds \geq at$ in (s, t). ☐

A terminology caution is worth stating explicitly, because the word "normal" is used differently in affine-semigroup theory. There, normality of a semigroup is measured relative to the group that the semigroup itself generates. By Observation 11.10 that group is $L_\mathcal{C}$, in which the congruence $t \equiv 0 \pmod{d}$ already holds identically, and $\mathcal{C}$ is the full set of cone points of $L_\mathcal{C}$ for every value of d: intrinsically, $\mathcal{C}$ is normal. Theorem 11.7 measures something different, namely saturation inside the larger ambient lattice L containing every $2^s w^t$. So d is an over-lattice saturation defect of finite index, not a failure of intrinsic normality. This is precisely parallel to its two other exact meanings established elsewhere in this report: d is the least rational-power index of Section 5 and the exact radical degree $[\mathbb{Q}(w^\vartheta) : \mathbb{Q}]$ of Section 6. One integer invariant, three characterizations — algebraic degree, least rational power, and lattice saturation defect.

11.5 Why lattice saturation is a good Lean target

Two points must be kept apart. The gcd/lcm closure of Theorem 11.2 and the saturation formula of Theorem 11.7 are compatible with failure by construction: they describe the counterexample world rather than obstruct it, and no contradiction can be extracted from them alone. They are structural information and search normalization. With that said, they are unusually attractive as the next formalization increment.

The mathematical inputs are already kernel-verified. The canonical primitive odd core, the rational-power index, the three-power denominator normalization, and the exact primitive generator with unique coordinates are all **[kernel-verified Lean]** and live in `ExponentialIdentities/TwoBaseIntegerExponent/`. Everything added in this section is finite combinatorics over those results: unique factorization in $\mathbb{N}$, one Bézout step that already exists in `PrimitiveOddCore` as `oddPrimeValuationGCD`, and elementary min/max arithmetic on $\mathbb{N}_0^2$. There is no analysis, no transcendence input, and no numerical certificate anywhere in the chain — a rare profile in this corpus, and the reason the work should be cheap relative to its yield. A single module, sitting beside `PrimitiveGenerator` and `RationalPowerIndex`, would carry the coordinate bijection of Lemma 11.1, the divisibility criterion (18), the lattice operations, the divisor count, the cone form (19), the saturation set, the exact value of $r_{\min}$, and the index characterization of d. The natural name is

`CandidateLattice`

and the blueprint conventions of Section 22 apply to it unchanged.

The contribution to the search is threefold. First, a kernel-checked normal form: candidate enumeration can be phrased over lattice coordinates rather than over raw integers, so rejection tests become integer inequalities of the form $ds \geq at$ and $d \mid t$ instead of factorizations recomputed for every scanned value. This is exactly the discipline recommended in the computational priority of Section 21, and it applies equally to the certificate architecture that any extension of the **[interval computation]** finite verifications of Section 19 would need; the kernel-verified region itself would still stop at $x \geq 12$ until new certificates are replayed. Second, a single invariant with three independently proved meanings: once Corollary 11.8 is formal alongside the radical-degree proposition of Section 6, any future argument that bounds d — from Kummer theory, from ramification, or from a determinant estimate — bounds all three at once, and the Lean statements make that transfer mechanical rather than a matter of prose. Third, and most concretely, root extraction is the step where descent arguments live. Every descent so far — no matched base depth, no common perfect-power degree, exact affine primitivity in Section 5 — proceeds by extracting a root of the output pair and contradicting leastness of β. Corollary 11.9 says exactly which integer roots of a candidate survive as candidates and which merely land in the saturation, so a formalized version supplies the reusable side condition that such arguments currently re-prove by hand. That is a modest but genuine improvement in the leverage of the conditional structure, and it costs no new mathematics.

12 A sufficient condition: independence of prime-log products

The reductions of Section 6 isolate the obstruction as a statement about one radical at a time. This section isolates it differently, as a single statement about $\mathbb{Q}$-linear independence of a natural family of real numbers. The resulting criterion is weaker in appearance than the four exponentials conjecture and is of a different kind: it does not mention exponentials at all.

12.1 Failure as a vanishing linear form in log-products

Suppose the conjecture fails, and let x be a nonintegral solution with outputs

$$M = 2^x \in \mathbb{N}, \qquad A = 3^x \in \mathbb{N}.$$

Taking logarithms of $M = 2^x$ and $A = 3^x$ and eliminating x gives the defining relation already recorded in (2):

$$\log M \, \log 3 - \log A \, \log 2 = 0. \tag{20}$$

Now expand both integers over their prime factorizations,

$$M = \prod_p p^{\alpha_p}, \qquad A = \prod_p p^{\gamma_p}, \qquad \alpha_p, \gamma_p \in \mathbb{N}_0,$$

so that $\log M = \sum_p \alpha_p \log p$ and $\log A = \sum_p \gamma_p \log p$. Substituting into (20),

$$\sum_p \alpha_p (\log p \, \log 3) - \sum_p \gamma_p (\log p \, \log 2) = 0. \tag{21}$$

Every term of (21) is an integer multiple of a product of exactly two logarithms of primes. Collecting coefficients in the family

$$\{\log p \, \log q \ : \ p \le q \text{ prime}\},$$

the unordered pair $\{p, 3\}$ with $p \notin \{2, 3\}$ receives the coefficient α_p; the pair $\{p, 2\}$ with $p \notin \{2, 3\}$ receives $-\gamma_p$; the diagonal pair $\{3, 3\}$ receives α_3; the diagonal pair $\{2, 2\}$ receives $-\gamma_2$; and the mixed pair $\{2, 3\}$ receives $\alpha_2 - \gamma_3$.

Thus *failure of the conjecture is exactly the existence of a nontrivial vanishing integer-coefficient linear form in products of two prime logarithms*, with the additional positivity constraint that the coefficients arise from prime factorizations.

12.2 The independence criterion

Definition 12.1 (Prime-log-product independence)**.** Say that *prime-log products are independent* if the real numbers $\log p \, \log q$, indexed by pairs of primes $p \le q$, are linearly independent over $\mathbb{Q}$.

Equivalently, in the integer-coefficient form convenient for formalization: for every finite set S of primes and every $c : S \times S \to \mathbb{Z}$,

$$\sum_{p \in S} \sum_{q \in S} c_{pq} \, \log p \, \log q = 0 \qquad \Longrightarrow \qquad c_{pq} + c_{qp} = 0 \ \text{ for all } p, q \in S.$$

(The diagonal case $p = q$ reads $2c_{pp} = 0$, hence $c_{pp} = 0$.) Clearing denominators shows the two formulations agree.

Theorem 12.2 (Prime-log-product independence implies the conjecture)**.** *If prime-log products are independent, then Conjecture 2.1 holds.*

Proof. Let x satisfy $2^x, 3^x \in \mathbb{Z}$. Both powers are positive, so $M := 2^x$ and $A := 3^x$ are positive integers, and $x \geq 0$. Relation (20) holds, hence so does (21).

Apply the independence hypothesis to the finite set $S = \operatorname{primes}(M) \cup \operatorname{primes}(A) \cup \{2, 3\}$ and to the coefficient family

$$c_{pq} = \begin{cases} \alpha_p, & q = 3, \\ -\gamma_p, & q = 2, \\ 0, & \text{otherwise}, \end{cases}$$

which reproduces the left-hand side of (21). The conclusion $c_{pq} + c_{qp} = 0$ yields, coefficient by coefficient:

- taking $q = 3$ and $p \in S \setminus \{2, 3\}$: $\alpha_p + 0 = 0$, so $\alpha_p = 0$;
- taking $p = q = 3$: $2\alpha_3 = 0$, so $\alpha_3 = 0$;
- taking $q = 2$ and $p \in S \setminus \{2, 3\}$: $-\gamma_p + 0 = 0$, so $\gamma_p = 0$;
- taking $p = q = 2$: $-2\gamma_2 = 0$, so $\gamma_2 = 0$;
- taking $p = 2$, $q = 3$: $\alpha_2 - \gamma_3 = 0$.

Primes outside S divide neither M nor A, so their exponents vanish as well. Hence $\alpha_p = 0$ for every $p \neq 2$, that is

$$M = 2^{\alpha_2},$$

and symmetrically $A = 3^{\gamma_3} = 3^{\alpha_2}$. Writing $n = \alpha_2 \in \mathbb{N}_0$ we get $2^x = M = 2^n$, and strict monotonicity of $t \mapsto 2^t$ gives $x = n \in \mathbb{Z}$. □

Status: [paper proof in this report] — a Lean module `LogProductIndependence` implementing this argument (with the independence hypothesis recorded as an explicit `Prop`, never asserted) is in preparation; the theorem statement there is `alaogluErdosConjecture_of_primeLogProductIndependence`.

12.3 What the criterion is, and is not

Three remarks place Theorem 12.2 correctly.

It is a consequence of Schanuel's conjecture. Schanuel's conjecture implies that logarithms of multiplicatively independent positive rationals are algebraically independent over $\mathbb{Q}$; in particular the numbers $\log p$, over all primes p, would be algebraically independent, and then the products $\log p \log q$ are $\mathbb{Q}$-linearly independent because distinct monomials of degree two in algebraically independent elements are linearly independent. So Definition 12.1 is strictly weaker than Schanuel and lies well inside the standard expectations.

Its strength beyond the conjecture lies in pairs that never occur. This calibration matters and is easy to miss. Collecting the relation (21) back up,

$$\sum_p \alpha_p (\log p \log 3) - \sum_p \gamma_p (\log p \log 2) = \log 3 \, \log M - \log 2 \, \log A,$$

so the only products ever exercised are those of the form $\log 2 \log p$ and $\log 3 \log p$. Write F for that sub-family. A vanishing rational relation over F factors as $\log 2\ \log u + \log 3\ \log v = 0$ for positive rationals u, v, i.e. says precisely that some positive rational $\neq 1$ has a rational ϑ-th power. Equivalently, $\mathbb{Q}$-linear independence of F is the *rational-output* form of the problem:

$$2^x \in \mathbb{Q} \ \text{ and } \ 3^x \in \mathbb{Q} \qquad \Longrightarrow \qquad x \in \mathbb{Z}. \tag{22}$$

This is worth separating from Conjecture 2.1. It implies the conjecture, since integers are rational, and it is *a priori strictly stronger*: a rational counterexample such as $2^x = 3/2$ is not an integer counterexample, and nothing in the corpus reduces one to the other. Both follow from four exponentials, because rationals are algebraic.

The proof of Theorem 12.2 establishes (22) verbatim: for rational outputs the prime exponents α_p, γ_p are integers of either sign, and the argument never used their positivity. So the criterion is not a restatement of the conjecture; it delivers the stronger rational-output form. What the calibration does show is where its surplus lies: independence of pairs $\log p \log q$ with $p, q \geq 5$ is never exercised, so one should not expect the full criterion to be easier than (22). Its unconditional dividend is the two-prime case of Section 12.5.

Status: **[kernel-verified Lean]** for the integer-output form (`alaogluErdosConjecture_of_primeLogProductIndependence`); the rational-output strengthening (22) is **[paper proof in this report]** and awaits a Lean version stated over `Rat` valuations.

It is nevertheless of a different kind from four exponentials. The four exponentials conjecture is a statement about 2×2 matrices of logarithms and their exponentials; the present criterion mentions no exponentials and is purely a linear-independence assertion about a countable family of real numbers. Neither is known, and neither is known to imply the other in general; what Theorem 12.2 shows is that *either* suffices for the Alaoglu–Erdős conjecture. This is worth recording because the two hypotheses invite different techniques: four exponentials is attacked by auxiliary functions and zero estimates, while linear independence of log-products is a question about the arithmetic of the multiplicative group of $\mathbb{Q}_{>0}$ and its second symmetric power.

The smallest instances are already open. Independence of the family includes, as its very first nontrivial case, the assertion that $(\log 3)^2$, $(\log 2)(\log 3)$ and $(\log 2)^2$ are $\mathbb{Q}$-linearly independent, equivalently that $\vartheta = \log_2 3$ is not a root of a rational quadratic — a statement about the irrationality of ϑ^2 modulo $\mathbb{Q}(\vartheta)$ that is not decided by present transcendence theory. Indeed the relation exhibited in Section 6 for a 3-smooth counterexample would already contradict the single independence

$$\alpha_2(\log 2)(\log 3) + \alpha_3(\log 3)^2 - \gamma_2(\log 2)^2 - \gamma_3(\log 2)(\log 3) \neq 0$$

for the relevant nonzero integer vector. Restricted to the two primes 2 and 3, the criterion needed is thus exactly: 1*,* ϑ *and* ϑ^2 *are linearly independent over* $\mathbb{Q}$, which is the 3-smooth shadow of Conjecture 6.3.

12.4 A finite form of the criterion

Because a counterexample has a fixed finite prime support, the full strength of Definition 12.1 is never needed at once.

Corollary 12.3 (Finite prime-support form)**.** *Let S be a finite set of primes containing 2 and 3. If the products $\log p \log q$ with $p \le q$ in S are $\mathbb{Q}$-linearly independent, then no counterexample to Conjecture 2.1 has both outputs supported on S.*

Proof. The proof of Theorem 12.2 used only the pairs drawn from the prime support of MA together with 2 and 3. □

12.5 The two-prime case is a theorem, and it is sharp

The finite form is more than a bookkeeping device, because its smallest instance is *provable*.

Proposition 12.4 (Two-prime log-product independence)**.** *Let $p \neq q$ be primes. Then $(\log p)^2$, $\log p \, \log q$ and $(\log q)^2$ are linearly independent over $\mathbb{Q}$.*

Proof. A vanishing relation $a(\log p)^2 + b \log p \log q + c(\log q)^2 = 0$ with rational a, b, c, divided by $(\log p)^2$, reads $a + bt + ct^2 = 0$ where $t = \log q/\log p$. If $(a, b, c) \neq 0$ this makes t a root of a nonzero rational polynomial, hence algebraic. But t is transcendental: it is irrational by unique factorization, and were it algebraic irrational then Gelfond–Schneider applied to $p^t = q$ would make the integer q transcendental. □

Consequently the hypothesis of Corollary 12.3 is *unconditionally true* for $S = \{2, 3\}$, and the corollary becomes an unconditional exclusion. Stated directly, and proved in Lean without any appeal to Definition 12.1:

Theorem 12.5 (Three-smooth outputs force an integral exponent)**.** *If $2^x \in \mathbb{Z}$ and $3^x \in \mathbb{Z}$ are both 3-smooth — divisible by no prime outside $\{2, 3\}$ — then $x \in \mathbb{Z}$.*

Proof. Write $2^x = 2^a 3^b$ and $3^x = 2^c 3^d$. Taking base-two logarithms gives the two relations $x = a + b\vartheta$ and $x\vartheta = c + d\vartheta$. Substituting the first into the second eliminates x and leaves

$$b\,\vartheta^2 + (a - d)\,\vartheta - c = 0.$$

If $b \neq 0$ this makes ϑ a root of a nonzero rational quadratic, contradicting its transcendence. Hence $b = 0$ and $x = a$. □

Corollary 12.6 (A smooth candidate has a rough partner)**.** *If x is a nonintegral solution and 2^x is 3-smooth, then 3^x has a prime factor at least 5. The two outputs of a counterexample can never both be built from $\{2, 3\}$.*

Status: [kernel-verified Lean] — `integer_of_threeSmooth_outputs` and `exists_prime_five_le_dvd_of_threeSmooth_candidate` in `SmoothOutputs`. The proofs use only transcendence of ϑ, itself kernel-verified in this corpus.

Two remarks fix the position of this result precisely.

The two-sided hypothesis is essential. Assuming only that 2^x is 3-smooth gives $x = a + b\vartheta$ and then requires $\left(3^\vartheta\right)^b \in \mathbb{Q}$, which is exactly the open localized-radical condition of Section 6. What the second smoothness hypothesis supplies is a *second* linear relation between x and ϑ; two such relations eliminate x and leave an algebraic equation in ϑ alone. This degree drop is the whole mechanism, and it is worth isolating as a template: *each independent multiplicative constraint on an output buys one linear relation, and two relations force algebraicity of ϑ.* The difficulty of the general problem is precisely that one never gets a second constraint for free.

Three primes is where the criterion becomes conjectural. Proposition 12.4 exhausts what transcendence of a single logarithmic ratio can give. For $S = \{2, 3, 5\}$ a vanishing relation among the six products, divided by $(\log 2)^2$, reads

$$a + b\vartheta + c\vartheta_5 + d\vartheta^2 + e\vartheta\vartheta_5 + f\vartheta_5^2 = 0, \qquad \vartheta_5 = \log_2 5,$$

that is, the point (ϑ, ϑ_5) lies on a rational conic. Both coordinates are transcendental and $1, \vartheta, \vartheta_5$ are $\mathbb{Q}$-independent by unique factorization, but no present method excludes a quadratic relation between them. So Definition 12.1 is a theorem for two primes, open from three onwards, and the Alaoglu–Erdős conjecture sits exactly on that boundary.

13 Finite-difference obstructions

The first local obstruction recorded in the corpus forbids three candidates in arithmetic progression under the convenient hypothesis $h^5 \le n$. That hypothesis is an artifact of the estimate, not of the mechanism: the sharp arbitrary-order curvature theorem is now kernel-verified for every $r \ge 2$, and it contains the three- and four-term cases as specializations.

For a function f and a step $h > 0$ write $\Delta_h f(t) = f(t+h) - f(t)$ and $\Delta_h^r = \Delta_h \circ \cdots \circ \Delta_h$ (r factors). Iterating the fundamental theorem of calculus gives the mean-value identity in integral form,

$$\Delta_h^r f(n) = \int_{[0,h]^r} f^{(r)}(n + t_1 + \cdots + t_r)\, dt_1 \cdots dt_r. \tag{23}$$

For $f(t) = t^\vartheta$ and $r \ge 2$ the derivative is

$$f^{(r)}(t) = (\vartheta)_{\underline{r}}\, t^{\vartheta - r}, \qquad (\vartheta)_{\underline{r}} := \prod_{j=0}^{r-1} (\vartheta - j) = \vartheta(\vartheta - 1)\cdots(\vartheta - r + 1).$$

Since $1 < \vartheta < 2$ the falling factorial never vanishes, so $f^{(r)}$ has a fixed nonzero sign on $(0, \infty)$, and $|f^{(r)}|$ decreases in t. These two facts — nonvanishing and decay — are the whole content of the obstruction.

Theorem 13.1 (Higher-difference progression obstruction)**.** *Let $r \ge 2$, let $n, h \in \mathbb{N}$ be positive, and set $c_r = |\,(\vartheta)_{\underline{r}}\,|$. If all of $n, n+h, \ldots, n+rh$ are candidates, then*

$$1 \le c_r\, h^r n^{\vartheta - r}.$$

Such a progression is therefore impossible whenever

$$c_r\, h^r < n^{\,r - \vartheta}. \tag{24}$$

Proof. If all $n + jh$ are candidates, every $(n + jh)^\vartheta$ is an integer, so the integer combination $\Delta_h^r f(n) = \sum_{j=0}^r (-1)^{r-j} \binom{r}{j} f(n + jh)$ is an integer. By (23) and the fixed sign of $f^{(r)}$ it is nonzero, hence of absolute value at least one. Bounding $|f^{(r)}|$ by its value at the left endpoint n,

$$1 \le |\Delta_h^r f(n)| \le c_r\, h^r n^{\vartheta - r}. \qquad \square$$

Status: [kernel-verified Lean] — `HigherDifference` proves the generic iterated mean-value identity `exists_iterated_fwdDiff_eq_pow_mul`, its real-power specialization `exists_iterated_fwdDiff_rpow_point`, integrality of the binomial forward difference, and the obstruction `not_arithmetic_progression_twoBaseNaturalCandidates_of_curvature` uniformly for every $r \geq 2$. The displayed iterated-integral identity (23) is not used by the Lean proof; the formal argument iterates the mean-value theorem instead, which avoids any integration theory.

13.1 The sharp three-term criterion

For $r = 2$ the criterion (24) reads

$$\vartheta(\vartheta - 1)h^2 < n^{2-\vartheta}, \qquad \text{i.e.} \qquad h < 1.038547802\ldots \cdot n^{0.2075187\ldots}, \tag{25}$$

strictly weaker as a hypothesis than $h \leq n^{1/5}$, whose exponent is only 0.2. The real form is `second_difference_logThreeDivLogTwo_ap_of_curvature` in `SharperProgressions`, built on the curvature estimate `second_difference_logThreeDivLogTwo_ap_bound`.

Almost all of the gain survives in a purely rational hypothesis, which is what a kernel can check without any real-number decision procedure. The step exponent $83/400 = 0.2075$ is admissible:

Theorem 13.2 (Sharp rational three-term criterion)**.** *Let $n, h \geq 1$ be integers with $h^{400} \leq n^{83}$. Then n, $n + h$, $n + 2h$ are not all candidates.*

Status: [kernel-verified Lean] — `SharpProgression` proves `second_difference_logThreeDivLogTwo_ap_sharp` and `not_three_arithmetic_progression_twoBaseNaturalCandidates_sharp`. The two numerical ingredients are the enclosure $\vartheta < 317/200$, replayed by the kernel from the exact power inequality $3^{200} < 2^{317}$ through `logThreeDivLogTwo_lt_317_div_200`, and the consequence $\vartheta(\vartheta - 1) < 37089/40000 < 1$. For $r = 3$, `ThirdDifference` kernel-checks the exact third-difference identity and the corresponding four-candidate obstruction.

13.2 Thresholds for higher order

Writing (24) as a bound on the step, a progression of length $r + 1$ starting at n is impossible once

$$h < C_r\, n^{1-\vartheta/r}, \qquad C_r = c_r^{-1/r}.$$

Table 2 lists the first thresholds.

13.3 Why longer progressions do not close the problem

As r grows the admissible step exponent $1 - \vartheta/r$ approaches one, which looks tantalizing: for large r the theorem forbids progressions with steps almost as large as the starting point itself. It does not contradict the exact candidate monoid, and it cannot be bootstrapped into a proof, for a purely combinatorial reason.

r	length $r+1$	exponent $1-\vartheta/r$	C_r
2	3	0.207518750	1.038547802
3	4	0.471679166	1.374849673
4	5	0.603759375	1.164124485
5	6	0.683007500	0.946705202
6	7	0.735839583	0.778537835
7	8	0.773576786	0.652646038
8	9	0.801879687	0.557368999

Table 2: Step thresholds for the higher-difference obstruction. A progression of length $r+1$ with step $h < C_r n^{1-\vartheta/r}$ cannot consist of candidates.

Observation 13.3 (Combinatorial limitation)**.** Conditionally on the structure described in Section 5, a dyadic block near X contains $O(\log X)$ candidates, so the average additive gap inside the block is of order $X/\log X$. For each fixed r this exceeds $X^{1-\vartheta/r}$ once X is large. A rank-two multiplicative monoid carries no reason whatsoever to contain long *additive* progressions; the hypothesis of Theorem 13.1 is therefore vacuous for the configurations that actually occur.

Extracting leverage from long progressions would require r growing with the height, control of the rapidly growing constant $c_r = |\,(\vartheta)_{\underline{r}}\,|$, and — decisively — a combinatorial source guaranteeing such progressions exist. None of the three is available. The productive response is to drop the additive-structure hypothesis altogether, which is what Section 14 does.

13.4 The optimized variable-order scale

Theorem 13.1 is uniform in r, so nothing forces the difference order to stay fixed while the starting point grows. Table 2 records the first fixed-order thresholds, but each row is only a snapshot of a one-parameter family that may be re-optimized at every height. Carrying out that optimization strengthens every fixed-order corollary at once and, more valuably, replaces the vague complaint that the candidate density "feels too small" by a single explicit constant. Throughout this subsection log denotes the natural logarithm.

The exact derivative coefficient. Fix $r \ge 2$ and recall the constant $c_r = |\,(\vartheta)_{\underline{r}}\,|$ of Theorem 13.1. Since $1 < \vartheta < 2$, the first two factors ϑ and $\vartheta - 1$ are positive while every later factor $\vartheta - j$ with $j \ge 2$ is negative, so taking absolute values termwise gives

$$c_r = \vartheta(\vartheta-1)\prod_{j=2}^{r-1}(j-\vartheta),$$

and the recursion $\Gamma(z+1) = z\Gamma(z)$ telescopes the product into closed form:

$$c_r = \frac{\vartheta(\vartheta-1)}{\Gamma(2-\vartheta)}\,\Gamma(r-\vartheta). \tag{26}$$

So c_r grows at factorial speed, which is exactly why no fixed order can be optimal for all n: the constant eventually defeats the improving exponent. Stirling's formula applied to (26) gives the root asymptotic that the optimization needs,

$$c_r^{1/r} = \frac{r}{e}\left(1 + O\!\left(\frac{\log r}{r}\right)\right). \tag{27}$$

Optimizing the step threshold over the order. The formalized condition (24) forbids a progression of length $r+1$ as soon as $h < c_r^{-1/r} n^{1-\vartheta/r}$, and substituting (27) rewrites that threshold as

$$c_r^{-1/r} n^{1-\vartheta/r} = \frac{en}{r} \exp\left(-\frac{\vartheta \log n}{r}\right)\left(1 + O\left(\frac{\log r}{r}\right)\right). \tag{28}$$

The two r-dependent factors pull in opposite directions: $c_r^{-1/r} \sim e/r$ decays in r, while $n^{-\vartheta/r}$ increases to 1. Discarding the lower-order factor and differentiating the logarithm of the leading term,

$$\frac{d}{dr}\left(1 + \log n - \log r - \frac{\vartheta \log n}{r}\right) = -\frac{1}{r} + \frac{\vartheta \log n}{r^2},$$

which is positive for $r < \vartheta \log n$, vanishes at $r = \vartheta \log n$, and is negative afterwards. The leading term is therefore unimodal in real $r > 0$ with a unique maximum at $r = \vartheta \log n$, and at that point the explicit factor e cancels against $\exp(-\vartheta \log n/r) = e^{-1}$, leaving the clean scale $n/(\vartheta \log n)$.

Theorem 13.4 (Optimized variable-order obstruction)**.** *Let $0 < \varepsilon < 1$. For all sufficiently large n choose an integer*

$$r_n = \vartheta \log n + O(1), \qquad r_n \geq 2.$$

If

$$1 \leq h \leq (1-\varepsilon)\,\frac{n}{\vartheta \log n},$$

then $n, n+h, \ldots, n+r_n h$ are not all candidates. Moreover, the largest leading-order step scale obtainable by optimizing the curvature condition (24) over r is

$$h_{\max}(n) \sim \frac{n}{\vartheta \log n} = \frac{\log 2}{\log 3} \cdot \frac{n}{\log n} = 0.6309297535\ldots \cdot \frac{n}{\log n}.$$

Proof. Take r_n to be either nearest integer to $\vartheta \log n$. Then $r_n \to \infty$, so (27) and hence (28) apply, and the quotient of $c_{r_n}^{-1/r_n} n^{1-\vartheta/r_n}$ by $n/(\vartheta \log n)$ tends to 1. The fixed margin $1-\varepsilon$ absorbs both the Stirling error $1 + O(\log r/r)$ and the $O(1)$ rounding of $\vartheta \log n$ to an integer, so (24) holds for every h in the stated range, and Theorem 13.1 excludes the progression.

For the optimality assertion, the leading approximation in (28) is unimodal in real $r > 0$ with its unique maximum at $r = \vartheta \log n$, where its value is $n/(\vartheta \log n)$. Any regime in which r stays bounded, or grows outside a constant multiple of $\log n$, returns a strictly smaller order or a strictly smaller leading constant. □

Status: **[paper proof in this report]** — the obstruction being optimized is **[kernel-verified Lean]** and holds uniformly for every $r \geq 2$ with no upper bound on the order, so substituting $r = r_n$ inside the formalized statement is legitimate; the relevant module is `HigherDifference`. The optimization itself, the Gamma evaluation (26), and the Stirling asymptotic (27) are paper arguments and have not been submitted to the kernel.

Comparison with the conditional candidate spacing. A dyadic block $[X, 2X)$ is an interval of length one in exponent space, so the exact block count of Section 5 applies verbatim: conditional on a counterexample with least nonintegral generator β,

$$\#(\mathcal{C} \cap [X, 2X)) = \frac{\log_2 X}{\beta} + O(1) = \frac{\log X}{\beta \log 2} + O_\beta(1). \tag{29}$$

The average additive spacing between consecutive candidates inside that block is therefore

$$\sim \beta \log 2 \cdot \frac{X}{\log X},$$

whereas Theorem 13.4 taken at $n \asymp X$ reaches only

$$h_{\max} \sim \frac{1}{\vartheta} \cdot \frac{X}{\log X}.$$

Both are of order $X/\log X$, so the comparison is a pure constant, and in it the transcendental ϑ cancels against the $\log 2$:

$$\frac{\beta \log 2}{1/\vartheta} = \beta\,\vartheta \log 2 = \beta \log 3. \tag{30}$$

The kernel-verified zero-free region $2^x < 4096$ forces $\beta > 12$ (Section 5), which already makes the ratio exceed $12 \log 3 = 13.1833\ldots$; the interval scan of Section 19 raises it past $20 \log 3 = 21.9722\ldots$, and the extended directed-rounding scan through 2^{26} raises it past $26 \log 3 = 28.5639\ldots$. The last two improvements are **[interval computation]** and do not enter any kernel-verified statement; the exponent bound the kernel actually certifies remains $x \geq 12$.

The same constant governs a second, purely cardinal comparison, which is the more telling of the two. The configuration forbidden by Theorem 13.4 consists of

$$r_n + 1 \sim \vartheta \log X$$

candidates arranged in one structured pattern, while the whole dyadic block contains only $\sim \log X/(\beta \log 2)$ candidates by (29). The ratio is again exactly $\beta \log 3$: the optimized theorem asks a block to supply about $\beta \log 3$ times as many candidates as it can possibly hold. Concretely, at $X = 2^{100}$ the block $[X, 2X)$ holds $100/\beta + O(1)$ candidates — about eight if β were as small as its kernel-certified floor, and about four under the 2^{26} scan — while the shortest progression the theorem forbids at that height already needs $r_n + 1 \approx 111$ terms.

Remark 13.5 (Quantified barrier). Optimizing over r materially strengthens the local theorem: the admissible step scale rises from $n^{1-\vartheta/r}$ at fixed r to $n^{1-o(1)}$, missing n itself only by the single logarithmic factor $\vartheta \log n$. It still cannot contradict the conditional monoid, because the conditional candidate density in a dyadic block is too small by the fixed factor $\beta \log 3 > 13.18$. No argument using only the number of candidates in a dyadic block can force the progression that the current finite-difference functional requires. A successful continuation must exploit additional arithmetic correlation — for example valuation congruences that force a progression of a special shape — or replace Δ_h^r by a different integer-valued functional with a better curvature-to-spacing constant.

This is a limitation of the method, not evidence against Conjecture 2.1. Its value is to replace "the density seems too weak" by a sharp asymptotic constant and a concrete numerical target that any improved functional must beat.

Agreement with the one-logarithm deficit. Section 16 audits the semigroup determinant in the same currency and identifies the same bottleneck. In both analyses the binding quantity is the conditional dyadic population $\asymp \log X$ of (29), and in both the archimedean inequality turns out to be comfortably satisfied by the exact monoid rather than violated by it. To that extent the two diagnoses agree. They differ, instructively, on the size of the shortfall. The semigroup determinant needs $r \asymp (\log X)^2$ nodes and is supplied $\asymp \log X$: a deficit of a full power of $\log X$. The optimized finite difference needs $\vartheta \log X$ nodes and is supplied $\log X/(\beta \log 2)$: a deficit of the bounded factor $\beta \log 3$. The finite difference is thus quantitatively much closer — but only because it buys that closeness with a hypothesis it cannot discharge, namely that the candidates lie in exact arithmetic progression, which Observation 13.3 says a rank-two multiplicative monoid has no reason to provide. Discarding that hypothesis is exactly what the determinants of Sections 14 and 15 do, and the one-logarithm deficit is the precise price charged for it. Read together, the two sections say that the present archimedean toolkit is short by one factor of $\log X$ in generality or by a factor of $\beta \log 3$ in strength, and that tuning constants converts neither shortfall into the other.

14 New result I: arbitrary-node lattice repulsion

The higher-difference theorem of Section 13 uses equally spaced nodes, because the forward-difference operator is built from an arithmetic progression. The same integer-versus-small-real principle extends to *arbitrary* nodes through interpolation determinants. The resulting statement constrains every configuration of candidates, not only the additively structured ones, which is exactly the gap identified in Observation 13.3.

14.1 The determinant identity

For distinct real numbers $x_0 < \cdots < x_r$ and a function f, write $f[x_0, \ldots, x_r]$ for the divided difference. The standard determinant identity is

$$\det \begin{pmatrix} 1 & x_0 & \cdots & x_0^{r-1} & f(x_0) \\ 1 & x_1 & \cdots & x_1^{r-1} & f(x_1) \\ \vdots & \vdots & & \vdots & \vdots \\ 1 & x_r & \cdots & x_r^{r-1} & f(x_r) \end{pmatrix} = V(x_0, \ldots, x_r)\, f[x_0, \ldots, x_r], \tag{31}$$

where

$$V(x_0, \ldots, x_r) = \prod_{0 \le i < j \le r} (x_j - x_i).$$

If $f \in C^r$ on the interval spanned by the nodes, the generalized mean-value theorem for divided differences supplies some $\xi \in (x_0, x_r)$ with

$$f[x_0, \ldots, x_r] = \frac{f^{(r)}(\xi)}{r!}. \tag{32}$$

For $f(t) = t^\vartheta$ this is again $f^{(r)}(t) = (\vartheta)_{\underline{r}}\, t^{\vartheta - r}$, and since ϑ is irrational — in particular not an integer — the coefficient $(\vartheta)_{\underline{r}}$ never vanishes.

14.2 The arbitrary-node theorem

Theorem 14.1 (Vandermonde repulsion for candidates)**.** *Let $r \ge 2$ and let*

$$1 \le m_0 < m_1 < \cdots < m_r$$

be natural candidates, so that $m_i^\vartheta \in \mathbb{N}$ for every i. Then

$$\prod_{0 \le i < j \le r} (m_j - m_i) \ge \frac{r!}{|\,(\vartheta)_{\underline{r}}\,|}\, m_0^{r-\vartheta}. \tag{33}$$

Proof. Apply (31) to $f(t) = t^{\vartheta}$ with the nodes m_i. Every entry of the matrix is an integer: the ordinary powers m_i^k are integral by construction, and the last column is integral precisely because the m_i are candidates. By (32) the determinant equals

$$D = V(m_0, \dots, m_r)\, \frac{(\vartheta)_{\underline{r}}}{r!}\, \xi^{\vartheta - r} \qquad (m_0 < \xi < m_r).$$

Each factor is nonzero, so $D \in \mathbb{Z} \setminus \{0\}$ and therefore $|D| \geq 1$. Because $r \geq 2 > \vartheta$ the exponent $\vartheta - r$ is negative, so $\xi^{\vartheta - r} \leq m_0^{\vartheta - r}$. Therefore

$$1 \leq |D| \leq V(m_0, \dots, m_r)\, \frac{|\,(\vartheta)_{\underline{r}}\,|}{r!}\, m_0^{\vartheta - r},$$

which rearranges to (33). □

Status: [paper proof in this report] — the argument is unconditional and elementary once the divided-difference mean-value theorem is available. It is not an instance of any committed Lean theorem: `HigherDifference` assumes arithmetic progressions. A formalization blueprint appears in Section 22.

Remark 14.2 (Formalization state). A Lean draft `ArbitraryNodeDeterminant` exists in the working tree. It supplies the algebraic half deliberately free of analysis — the node matrix, an explicit integral model, integrality of the determinant with the resulting bound $1 \leq |\det|$, the Vandermonde product and its positivity on strictly increasing nodes. **[Lean draft, not yet accepted]**: it has not yet been accepted by the kernel, and the analytic half (the divided-difference mean-value theorem for $t \mapsto t^{\vartheta}$) is still an input rather than a proved ingredient. Nothing in this section may be cited as machine-checked.

14.3 Three points: a lattice-triangle inequality

For $r = 2$ the determinant is

$$D = \det \begin{pmatrix} 1 & m_0 & m_0^{\vartheta} \\ 1 & m_1 & m_1^{\vartheta} \\ 1 & m_2 & m_2^{\vartheta} \end{pmatrix},$$

and $|D|$ is twice the area of the lattice triangle with vertices (m_i, m_i^{ϑ}). Strict convexity of $t \mapsto t^{\vartheta}$ makes the area positive; lattice integrality makes it at least $1/2$. The bound below is the quantitative form of that picture.

Corollary 14.3 (Three-candidate span)**.** *If three candidates lie in $[N, N+H]$ with $N \geq 1$, then*

$$H \geq \left(\frac{8}{\vartheta(\vartheta - 1)} \right)^{1/3} N^{(2-\vartheta)/3}. \tag{34}$$

Rounded to ten decimal places the constant is 2.0510723957 *and the exponent* 0.1383458331*; in particular* $H \geq 2.0510723956\, N^{0.1383458330}$.

Proof. Write $a = m_1 - m_0$ and $b = m_2 - m_1$. Then $a, b > 0$ and $a + b \leq H$, so the Vandermonde product satisfies

$$V = ab(a+b) \leq \frac{H^3}{4},$$

the continuous maximum being attained at $a = b = H/2$. Theorem 14.1 at $r = 2$ gives

$$V \geq \frac{2}{\vartheta(\vartheta-1)} N^{2-\vartheta}.$$

Combining the two inequalities yields (34). □

14.4 General local packing bound

Put

$$R_r := \binom{r+1}{2} = \frac{r(r+1)}{2}, \qquad C_r := \left(\frac{r!}{|\,(\vartheta)_{\underline{r}}\,|}\right)^{1/R_r}, \qquad \alpha_r := \frac{r-\vartheta}{R_r} = \frac{2(r-\vartheta)}{r(r+1)}.$$

Every pairwise gap among points of $[N, N+H]$ is at most H, so the Vandermonde product is at most H^{R_r}, and Theorem 14.1 immediately gives the following.

Corollary 14.4 (At most r points below the repulsion scale)**.** *If $r+1$ candidates lie in $[N, N+H]$, then*

$$H \geq C_r N^{\alpha_r}. \tag{35}$$

Equivalently, every interval of length strictly less than $C_r N^{\alpha_r}$ beginning at or after N contains at most r candidates.

The exponent can be optimized in r. As a function of a real variable $r > \vartheta$,

$$\alpha(r) = \frac{2(r-\vartheta)}{r(r+1)}$$

attains its maximum at

$$r_* = \vartheta + \sqrt{\vartheta^2 + \vartheta} = 3.6090841941\ldots,$$

so the best integer order in this ordinary-power family is $r = 4$ — not the three-point case that geometry makes most vivid. Table 3 lists the constants.

r	R_r	$\lvert\,(\vartheta)_{\underline{r}}\,\rvert$	α_r	C_r
2	3	0.9271436280	0.1383458331	1.2920946431
3	6	0.3847993728	0.2358395832	1.5805909191
4	10	0.5445055422	0.2415037499	1.4602287461
5	15	1.3150013031	0.2276691666	1.3510881062
6	21	4.4907787617	0.2102398809	1.2735045942
7	28	19.8269566337	0.1933941964	1.2187063909
8	36	107.3637136680	0.1781954861	1.1790125192

Table 3: Ordinary-power arbitrary-node repulsion constants. For $r = 2$ the sharper three-point constant of Corollary 14.3 uses the exact maximum of the three-gap product rather than the crude bound H^{R_2}.

Corollary 14.5 (Best ordinary-power local statement)**.** *Every interval*

$$[N, N+H], \qquad H < 1.4602287460\, N^{0.2415037499},$$

contains at most four natural candidates. The exact constant is $C_4 = (4!/|\,(\vartheta)_{\underline{4}}\,|)^{1/10} = 1.4602287461$ to ten decimal places, and the exact exponent is $\alpha_4 = (4-\vartheta)/10 = 0.2415037499\ldots$

14.5 A bound for any longer interval

The span estimate converts into a counting inequality. Let $m_1 < \cdots < m_s$ be all candidates in $[N, N+H]$. For each $i \le s-r$ the consecutive window $m_i, \ldots, m_{i+r}$ has span at least $C_r N^{\alpha_r}$ by Corollary 14.4. Summing these spans, each adjacent gap $m_{j+1} - m_j$ is counted at most r times, so

$$(s-r)C_r N^{\alpha_r} \le r(m_s - m_1) \le rH.$$

Corollary 14.6 (Arbitrary-node counting inequality)**.** *For every $r \ge 2$,*

$$\#(\mathcal{C} \cap [N, N+H]) \le r + \frac{r}{C_r} HN^{-\alpha_r}. \tag{36}$$

This is a genuine strengthening of progression avoidance: it constrains every possible configuration of candidates in an interval, with no additive hypothesis at all. It remains far from the conditional benchmark of Section 5. At the optimal $r = 4$ and $H = N$, the right-hand side of (36) is $O(N^{0.7585\cdots})$, whereas the exact population of a failed conjecture would be only $O(\log N)$. The gap is enormous, and closing it by ordinary powers alone is hopeless — which motivates enlarging the column space, the subject of Section 15.

14.6 Relation to higher differences

For equally spaced nodes $m_i = n + ih$ the Vandermonde factor is explicit:

$$V = h^{R_r} \prod_{j=1}^{r} j!.$$

Substituting in (31) recovers an integer interpolation remainder closely related to the rth forward difference of Section 13. The committed repository theorem is sharper for this special configuration, because it uses the exact binomial combination rather than bounding all pairwise gaps by a common span. Conversely, Theorem 14.1 applies when the nodes carry no additive structure whatsoever. The two results are complementary, and only the arbitrary-node form escapes the vacuity diagnosed in Observation 13.3.

15 New result II: semigroup generalized Vandermonde determinants

The determinant of Section 14 used only the integral columns $1, m, m^2, \ldots$ and one extra integral column m^ϑ. But a candidate carries much more integral data. If $m^\vartheta = n \in \mathbb{N}$, then for every $a, b \in \mathbb{N}_0$,

$$m^{a+b\vartheta} = m^a n^b \in \mathbb{N}. \tag{37}$$

The full set of available exponents is therefore

$$\Lambda_\vartheta := \{a + b\vartheta : a, b \in \mathbb{N}_0\}.$$

Since ϑ is irrational, all pairs (a, b) give distinct exponents.

15.1 Positivity of generalized Vandermonde determinants

Let

$$0 \le \lambda_0 < \lambda_1 < \cdots < \lambda_r \qquad \text{and} \quad 0 < x_0 < x_1 < \cdots < x_r.$$

The functions x^{λ_j} form an extended complete Chebyshev system on $(0, \infty)$. Their Wronskian is classical in the Karlin–Studden theory of Tchebycheff systems:

$$W(x) = x^{\sum_j \lambda_j - R_r} \prod_{0 \le i < j \le r} (\lambda_j - \lambda_i) > 0. \tag{38}$$

Indeed, after factoring x^{λ_j} from column j and x^{-i} from derivative row i, the remaining determinant is the Vandermonde determinant of the λ_j, expressed in the falling-factorial basis. Positivity of all initial Wronskians implies strict positivity of every ordered evaluation determinant.

For completeness, there is also an elementary zero-count proof. Put $x = e^t$. A nonzero linear combination of x^{λ_j} becomes an exponential polynomial $\sum c_j e^{\lambda_j t}$. After factoring the lowest exponential, differentiation removes the constant term and leaves one fewer exponential. Induction with Rolle's theorem shows that a combination of $r+1$ such functions has at most r real zeros counting multiplicity. Consequently an evaluation determinant at $r+1$ ordered nodes cannot vanish. Its sign is constant on the connected chamber $x_0 < \cdots < x_r$ and is positive in the coalescing-node limit by (38).

Lemma 15.1 (Generalized Vandermonde positivity)**.** *For strictly increasing nonnegative exponents λ_j and strictly increasing positive nodes x_i,*

$$\det(x_i^{\lambda_j})_{0 \le i,j \le r} > 0. \tag{39}$$

The two arguments just sketched are carried out in full, with all induction bookkeeping made explicit, in Appendix D.

15.2 Integer determinants from candidates

Theorem 15.2 (Semigroup integer determinant)**.** *Let $m_0 < \cdots < m_r$ be natural candidates. Choose distinct pairs $(a_j, b_j) \in \mathbb{N}_0^2$ and order*

$$\lambda_j = a_j + b_j \vartheta$$

so that $\lambda_0 < \cdots < \lambda_r$. Then

$$D_{\boldsymbol{\lambda}}(\boldsymbol{m}) := \det(m_i^{\lambda_j})_{0 \le i,j \le r} \tag{40}$$

is a positive integer. In particular, $D_{\boldsymbol{\lambda}}(\boldsymbol{m}) \ge 1$.

Proof. By (37), each entry equals

$$m_i^{a_j} (m_i^{\vartheta})^{b_j} \in \mathbb{N},$$

so the determinant is an integer. It is strictly positive by Lemma 15.1. □

Status: [paper proof in this report]

This theorem packages all mixed integrality consequences into one reusable object. It is not the same as the repository's exponential interpolation determinant in `MultiplicativeRank.lean`: here the rows are arbitrary natural candidates and the columns are chosen from the two-dimensional exponent semigroup.

15.3 Divided-difference factorization

Let $f_j(x) = x^{\lambda_j}$ and let

$$V(\boldsymbol{m}) = \prod_{i<j}(m_j - m_i).$$

Applying the Newton divided-difference row transformation gives

$$D_{\boldsymbol{\lambda}}(\boldsymbol{m}) = V(\boldsymbol{m}) \det\big(f_j[m_0, \dots, m_i]\big)_{0 \le i,j \le r}. \tag{41}$$

The transformation is exact; its determinant is $V(\boldsymbol{m})^{-1}$.

Suppose

$$N \le m_0 < \cdots < m_r \le N + H \le 2N.$$

For each divided difference, the mean-value theorem gives a point in $[N, 2N]$ and

$$|f_j[m_0, \dots, m_i]| \le \frac{|\,(\lambda_j)_{\underline{i}}\,|}{i!} \sup_{N \le x \le 2N} x^{\lambda_j - i}. \tag{42}$$

We need a uniform elementary coefficient estimate.

Lemma 15.3 (Falling-factorial coefficient bound)**.** *For $\lambda \ge 0$ and $i \in \mathbb{N}_0$,*

$$\frac{|\,(\lambda)_{\underline{i}}\,|}{i!} \le 2^{\lambda + i}. \tag{43}$$

Proof. If $0 \le \lambda \le 1$, each factor satisfies $|\lambda - k| \le k + 1$, so the quotient is at most 1. If $\lambda \ge 1$, let $n = \lceil \lambda \rceil$. Then

$$\frac{|\,(\lambda)_{\underline{i}}\,|}{i!} \le \prod_{k=0}^{i-1} \frac{\lambda + k}{k+1} \le \prod_{k=0}^{i-1} \frac{n+k}{k+1} = \binom{n+i-1}{i}.$$

The last binomial coefficient is at most the sum of its row, 2^{n+i-1}, and $n - 1 \le \lambda$. Hence it is at most $2^{\lambda+i}$. □

For $x \in [N, 2N]$,

$$x^{\lambda - i} \le 2^{\lambda} N^{\lambda - i}. \tag{44}$$

If $\lambda - i \ge 0$, use $x \le 2N$; if it is negative, use $x \ge N$. Combining (42), (43), and (44),

$$|f_j[m_0, \dots, m_i]| \le 2^{2\lambda_j + i} N^{\lambda_j - i}. \tag{45}$$

15.4 Quantitative semigroup gap theorem

Set

$$R := R_r = \frac{r(r+1)}{2}, \qquad \Sigma := \sum_{j=0}^{r} \lambda_j.$$

By the Leibniz formula and (45), every permutation term in the small determinant in (41) is at most

$$2^{2\Sigma + R} N^{\Sigma - R},$$

and there are $(r+1)!$ terms. Since every pairwise node difference is at most H, $V(\boldsymbol{m}) \le H^R$. The positive integer lower bound from Theorem 15.2 now gives the following.

Theorem 15.4 (Semigroup determinant gap)**.** *Let $r \geq 1$. Let $\lambda_0 < \cdots < \lambda_r$ be any $r+1$ exponents in Λ_ϑ, and put $R = r(r+1)/2$ and $\Sigma = \sum_j \lambda_j$. If $r+1$ natural candidates lie in*

$$[N, N+H] \subseteq [N, 2N], \qquad N \geq 1,$$

then

$$H \geq ((r+1)!)^{-1/R} 2^{-1-2\Sigma/R} N^{1-\Sigma/R}. \tag{46}$$

Proof. Equations (41) and (45) imply

$$1 \leq D_{\boldsymbol{\lambda}}(\boldsymbol{m}) \leq H^R (r+1)! 2^{R+2\Sigma} N^{\Sigma - R}.$$

Taking Rth roots and rearranging proves (46). □

The exponent $1 - \Sigma/R$ is the key. We should choose the $r+1$ smallest available semigroup exponents to minimize Σ.

15.5 The ordered exponent semigroup

Write the elements of Λ_ϑ in increasing order:

$$0 = \lambda_0 < \lambda_1 < \lambda_2 < \cdots.$$

The beginning is

$$0,\ 1,\ \vartheta,\ 2,\ 1+\vartheta,\ 3,\ 2\vartheta,\ 2+\vartheta,\ 4,\ 1+2\vartheta,\ 3+\vartheta,\ 3\vartheta,\ 5, \ldots$$

The counting function is elementary:

$$\begin{aligned} L(T) &:= \#\{\lambda \in \Lambda_\vartheta : \lambda \leq T\} \\ &= \sum_{0 \leq b \leq T/\vartheta} (\lfloor T - b\vartheta \rfloor + 1) = \frac{T^2}{2\vartheta} + O(T). \end{aligned} \tag{47}$$

Irrationality of ϑ ensures that different pairs (a, b) are counted distinctly. Inverting (47) and summing the order statistics gives

$$\lambda_r = \sqrt{2\vartheta r} + O(1), \tag{48}$$

$$\Sigma_r := \sum_{j=0}^{r} \lambda_j = \frac{2}{3}\sqrt{2\vartheta}\, r^{3/2} + O(r), \tag{49}$$

$$\frac{\Sigma_r}{R_r} = \frac{4}{3}\sqrt{\frac{2\vartheta}{r}} + O(r^{-1}). \tag{50}$$

Consequently the power of N in (46) tends to 1:

$$1 - \frac{\Sigma_r}{R_r} = 1 - \frac{4}{3}\sqrt{\frac{2\vartheta}{r}} + O(r^{-1}). \tag{51}$$

r	Σ_r	$1-\Sigma_r/R_r$
2	2.5849625007	0.1383458331
3	4.5849625007	0.2358395832
4	7.1699250014	0.2830074999
5	10.1699250014	0.3220049999
6	13.3398500029	0.3647690475
7	16.9248125036	0.3955424106
8	20.9248125036	0.4187552082
9	25.0947375050	0.4423391666
10	29.6797000058	0.4603690908
12	39.4345875079	0.4944283653
20	87.7939875195	0.5819333928

Table 4: Exponent gain from using the first $r+1$ mixed exponents in $\mathbb{N}_0+\mathbb{N}_0\vartheta$. The ordinary-power family peaks and then decays; the semigroup family improves monotonically in these data and tends to exponent 1.

15.6 An explicit coarse form

For the global counting corollary we need no delicate lattice-point error term. A simple bound suffices.

Lemma 15.5 (Crude semigroup weight bound)**.** *For $r\ge 2$, the first $r+1$ semigroup exponents satisfy*

$$\lambda_r \le 6\sqrt{r}, \qquad \frac{\Sigma_r}{R_r} \le \frac{12}{\sqrt{r}}. \tag{52}$$

Proof. Let $q=\lceil\sqrt{r+1}\rceil$. The $(q+1)^2$ pairs $0\le a,b\le q$ give distinct exponents and each is at most $q(1+\vartheta)<3q$. Since $(q+1)^2>r+1$, the rth ordered exponent is below $3q$. For $r\ge 2$, $q\le\sqrt{r+1}+1\le 2\sqrt{r}$, hence $\lambda_r\le 6\sqrt{r}$. Then

$$\Sigma_r \le (r+1)\lambda_r \le 6(r+1)\sqrt{r},$$

and division by $R_r=r(r+1)/2$ gives the second inequality. □

For $r\ge 144$, (52) gives $\Sigma_r/R_r\le 1$. Also

$$((r+1)!)^{1/R_r} \le (r+1)^{(r+1)/R_r} = (r+1)^{2/r} \le 3$$

for $r\ge 2$. Therefore Theorem 15.4 yields a particularly clean form.

Corollary 15.6 (Explicit near-linear span)**.** *If $r\ge 144$ and $r+1$ natural candidates lie in $[N,N+H]\subseteq[N,2N]$, then*

$$H \ge \frac{1}{24}N^{1-12/\sqrt{r}}. \tag{53}$$

Proof. In (46), use $((r+1)!)^{-1/R}\ge 1/3$, $2^{-1-2\Sigma/R}\ge 2^{-3}=1/8$, and, since $N\ge 1$, $N^{1-\Sigma/R}\ge N^{1-12/\sqrt{r}}$. □

15.7 Counting candidates in a general interval

As in the ordinary-power count of Section 14, sum the spans of consecutive windows of $r+1$ candidates. Every adjacent gap is counted at most r times.

Corollary 15.7 (Semigroup interval count). *For $r \ge 144$ and $1 \le H \le N$,*

$$\#(\mathcal{C} \cap [N, N+H]) \le r + 24rHN^{-1+12/\sqrt{r}}. \tag{54}$$

Proof. If s is the number of candidates and $s \le r$, there is nothing to prove. Otherwise, Corollary 15.6 gives a lower bound $L = N^{1-12/\sqrt{r}}/24$ for each span $m_{i+r} - m_i$. Thus

$$(s-r)L \le rH,$$

which is equivalent to (54). □

15.8 An independent dyadic logarithmic-square bound

Corollary 15.8 (Explicit $O((\log X)^2)$ dyadic bound). *For every real $X \ge 1$,*

$$\#(\mathcal{C} \cap [X, 2X]) \le 10000 \max\{1, \log X\}^2. \tag{55}$$

Proof. Put $L = \max\{1, \log X\}$ and

$$r = \left\lceil 144L^2 \right\rceil.$$

Then $r \ge 144$, $\sqrt{r} \ge 12L$, and

$$X^{12/\sqrt{r}} = \exp\left(\frac{12 \log X}{\sqrt{r}}\right) \le e.$$

Apply (54) with $N = H = X$:

$$\#(\mathcal{C} \cap [X, 2X]) \le r + 24rX^{12/\sqrt{r}} \le (1+24e)r.$$

Since $L \ge 1$, $r \le 145L^2$, and $(1+24e)145 < 10000$. □

Status: [paper proof in this report]

The bound is numerically coarse by design. Its value is conceptual independence: it comes from positivity and integrality of generalized Vandermonde determinants, not from the prime-factor valuation rank of candidates. Refining constants is much less important than finding an additional source of determinant lower bound.

Status: [Lean draft, not yet accepted]

Lean drafts of this hierarchy exist in the repository but have not yet been accepted by the kernel: `MixedExponentSemigroup` for (37) and the ordered exponent semigroup, `SemigroupDeterminant` for Theorem 15.2 and Theorem 15.4, `FallingFactorialBound` for Lemma 15.3, and `SemigroupWeightBound` for Lemma 15.5 and its explicit consequences. Until those files compile, every statement in this section rests on the paper proofs given above.

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16 Why the new hierarchy still stops short

The semigroup determinant is much stronger locally than the ordinary-power determinant: its repulsion exponent approaches 1. It is therefore important to identify exactly why it does not prove the conjecture.

16.1 How many nodes are needed to approach scale X?

The asymptotic exponent in (51) is

$$\delta_r := 1 - \frac{\Sigma_r}{R_r} = 1 - \frac{c_\vartheta}{\sqrt{r}} + O(r^{-1}), \qquad c_\vartheta = \frac{4}{3}\sqrt{2\vartheta} = 2.3739044262\ldots.$$

For nodes near X, the lower span has the shape

$$X^{\delta_r} = X \exp\left(-\frac{c_\vartheta \log X}{\sqrt{r}} + O\left(\frac{\log X}{r}\right)\right). \tag{56}$$

To make the exponential loss in (56) bounded independently of X, one needs

$$\sqrt{r} \asymp \log X, \qquad \text{that is,} \qquad r \asymp (\log X)^2. \tag{57}$$

This is precisely why Corollary 15.8 has a logarithmic square.

16.2 How many nodes a counterexample supplies

Conditional on a counterexample, the exact dyadic block count established in Section 5 gives

$$\#(\mathcal{C} \cap [X, 2X]) = \frac{\log_2 X}{\beta} + O(1) \asymp \log X. \tag{58}$$

Thus the actual conditional population is smaller than the determinant threshold (57) by one factor of $\log X$.

Taking $r \asymp \log X$ in (56) yields only

$$X \exp(-C\sqrt{\log X}),$$

which is much smaller than the average spacing $X/\log X$ among the $\asymp \log X$ conditional candidates. The inequality is therefore compatible with the exact monoid even at the level of expected spacing.

16.3 Broader multiplicative intervals do not help

The cumulative conditional count below X is $\asymp (\log X)^2$, so one might try to use all those points at once. But their nodes range from 1 to X, and the gap theorem depends on the smallest node. Restricting to $[X^\gamma, X]$ with fixed $0 < \gamma < 1$ does retain $\asymp (\log X)^2$ points, but its additive length is $\asymp X$, while the lower scale is at most $X^{\gamma+o(1)}$. Again there is no contradiction.

This is not a mere defect of coarse constants. It reflects a mismatch between additive node geometry and the logarithmic triangular lattice of exponents (n, k) in $m = 2^n M^k$.

16.4 The determinant needs one more source of size

The archimedean argument uses only

$$D \in \mathbb{Z}_{>0} \quad \Longrightarrow \quad D \geq 1.$$

If the exact monoid coordinates forced

$$D \equiv 0 \pmod{Q} \qquad \text{with} \quad \log Q \gg R \log X/\sqrt{r},$$

then the lower bound $D \geq Q$ could remove the one-logarithm deficit. Alternatively, one could seek a determinant with total exponent weight $\Sigma = O(r)$ rather than $\Sigma \asymp r^{3/2}$, but the two-dimensional semigroup counting law $L(T) \asymp T^2$ makes that impossible using only distinct nonnegative mixed monomials.

This gives a concrete design specification:

> A successful determinant proof must add nonarchimedean divisibility, introduce lower-weight functions outside the mixed-monomial semigroup, or exploit the exact two-dimensional row grid so efficiently that $O(\log X)$ nodes suffice.

Status: [paper proof in this report]

The deficit analysis of this section is a paper argument about the limits of the method, not a formalized theorem. It does not weaken any statement proved above; it explains why the explicit $O((\log X)^2)$ bound of Corollary 15.8 cannot be pushed to the conditional $\asymp \log X$ population by tuning constants alone. The conjecture remains open.

17 Further equivalent reformulations

Several equivalent forms are already formalized in the repository and were catalogued in Section 4. The exact conditional structure of Section 5 also yields useful paper-level reformulations that expose different possible proof technologies. Each restatement below is an exact equivalence, not a heuristic analogy: a proof of any one of them is a proof of Conjecture 2.1.

17.1 Integer points on a transcendental lattice curve

The candidate formulation of Section 2 says that

$$\{(m,n) \in \mathbb{N}^2 : n = m^{\vartheta}\} = \{(2^k, 3^k) : k \in \mathbb{N}_0\}. \tag{59}$$

The curve $n = m^{\vartheta}$ is strictly convex and transcendental, and (59) asserts that its only lattice points are the trivial exponential ones. The arbitrary-node repulsion theorem of Section 14 can be read as a lower bound for the volumes of lattice simplices cut from this curve; the deficit analysis of Section 16 measures exactly how far the current bound is from excluding the nontrivial points.

17.2 Vanishing determinant of four logarithms

Writing $M = 2^x$ and $A = 3^x$, a solution is a positive integer pair with

$$\log M \ \log 3 - \log A \ \log 2 = 0, \quad M, A \in \mathbb{N}, \quad \Longrightarrow \quad (M, A) = (2^k, 3^k). \tag{60}$$

The hypothesis of (60) is a vanishing 2×2 determinant of logarithms of positive integers. This is the most direct bridge to the four-exponentials circle of ideas and to algebraic independence of logarithms, and it is the form in which the kernel-verified four-exponentials reduction is stated.

17.3 Primitive localized-radical exclusion

The kernel-verified radical form of Section 6 is

$$\forall\, w > 1 \text{ odd and power-primitive}, \quad \forall\, d > 0,\ a \geq 0, \qquad 3^a (w^\vartheta)^d \notin \mathbb{N}. \tag{61}$$

If (61) fails with least rational-power index d, then $X^d - c/3^a$ is the exact minimal polynomial of w^ϑ over $\mathbb{Q}$ for the corresponding rational c.

This formulation is well suited to Kummer-theoretic, norm, and ramification investigations, because all redundant proper powers have been removed by primitivity and the algebraic degree of the hypothetical radical is known exactly rather than merely bounded.

17.4 Exact growth dichotomy

Write

$$B(T) := \#(\mathcal{C} \cap [2^T, 2^{T+1}))$$

for the Tth dyadic block count. The repository formalizes the cumulative subquadratic equivalence of Section 5; the exact block formula there, namely

$$B(T) = 1 + \left\lfloor \frac{T+1}{\beta} \right\rfloor \qquad \text{under failure with least generator } \beta, \tag{62}$$

gives several immediate variants.

Proposition 17.1 (Block-growth reformulations). *Conjecture 2.1 is equivalent to each of the following:*

1. *the dyadic block counts $B(T)$ are bounded;*
2. $B(T) = o(T)$*;*
3. $\liminf_{T\to\infty} B(T)/T = 0$*;*
4. $B(T) = 1$ *for every* $T \in \mathbb{N}_0$.

Proof. If the conjecture holds, every block contains only 2^T, so all four properties hold. If it fails, (62) gives $B(T) = 1 + \lfloor (T+1)/\beta \rfloor$, so $B(T)/T \to 1/\beta > 0$ and none of the first three properties holds. The fourth is visibly equivalent to the candidate statement. □

This is useful because a future analytic theorem need not classify candidates individually. Any uniform $o(T)$ bound for the number of candidates in the Tth dyadic block would prove the conjecture.

The exactness of (62) permits a sharpening of item (4) that is, at first sight, dramatically weaker than the conjecture and yet still equivalent to it.

Proposition 17.2 (Infinitely many singleton blocks suffice)**.** *Conjecture 2.1 is equivalent to the statement*

$$B(T) = 1 \quad \textit{for infinitely many } T \in \mathbb{N}_0.$$

Proof. If the conjecture holds then $B(T) = 1$ for every T, so certainly for infinitely many. Conversely, suppose the conjecture fails, and let $\beta > 0$ be the irrational least generator of the exact solution monoid. By (62),

$$B(T) = 1 \iff \left\lfloor \frac{T+1}{\beta} \right\rfloor = 0 \iff T + 1 < \beta,$$

which holds for at most the finitely many indices $T < \beta - 1$. Hence failure forces $B(T) = 1$ for only finitely many T, and the contrapositive gives the equivalence. □

Remark 17.3. Proposition 17.2 cannot be weakened further to "$B(T) = 1$ for some T". That statement is already known unconditionally and carries no information: the kernel-verified zero-free region $2^x < 4096$ gives $\beta \geq 12$, hence $B(T) = 1$ for $T = 0, \ldots, 10$, and the interval computation of Section 19 pushes the same conclusion to $T \leq 19$ at the weaker **[interval computation]** trust level. The gap between "for some T" and "for infinitely many T" is exactly the gap between a finite check and the conjecture, because none of the present results bounds β from above: a counterexample with a very large least generator would have $B(T) = 1$ for every $T < \beta - 1$, that is, on any range a fixed finite computation could ever reach. Only the infinitude of singleton blocks is decisive.

Status: **[paper proof in this report]** — Proposition 17.1 and Proposition 17.2 are paper proofs in this report, resting on the kernel-verified exact dyadic block counts. A Lean draft `BlockGrowthReformulations` exists but has not yet been accepted by the kernel. **[Lean draft, not yet accepted]**

17.5 Compact synchronized S-integral orbit

The S-integral orbit of Section 5 yields another equivalence, this time one that isolates the synchronization of the two denominators.

Proposition 17.4 (Compact-orbit reformulation)**.** *Conjecture 2.1 fails if and only if there exist integers $M, A > 1$ and an irrational $\beta > 0$ such that, for every $k \geq 1$, the point*

$$\left(\frac{M^k}{2^{\lfloor k\beta \rfloor}}, \frac{A^k}{3^{\lfloor k\beta \rfloor}} \right) \tag{63}$$

lies on the compact arc

$$\mathcal{A} = \{(2^t, 3^t) : 0 \leq t < 1\}.$$

Under these conditions the orbit is dense in $\mathcal{A}$.

Proof. Failure supplies the least generator β, with $M = 2^\beta$ and $A = 3^\beta$; then (63) is exactly $(2^{\{k\beta\}}, 3^{\{k\beta\}})$, and density follows from irrational rotation.

Conversely, arc membership gives $t_k \in [0,1)$ such that

$$k \log_2 M - \lfloor k\beta \rfloor = t_k = k \log_3 A - \lfloor k\beta \rfloor.$$

Thus $\log_2 M = \log_3 A =: \gamma$, and $0 \le k\gamma - \lfloor k\beta \rfloor < 1$ for every k. Hence $\lfloor k\gamma \rfloor = \lfloor k\beta \rfloor$, so $|k(\gamma-\beta)| < 1$ for every k; therefore $\gamma = \beta$. Consequently $M = 2^\beta$ and $A = 3^\beta$ are integers while β is nonintegral, so the conjecture fails. The same identity now shows that the orbit equals the irrational-rotation orbit and is dense. ☐

Status: [paper proof in this report] — paper proof in this report; the Lean draft `CompactOrbit` has not yet been accepted by the kernel. **[Lean draft, not yet accepted]**

This formulation isolates the synchronized-denominator feature that a specialized S-integral counting theorem would need to exploit: the two coordinates are S-integral for disjoint sets of primes, yet their denominator exponents are forced to agree exactly at every height.

17.6 Two-dimensional exponential grid

Under failure, every solution coordinate (n, k) gives

$$m_{n,k} = 2^n M^k, \qquad m_{n,k}^{\vartheta} = 3^n A^k,$$

and for every mixed exponent $a + b\vartheta$ with $a, b \in \mathbb{N}_0$,

$$m_{n,k}^{a+b\vartheta} = (2^a 3^b)^n (M^a A^b)^k. \tag{64}$$

Thus the candidate evaluation matrix factors through a two-dimensional row lattice and a two-dimensional column lattice. The conjecture can be viewed as the assertion that this integer exponential grid cannot exist unless (M, A) is a common integral power of $(2, 3)$. This is the most natural starting point for a bespoke 2×2 interpolation determinant that uses more than the arbitrary-node geometry of Section 14, and it is the structure the semigroup hierarchy of Section 15 exploits only partially.

17.7 An operator formulation on the prime-logarithm space

All the reformulations above are Diophantine. There is also a purely linear-algebraic one, which recasts the problem as a statement about multiplication by ϑ acting on a single $\mathbb{Q}$-vector space.

Let

$$W := \operatorname{span}_{\mathbb{Q}}\{\log p \ : \ p \text{ prime}\} \subset \mathbb{R},$$

a $\mathbb{Q}$-vector space of countably infinite dimension, with the $\log p$ as a basis by unique factorization. Multiplication by ϑ does not preserve W, so consider the largest part of W that it does return to W:

$$U := \{\, w \in W \ : \ \vartheta\, w \in W \,\}.$$

This is a $\mathbb{Q}$-subspace, and it is never zero: $\vartheta \log 2 = \log 3 \in W$, so $\mathbb{Q} \log 2 \subseteq U$.

Proposition 17.5 (Operator form)**.** *$\dim_{\mathbb{Q}} U = \operatorname{rank} K$, where $K = \{v \in \mathbb{Q}_{>0} : v^{\vartheta} \in \mathbb{Q}\}$ is the multiplicative group of Section 12. Consequently the rational-output form (22) — and hence Conjecture 2.1 — is equivalent to*

$$\boxed{\dim_{\mathbb{Q}}\{\, w \in W : \vartheta\, w \in W \,\} = 1.}$$

Proof. An element of W is $w = \sum_p c_p \log p$ with rational c_p, almost all zero. Clearing a common denominator N gives $Nw = \log v$ for a positive rational v. Now $\vartheta w \in W$ says $\vartheta \log v = \sum_p d_p \log p$ with rational d_p; clearing a denominator M turns this into $\vartheta \log(v^M) = \log u$ with $u \in \mathbb{Q}_{>0}$, that is $(v^M)^{\vartheta} = u$, i.e. $v^M \in K$. So U is exactly the $\mathbb{Q}$-span of $\log K$ inside W, whence $\dim_{\mathbb{Q}} U = \operatorname{rank} K$. The conjecture in its rational-output form says $K = \langle 2 \rangle$, of rank one. ☐

Status: **[paper proof in this report]** — an elementary translation, but a genuinely different framing: it replaces a Diophantine assertion by the statement that the operator "multiply by ϑ" carries only a single line of W back into W.

Two remarks on what this framing does and does not give.

It explains why finite-dimensional arguments cannot apply directly. If some nonzero finite-dimensional subspace $V \subseteq W$ were *ϑ-stable*, then ϑ would be an eigenvalue of the rational matrix representing multiplication by ϑ on V, hence algebraic, contradicting transcendence. So a proof cannot proceed by exhibiting an invariant finite-dimensional subspace: none exists, unconditionally. The subspace U above is not ϑ-stable — $\vartheta U \subseteq W$ but generally $\vartheta U \not\subseteq U$ — and this failure of stability is precisely what blocks the eigenvalue argument.

The natural stable candidate is empty for the same reason. Setting $U_\infty = \{w \in W : \vartheta^k w \in W \text{ for all } k \geq 0\}$ does produce a ϑ-stable space, but it is either zero or infinite-dimensional, and infinite dimension is automatic whenever it is nonzero: if $0 \neq w \in U_\infty$, the powers $\vartheta^k w$ are $\mathbb{Q}$-linearly independent, since a relation $\sum_k c_k \vartheta^k w = 0$ with $w \neq 0$ would make ϑ algebraic. So the operator picture reproduces the transcendence barrier exactly, rather than circumventing it. Its value is diagnostic: it shows the obstruction is not a missing estimate but the absence of any finite-dimensional invariant structure to work with.

18 A consolidated catalogue of equivalent formulations

The preceding sections restate Conjecture 2.1 many times over: as a candidate-base identity, as a localized radical exclusion, as a counting dichotomy, as a statement about fractional parts, and as a closure property of the solution monoid. Because these restatements were introduced where they were needed rather than where they belong together, this section collects them in one place.

Throughout, $\vartheta = \log_2 3$,

$$\mathcal{S} = \{x \in \mathbb{R} : 2^x, 3^x \in \mathbb{Z}\}, \qquad \mathcal{C} = \{m \in \mathbb{N} : m > 0,\ m^{\vartheta} \in \mathbb{N}\},$$

which correspond under $x \mapsto 2^x$ by the exponent–candidate bijection of Section 2, and

$$B(T) := \#(\mathcal{C} \cap [2^T, 2^{T+1}))$$

is the Tth dyadic block count.

18.1 How to read the catalogue

Every item (E1)–(E26) below is an *exact biconditional*: a proof of any one of them is a proof of Conjecture 2.1, and a disproof of any one of them is a counterexample. None of them is proved. The status tag on each item records the strength of the evidence for the *equivalence*, never for the conjecture: **[kernel-verified Lean]** means the biconditional itself has been accepted by the Lean kernel, **[paper proof in this report]** means it is a paper proof in this report, and **[classical/open background]** means the equivalence is obtained by combining a paper proof here with a classical transcendence theorem.

Two adjacent notions are deliberately excluded from the catalogue and treated separately in Section 18.8 below: *sufficient conditions*, which are one-way implications from hypotheses that are themselves unproved, and *unconditional partial results*, which settle a bounded region rather than the statement. Conflating the three is the most common way to overstate what the corpus establishes.

18.2 Cluster A: normalized restatements

These three items are the conjecture rewritten in the coordinates in which the rest of the corpus works. They carry no arithmetic content beyond that bijection, and their value is that each suggests a different toolkit.

(E1) **[kernel-verified Lean]** *Candidate form.* $\mathcal{C} = \{2^n : n \in \mathbb{N}_0\}$; equivalently, the only positive integers m with $m^\vartheta \in \mathbb{N}$ are the powers of two (Section 2).

(E2) **[kernel-verified Lean]** *Vanishing four-log determinant.* The only positive integer pairs (M, A) with $\log M \, \log 3 - \log A \, \log 2 = 0$ are $(M, A) = (2^n, 3^n)$, $n \in \mathbb{N}_0$ (Section 2); this is the 2×2 logarithmic determinant whose entries include two logarithms of unknown integers, and the form in which the four-exponentials reduction of Section 7 is stated.

(E3) **[kernel-verified Lean]** *One-parameter subgroup intersection.* The real one-parameter subgroup $\Gamma = \{(2^t, 3^t) : t \in \mathbb{R}\}$ of the multiplicative torus meets $\mathbb{N}^2$ exactly in $\{(2^n, 3^n) : n \in \mathbb{N}_0\}$ (Section 2); a geometric restatement of (E2), with the difficulty that the direction vector $(\log 2, \log 3)$ is transcendental rather than algebraic.

Status: **[kernel-verified Lean]** — the exponent–candidate bijection is `exists_twoBaseNaturalCandidate_of_two_three_rpow_integer` together with `TwoBaseNaturalCandidate.integer_powers`, and the equivalence (E1) is

`alaogluErdosConjecture_iff_candidates_are_powers_of_two`.

Items (E2) and (E3) are the same content transported along that bijection and along $m \mapsto (m, m^\vartheta)$; they carry no separate Lean identifier.

18.3 Cluster B: localized radicals

This cluster is the most compressed form of the problem known: a statement quantified over all real x becomes a statement about one integer radical at a time, with all 2-adic and all but one 3-adic degree of freedom eliminated. It is also the cluster closest to established transcendence theory.

(E4) [**kernel-verified Lean**] *Localized radical exclusion.* There is no odd integer $u > 1$ with $u^{\vartheta} \in \mathbb{Z}[1/3]$ (Section 6).

(E5) [**kernel-verified Lean**] *Integral localized form.* There is no pair (u, a) with $u > 1$ odd, $a \in \mathbb{N}_0$ and $3^a u^{\vartheta} \in \mathbb{N}$; equivalently, no candidate has odd part greater than one (Section 6).

(E6) [**kernel-verified Lean**] *Primitive localized radical exclusion.* There is no power-primitive odd $w > 1$ and $d \geq 1$ with $(w^{\vartheta})^d \in \mathbb{Z}[1/3]$ (Section 6); primitivity removes every proper-power redundancy, and under failure the minimal polynomial of w^{ϑ} is known exactly.

(E7) [**kernel-verified Lean**] *Symmetric base-swapped form.* With $\eta = \log_3 2 = 1/\vartheta$, there is no integer $v > 1$ with $3 \nmid v$ and $v^{\eta} \in \mathbb{Z}[1/2]$ (Appendix C); the two normalizations produce the same least exponent, so the reduction is not an artifact of privileging the base 2.

Status: [**kernel-verified Lean**] — (E4) is

`alaogluErdosConjecture_iff_odd_rpow_not_localized`

in `Localization`, restated as

`alaogluErdosConjecture_iff_oddLocalizedRadicalExclusion`

in `RadicalReformulation`; (E5) is the independently proved candidate-level variant

`alaogluErdosConjecture_iff_no_odd_multiple_candidate`;

(E6) is

`alaogluErdosConjecture_iff_primitiveOddLocalizedRadicalExclusion`

in `PrimitiveRadical`; and (E7) is

`alaogluErdosConjecture_iff_threeFreeLocalizedRadicalExclusion`

in `RadicalReformulation`. The equivalences are kernel-verified; the exclusions on their right-hand sides are the open conjecture.

18.4 Cluster C: counting and block growth

The counting cluster replaces an integrality statement about individual candidates by an upper bound on how many of them there can be. It is attractive because an analytic theorem need never identify a candidate, and unattractive because the target quantities are extremely far from the arithmetic conclusion; Section 16 measures that distance precisely.

(E8) [**kernel-verified Lean**] *Subquadratic cumulative growth.* $\#(\mathcal{C} \cap [1, 2^T])$ is $o(T^2)$ as $T \to \infty$ (Section 5).

(E9) [**paper proof in this report**] *Bounded blocks.* The dyadic block counts $B(T)$ are bounded (Section 17).

(E10) **[paper proof in this report]** *Sublinear blocks.* $B(T) = o(T)$ (Section 17).

(E11) **[paper proof in this report]** *Sublinear along a subsequence.* $\liminf_{T\to\infty} B(T)/T = 0$ (Section 17).

(E12) **[paper proof in this report]** *Singleton blocks.* $B(T) = 1$ for every $T \in \mathbb{N}_0$ (Section 17); an immediate restatement of (E1) blockwise.

(E13) **[paper proof in this report]** *Infinitely many singleton blocks.* $B(T) = 1$ for infinitely many $T \in \mathbb{N}_0$ (Section 17).

Remark 18.1. Item (E13) is the sharpest illustration of how exact the conditional structure theory is. It looks dramatically weaker than (E12), yet the exact block formula $B(T) = 1 + \lfloor (T+1)/\beta \rfloor$ under failure with least generator β forces $B(T) = 1$ to hold only for the finitely many $T < \beta - 1$. It cannot be weakened further to "$B(T) = 1$ for some T": that statement is already known unconditionally and carries no information, since the kernel-verified zero-free region gives $\beta \geq 12$. The gap between "for some" and "for infinitely many" is exactly the gap between a finite check and the conjecture.

Status: **[kernel-verified Lean]** for (E8) —

`alaogluErdosConjecture_iff_candidate_count_below_two_pow_subquadratic`.

[paper proof in this report] for (E9)–(E13) — paper proofs in this report, resting on the kernel-verified exact dyadic block counts of `ExactCounting` and the quadratic asymptotic of `QuadraticAsymptotic`. A Lean draft `BlockGrowthReformulations` exists but has not been accepted by the kernel. **[Lean draft, not yet accepted]**

18.5 Cluster D: orbits and fractional parts

Under failure the solution set is an irrational-rotation orbit in disguise. This cluster converts the conjecture into the assertion that no such orbit can be arithmetically synchronized, and into three gap statements on the unit interval.

(E14) **[paper proof in this report]** *Compact synchronized S-integral orbit.* There are no integers $M, A > 1$ and irrational $\beta > 0$ such that $(M^k/2^{\lfloor k\beta \rfloor}, A^k/3^{\lfloor k\beta \rfloor})$ lies on the compact arc $\{(2^t, 3^t) : 0 \leq t < 1\}$ for every $k \geq 1$ (Section 17). The two coordinates are S-integral for disjoint prime sets, yet their denominator exponents must agree exactly at every height.

(E15) **[kernel-verified Lean]** *Avoided interval.* Some nonempty open interval $(u, v) \subseteq (0, 1)$ contains no fractional part $\{y\}$ of a nonintegral solution y (Section 4).

(E16) **[kernel-verified Lean]** *Gap at the top.* The fractional parts of nonintegral solutions are bounded away from 1 (Section 4).

(E17) **[kernel-verified Lean]** *Gap at the bottom.* The fractional parts of nonintegral solutions are bounded away from 0 (Section 4).

Items (E15)–(E17) are unconditional biconditionals, and they say something quantitative about the elementary gap bounds of Section 4: those bounds place $\{x\}$ away from 0 and 1 by margins that decay like $2^{-\lfloor x \rfloor}$, and the equivalences show that *any* height-independent margin, however small and wherever located, is already the full conjecture. The decay is therefore essential rather than an artifact of the estimate.

Status: [paper proof in this report] for (E14) — paper proof in this report; the Lean draft `CompactOrbit` has not been accepted by the kernel ([Lean draft, not yet accepted]). [kernel-verified Lean] for (E15)–(E17) —

`alaogluErdosConjecture_iff_fract_avoids_interval`,

`alaogluErdosConjecture_iff_fract_bounded_away_one`,

`alaogluErdosConjecture_iff_fract_bounded_away_zero`

in `FractionalPartDensity`.

18.6 Cluster E: closure, smoothness, and algebraicity

The last cluster is the least classical and, for that reason, the most interesting as a source of new attacks. It says that the conjecture is equivalent to the failure of a single nonlinear closure property: the solution monoid, which is trivially closed under addition, must not also be closed under multiplication.

(E18) [paper proof in this report] *Multiplicative closure.* $\mathcal{S}$ is closed under multiplication.

(E19) [paper proof in this report] *Square closure.* $\mathcal{S}$ is closed under squaring.

(E20) [paper proof in this report] *Star closure.* $\mathcal{C}$ is closed under the operation $m \star n = 2^{\log_2 m \log_2 n}$.

(E21) [paper proof in this report] *Star idempotence.* Every $m \in \mathcal{C}$ satisfies $m \star m \in \mathcal{C}$.

(E22) [paper proof in this report] *Smooth outputs.* Every $x \in \mathcal{S}$ has both 2^x and 3^x 3-smooth.

(E23) [paper proof in this report] *Smooth product output.* Every $x \in \mathcal{S}$ has 6^x 3-smooth.

(E24) [classical/open background] *Square-exponent algebraicity.* Every $x \in \mathcal{S}$ has $6^{x^2} \in \overline{\mathbb{Q}}$.

(E25) [classical/open background] *Split square-exponent algebraicity.* Every $x \in \mathcal{S}$ has both $2^{x^2} \in \overline{\mathbb{Q}}$ and $3^{x^2} \in \overline{\mathbb{Q}}$.

(E26) [classical/open background] *Mixed-product algebraicity.* All $x, y \in \mathcal{S}$ satisfy $6^{xy} \in \overline{\mathbb{Q}}$.

Remark 18.2 (Manufacturing a third base). Items (E22)–(E26) admit a concrete reading. By the kernel-verified 3-smooth classification of Section 4, a hypothetical nonintegral solution β admits an integer base B with $B^\beta \in \mathbb{Z}$ only when B is 3-smooth. It would therefore suffice to manufacture *one* non-3-smooth positive integer B with $B^\beta \in \mathbb{Z}$. The obvious candidates $B = 2^\beta$, $B = 3^\beta$, $B = 6^\beta$ have β-th powers 2^{β^2}, 3^{β^2}, 6^{β^2}, so "manufacture a third base" is exactly the square-closure problem of (E19) in different language. Whenever the chosen base is not 3-smooth its β-th power is already known to be transcendental by six exponentials, so any proof of algebraicity would finish the conjecture at once. What is missing is a single bridge:

$$\boxed{\text{derive one nonlinear algebraic value from the two linear integer values.}}$$

Neither unique factorization, nor the ordinary exponent laws, nor linear forms in logarithms supplies it.

Status: [paper proof in this report] for (E18)–(E23) — paper proofs in this report, from the closure equivalences and the output-smoothness theorem; no Lean formalization yet. [classical/open background] for (E24)–(E26) — the equivalences combine those paper proofs with the classical six exponentials theorem, which supplies transcendence of a non-3-smooth base raised to the hypothetical exponent. None of the nine items has been accepted by the kernel.

18.7 Which formulations are worth attacking

The twenty-six items are not equally promising, and it is worth saying so plainly. Cluster C is exact but analytically distant: the deficit analysis of Section 16 shows the determinant hierarchy reaching $O((\log X)^2)$ while a counterexample supplies $\asymp \log X$ candidates, a full logarithm short of the $o(T)$ that (E10) would need. Cluster A is a change of coordinates, useful for orientation and for locating the four-exponentials barrier, but not itself a method. Cluster B is closest to established transcendence theory and is where a Kummer-theoretic, norm, or ramification argument would land; its smallest open instance is the irrationality of $3^{\log_2 3}$. Cluster D isolates a synchronization phenomenon that a bespoke S-integral counting theorem might exploit. Cluster E is the only one whose hypotheses are elementary and nonlinear at the same time, and it is the natural home of any attempt to manufacture a third base.

18.8 Equivalences versus sufficient conditions

An equivalence and a sufficient condition play very different roles, and the distinction is easy to lose in a corpus this size. Items (E1)–(E26) are biconditionals: each is exactly as strong as Conjecture 2.1, so proving any of them proves the conjecture and no weakening of any of them is safe without checking that the reverse implication survives. The statements below are strictly one-way. Each is a hypothesis that is itself open, each implies the conjecture, and none is known to follow from it — so none belongs in the catalogue.

(S1) *The real four exponentials conjecture.* If for all $\mathbb{Q}$-linearly independent x_1, x_2 and $\mathbb{Q}$-linearly independent y_1, y_2 some $e^{x_i y_j}$ is transcendental, then Conjecture 2.1 holds. The implication is kernel-verified (Section 7); the hypothesis is open, and is the reason no routine combination of present methods closes the problem.

(S2) *Schanuel's conjecture.* It implies algebraic independence of the logarithms of the primes, hence (S3), hence Conjecture 2.1 (Section 12).

(S3) *Prime-log-product independence.* If the real numbers $\log p \, \log q$, indexed by pairs of primes $p \leq q$, are linearly independent over $\mathbb{Q}$, then Conjecture 2.1 holds (Section 12). This criterion mentions no exponentials at all, is strictly weaker than Schanuel, and is not known to imply or to follow from (S1).

Status: [**kernel-verified Lean**] for (S1) —

`alaogluErdosConjecture_of_realFourExponentialsConjecture`

in `FourExponentialsReduction`, which formalizes the implication without assuming its hypothesis. [**paper proof in this report**] for (S2) and (S3) — paper proofs in this report; a Lean module `LogProductIndependence` recording the independence hypothesis as an explicit `Prop`, never asserted, is in preparation, with headline statement

`alaogluErdosConjecture_of_primeLogProductIndependence`.

The finite prime-support form of (S3) is weaker still and correspondingly more approachable: if the products $\log p \log q$ over a fixed finite prime set $S \ni 2, 3$ are $\mathbb{Q}$-linearly independent, then no counterexample has both outputs supported on S. For $S = \{2, 3\}$ this reduces to $\mathbb{Q}$-linear independence of 1, ϑ and ϑ^2, which would exclude every 3-smooth counterexample. That is a partial exclusion, not an equivalence.

A third category deserves the same care. Unconditional partial results settle a bounded region and are neither equivalences nor sufficient conditions: the kernel-verified zero-free region gives $2^x \geq 4096$, hence $x \geq 12$, for every nonintegral solution (Section 19), [**kernel-verified Lean**]; the independent scans through 2^{20} and 2^{26} extend the same conclusion at the weaker trust level of an interval computation, [**interval computation**], and are not kernel proofs. The kernel-verified exponent bound remains $x \geq 12$. No item in this section, and no combination of them, proves Conjecture 2.1; the catalogue records where a proof could enter, not that one has.

19 Independent finite verification

Exactly one zero-free region in this report is a kernel certificate. Lean proves that a solution with $2^x < 4096$ forces $x \in \mathbb{Z}$, so that a nonintegral solution satisfies $x \geq 12$; the statement lives in `FiniteCheck4096` as

`twelve_le_of_not_integer_of_two_three_rpow_integer`

and it is the only finite exclusion that any other statement here may use without adding trust assumptions. [**kernel-verified Lean**]

Two further finite computations push the excluded range far past 4096: an earlier interval-arithmetic scan through 2^{20}, and a newer directed-rounding exclusion through 2^{26}. Both are software results. Neither replays in the Lean kernel, neither is interchangeable with `FiniteCheck4096`, and neither raises the kernel-verified exponent bound above $x \geq 12$. They are recorded here because they strengthen the explicit conditional picture, test the conditional models of Section 5 against data, and supply the extremal cases that a future generated certificate must clear.

19.1 What each scan must certify

Write $\vartheta = \log 3/\log 2$, and let m be a positive integer that is not a power of two. Excluding m means proving $m^\vartheta \notin \mathbb{N}$, and it suffices to exhibit an integer n with $n < m^\vartheta < n+1$. Exponentiation is never performed: the pair of strict inequalities is equivalent, after taking logarithms and clearing the positive factor $\log 2$, to

$$\log m \,\log 3 - \log n \,\log 2 > 0, \tag{65}$$
$$\log(n+1) \,\log 2 - \log m \,\log 3 > 0. \tag{66}$$

Every logarithm and every product occurring in the scanned ranges is positive, so the required rounding directions are monotone and unambiguous.

In both scans an ordinary floating-point evaluation merely *proposes* $n \approx \lfloor m^{\vartheta} \rfloor$. Once certified lower bounds for (65) and (66) are both strictly positive, the floating proposal has no remaining logical role: it is a witness generator, not a step of the argument. A locator error can therefore create a spurious failure report, but it cannot create a spurious exclusion. Powers of two are skipped rather than tested, since 2^q has the integral value $(2^q)^{\vartheta} = 3^q$; an exactly integral value at a non-power of two would satisfy neither strict inequality for any n and would be reported as a failure.

19.2 Interval arithmetic through 2^{20}

Proposition 19.1 (Finite exclusion under interval-software trust)**.** *For every integer $1 \le m < 2^{20}$: if $m^{\vartheta} \in \mathbb{Z}$ then m is a power of two. Equivalently, if $2^x, 3^x \in \mathbb{Z}$ and $2^x < 2^{20}$ then $x \in \mathbb{Z}$; a nonintegral solution satisfies $x > 20$.*

Status: [interval computation] — CPython 3.13.5, `mpmath` 1.3.0 interval arithmetic at 40 decimal digits, outward endpoints. Reproducible but *not* replayed by the Lean kernel; not interchangeable with `twelve_le_of_not_integer_of_two_three_rpow_integer`.

Each candidate n is certified by evaluating (65) and (66) in `mpmath` interval arithmetic and testing the lower endpoint of each resulting interval for strict positivity. All 1,048,555 non-powers of two below 2^{20} passed, in two ranges:

Range	Checked	Failures	Smallest certified log margins
$1 \le m < 2^{19}$	524,268	0	lower 1.4147×10^{-15} at $(m, n) = (189427, 231498127)$; upper 3.0841×10^{-15} at $(306982, 497586055)$.
$2^{19} \le m < 2^{20}$	524,287	0	lower 9.4902×10^{-16} at $(958460, 3023921230)$; upper 2.3020×10^{-15} at $(556061, 1275861767)$.

At the globally tightest case the 40-digit interval had width about 1.3×10^{-54}, some thirty-nine orders of magnitude below the certified positive margin; a separate 100-digit computation gives $958460^{\vartheta} - 3023921230 = 4.14020498680340526 \times 10^{-6}$. The two runs report the SHA-256 manifests `02b589461fa8c1721bc251575b98b1e3d58001f54cb7faa2cd6c59edb8c819a5` (lower half) and `fccccf5de8673379d37ff0fc840ab091d9ff7a2352ff64a93091f01c21bf4cb4` (upper half) of the ASCII stream of proposed-and-certified pairs, which makes accidental changes to the candidate sequence detectable across reruns. The verification script and the full outputs are reproduced in Appendix F.

19.3 Directed rounding through 2^{26}

The second scan replaces interval arithmetic by explicit directed rounding in MPFR, which makes the numerical contract sharper — every elementary operation is correctly rounded in a named direction — without removing any of the software assumptions.

Directed enclosures. At precision $p = 192$ bits the verifier constructs

$$\begin{aligned} &L_m^- \le \log m \le L_m^+, \quad L_2^- \le \log 2 \le L_2^+, \qquad L_3^- \le \log 3 \le L_3^+, \\ &L_n^+ \ge \log n, \qquad L_{n+1}^- \qquad \le \log(n+1), \end{aligned}$$

and then evaluates, rounding downward throughout,

$$\delta_- = L_m^- L_3^- - L_n^+ L_2^+, \qquad \delta_+ = L_{n+1}^- L_2^- - L_m^+ L_3^+.$$

Thus $\delta_- > 0$ proves (65) and $\delta_+ > 0$ proves (66). The integers occurring on the n-side stay below 2^{42} and are exactly representable at 192 bits. The locator proposes $n \approx \lfloor m^{\vartheta} \rfloor$ using ordinary `long double` and tries the offsets $0, -1, 1, -2, 2, -3, 3$; a proposal is accepted only when both directed lower bounds are strictly positive, and a failure to locate any admissible n is reported as a failure rather than being silently classified as nonintegral.

Execution environment and source identity. The recorded run used MPFR 4.2.2 (linked as `libmpfr.so.6`), GCC 14.2.0 with `-O3 -fopenmp`, GNU OpenMP with five worker threads, Linux x86-64, 192-bit precision, and the directed modes `MPFR_RNDD` and `MPFR_RNDU`. The executed C source is archived with the run and has SHA-256 digest

`f888c0d89be9366f01e2d26f339dcbe446556d39d4c5dbcf8f579ec0d0052e1d`.

The execution image contained the MPFR shared library but not its development header, so the source carries a guarded minimal declaration of the MPFR 4.x ABI; on an ordinary system with `mpfr.h` the standard header branch is taken instead. This ABI detail is part of the software trust boundary and is disclosed rather than hidden.

Coverage and chunking. There are $2^{26} - 1 = 67{,}108{,}863$ positive integers below 2^{26}, of which exactly 26 are powers of two, namely 2^0 through 2^{25}. The verifier therefore tests

$$67{,}108{,}863 - 26 = 67{,}108{,}837$$

non-powers. The range was split at multiples of 2^{22} into one initial chunk $[1, 2^{22})$ and fifteen further chunks of width 2^{22}, so chunk k covers $[\max(1, k \cdot 2^{22}), (k+1) \cdot 2^{22})$ for $0 \le k \le 15$. All sixteen invocations returned `failures=0`. The canonical chunk log, containing the full standard output of every invocation, has SHA-256 digest

`dcaa7f3b7e5be44acaee9015a2045da14ee42d8fa2d96b9619cdd44c072a6fd7`.

Table 5: Directed-rounding audit through 2^{26}; every failure count was zero.

chunk	integer range	checked	min. lower gap	min. upper gap
0	1–4,194,303	4,194,281	1.1539×10^{-17}	3.0076×10^{-17}
1	4,194,304–8,388,607	4,194,303	4.6319×10^{-20}	1.3515×10^{-19}
2	8,388,608–12,582,911	4,194,303	1.2811×10^{-18}	1.3515×10^{-19}
3	12,582,912–16,777,215	4,194,304	4.6319×10^{-20}	2.9082×10^{-19}
4	16,777,216–20,971,519	4,194,303	1.1898×10^{-18}	1.3515×10^{-19}
5	20,971,520–25,165,823	4,194,304	5.4212×10^{-19}	1.5111×10^{-19}
6	25,165,824–29,360,127	4,194,304	4.6319×10^{-20}	1.7669×10^{-19}
7	29,360,128–33,554,431	4,194,304	1.4439×10^{-19}	5.4132×10^{-21}
8	33,554,432–37,748,735	4,194,303	4.2839×10^{-19}	1.3515×10^{-19}
9	37,748,736–41,943,039	4,194,304	3.9292×10^{-19}	1.0460×10^{-19}
10	41,943,040–46,137,343	4,194,304	1.9177×10^{-19}	1.5111×10^{-19}
11	46,137,344–50,331,647	4,194,304	1.0117×10^{-19}	3.4145×10^{-19}
12	50,331,648–54,525,951	4,194,304	5.6927×10^{-20}	1.2322×10^{-19}
13	54,525,952–58,720,255	4,194,304	3.6447×10^{-20}	1.1345×10^{-19}
14	58,720,256–62,914,559	4,194,304	6.7779×10^{-20}	5.4132×10^{-21}
15	62,914,560–67,108,863	4,194,304	5.0923×10^{-20}	3.9197×10^{-20}
	total	**67,108,837**		

Theorem 19.2 (Software-level finite exclusion). *For every integer m with $1 \le m < 2^{26}$, if $m^\vartheta \in \mathbb{N}$ then m is a power of two. Consequently, at the stated software trust level,*

$$x \in \mathcal{S} \setminus \mathbb{N}_0 \quad\Longrightarrow\quad x > 26.$$

Status: [interval computation] — directed-rounding computation in MPFR 4.2.2 at 192 bits. A software trust boundary, not a kernel certificate. The kernel-verified zero-free region is still $2^x < 4096$ in `FiniteCheck4096`, and the kernel-verified exponent bound is still $x \ge 12$.

Explanation of the consequence. The verifier establishes the first implication for every non-power input, and powers of two are the known integral candidates. If x were a nonintegral solution, then $m = 2^x$ would be an integer candidate that is not a power of two, so the finite exclusion forces $m \ge 2^{26}$. Equality would make m a power of two and force $x = 26$, hence $m > 2^{26}$ and $x > 26$. □

Closest approaches. The smallest certified lower logarithmic margin over the whole scan is $3.6447144482202515 \times 10^{-20}$, attained at

$$(m, n) = (58{,}570{,}438,\ 2{,}048{,}703{,}434{,}093),$$

where a separate 100-decimal evaluation gives $m^\vartheta - n = 1.07725158750919017548\ldots \times 10^{-7}$. The smallest certified upper logarithmic margin is $5.4132120119800561 \times 10^{-21}$, attained at

$$(m, n) = (29{,}411{,}704,\ 687{,}581{,}893{,}443),$$

where $n + 1 - m^\vartheta = 5.36974926710988874953\ldots \times 10^{-9}$. The same upper margin recurs at the scaled point $(58{,}823{,}408,\ 2{,}062{,}745{,}680{,}331)$, exactly as expected from doubling m and the identity $(2m)^\vartheta = 3m^\vartheta$. Both extremal inequalities were reevaluated independently at 100 decimal digits;

that secondary check is not an independent proof system, but it guards against transcription and aggregation errors. Every certified margin is many orders of magnitude above the nominal 192-bit rounding scale of the logarithmic operations.

Reproducibility data. The finite statement is deterministic: thread count changes execution time and the reduction order for the recorded minimum metadata, but not whether an individual directed inequality is certified. To reproduce the claim independently one obtains MPFR 4.2.x and GMP on a 64-bit platform, restores the archived C source and checks its SHA-256 digest, compiles with optimization and OpenMP (or removes the OpenMP pragmas for a serial run), runs the sixteen half-open intervals of Table 5 at 192 bits, verifies that every invocation reports `failures=0`, sums the per-chunk counts to 67,108,837, compares the two global extremal records above, and optionally reruns at 256 or 320 bits and against a second MPFR build. For a theorem-grade result one instead translates the enclosures into exact rational certificates and replays them in Lean.

19.4 The trust boundary, stated plainly

Kernel. $2^x < 4096$ and $2^x, 3^x \in \mathbb{Z}$ imply $x \in \mathbb{Z}$; a nonintegral solution has $x \geq 12$. **[kernel-verified Lean]** `FiniteCheck4096`.
Interval scan. Under CPython and `mpmath` interval arithmetic, a nonintegral solution has $x > 20$. **[interval computation]**
Directed-rounding scan. Under the C source, GCC code generation, OpenMP scheduling, the MPFR and GMP shared libraries, the x86-64 arithmetic environment, and the operating system, a nonintegral solution has $x > 26$. **[interval computation]**

MPFR's correct-rounding contract makes the numerical reasoning crisp, and the interval scan is reproducible from a short script, but neither removes the software assumptions listed above. Nothing in Section 5 or Section 21 that is tagged **[kernel-verified Lean]** depends on either scan; wherever a numeric threshold is used inside a kernel-verified statement, that threshold is 12.

A drafted extension of the kernel region. An attempt to raise the kernel-checked region from $2^x < 4096$ to $2^x < 16384$ exists in draft. It follows the `FiniteCheck4096` pattern, splitting the enlarged table into four chunks so that no single `decide` call carries the whole certificate. The certificate rows themselves verify: each chunk's Boolean recursion evaluates and each row's exact integer comparison is accepted. What has not been accepted is the assembly — the per-chunk length lemmas that glue the four tables into a single covering statement exceeded the elaborator's recursion limit, so the module does not compile end to end and no theorem is available for use. Until that is resolved the kernel bound is unchanged at $x \geq 12$. **[Lean draft, not yet accepted]**

From scan to kernel certificate. The repository's own certificate architecture (Section 4) shows the stronger trust model: generate rational certificates externally, revalidate them independently, and replay only exact integer comparisons in Lean by `decide`. The next step is certificate extraction, not a larger blind scan. For each block one records rational lower and upper approximants to ϑ together with integer inequalities sufficient to replay every exclusion, exploiting monotonicity and convexity so that a single certificate covers a run of consecutive m rather than storing one record per input. A scalable formal certificate should share convergent enclosures across ranges, chunk the data modules to bound elaboration memory — the failure mode of the 16384 draft is precisely that the chunk bookkeeping, not the arithmetic, is the expensive part — and rely on verified binary powering rather than giant literals. The extremal pairs identified by both scans predict the required tier depth. A practical intermediate target is 2^{16}, followed by a generated hierarchical certificate through 2^{26}.

20 Attempts at a full proof and their exact failure points

This section records the principal avenues that were pushed until they broke, together with the exact place where each one breaks. The purpose is not merely negative. Every failure point names a quantitative or structural feature that a successful proof must improve, and several of them turn out to be the same feature seen from different sides; the last subsection gives that common feature its exact name.

20.1 Collapsing rank two to rank one

The multiplicative-rank bound is kernel-verified: the exponent vectors of the candidate set span a subgroup of rank at most two. Under failure the rank-two possibility is not vague but known exactly, $\mathcal{C} = \langle 2, M\rangle$, with M the primitive generator of Section 5. Nothing in the rank machinery distinguishes this free rank-two monoid from the allowed monoid of powers of two.

Failure point. A rank bound is an upper bound on dimension; it supplies no mechanism for forcing the dimension down to one. Excluding the second generator *is* the original transcendence problem, restated.

20.2 Rational approximation to a least generator

Assume failure and let β be the least nonintegral solution, with $M = 2^\beta$ and $A = 3^\beta$. Let p/q approximate β and put $\varepsilon = q\beta - p$. Then

$$M^q - 2^p = 2^p\big(e^{\varepsilon \log 2} - 1\big), \tag{67}$$
$$A^q - 3^p = 3^p\big(e^{\varepsilon \log 3} - 1\big). \tag{68}$$

Both left-hand sides are integers, so a nonzero value has absolute value at least 1, which for small ε gives bounds of the shape

$$|\varepsilon| \gtrsim 2^{-p} \qquad \text{and} \qquad |\varepsilon| \gtrsim 3^{-p}.$$

These are exponentially tiny in q. Continued fractions only guarantee infinitely many approximants with $|\varepsilon| \ll q^{-1}$, and Baker-type lower bounds for ordinary linear forms in logarithms are polynomial in the relevant heights [3, 10]. Neither scale conflicts with a lower bound as weak as M^{-q} or A^{-q}. Multiplying (67) by (68), or eliminating ε between them, only recovers $\log 3/\log 2$ to first order; the higher terms are far too small to create an integer contradiction.

Failure point. The two integer gaps are correlated rather than independent, so the two exponential inequalities carry essentially one piece of information, not two.

Observation 20.1. Any rational-approximation proof needs more than "a nonzero integer has absolute value at least one." It needs a divisibility lower bound growing exponentially with q, or an approximation to β exponentially better than continued fractions supply in general.

20.3 Equidistribution against exponentially shrinking targets

At paper level the orbit $\{k\beta\}$ is equidistributed modulo 1 (Section 5); the finite-Fourier convergence needed for Weyl's criterion, covering every fixed mode and every finite trigonometric polynomial, is kernel-verified. Meanwhile, integrality of 2^{n+t} between adjacent integral powers pushes the fractional part t away from both endpoints by an amount comparable with 2^{-n}, and the base-3 condition imposes a comparable 3^{-n} restriction; at height k the admissible target width is of exponential scale M^{-k}, a bound also kernel-verified in the shrinking-target module.

Equidistribution controls visits to fixed intervals and, with effective discrepancy, to polynomially shrinking ones along subsequences. It does not force visits to a summable family of targets of length e^{-ck}: the Borel–Cantelli heuristic predicts only finitely many such visits for a typical rotation, and numbers with bounded partial quotients show that equidistributed multiples can avoid exponentially small targets forever. The exact conditional monoid is perfectly compatible with that behaviour.

Failure point. The blockwise Weyl theorems are structurally correct but point the wrong way: they describe how already existing solutions distribute inside each block, not how likely arbitrary exponents are to hit the integer lattice. Closing the route needs Diophantine information specific to the transcendental β with $2^\beta, 3^\beta \in \mathbb{N}$, not generic metric theory.

20.4 Finite differences and additive progressions

For $f(t) = t^\vartheta$ the repository proves the exact mean-value formula for every forward difference order r,

$$\Delta_h^r f(n) = (\vartheta)_{\underline{r}}\; h^r \xi^{\vartheta - r} \qquad (n < \xi < n + rh),$$

so that if all $r+1$ values $f(n), f(n+h), \ldots, f(n+rh)$ are integers, the left side is an integer; under $|\,(\vartheta)_{\underline{r}}\,| h^r < n^{r-\vartheta}$ its absolute value lies strictly between 0 and 1. This is a sharp uniform obstruction to candidate arithmetic progressions, and its rational effective three-term shadow $h^{400} \le n^{83}$ is kernel-verified together with the arbitrary-order form.

The difficulty is combinatorial rather than analytic. Conditional failure produces only $O(\log X)$ candidates per multiplicative block, far below the positive-density regime of Roth-type theorems, and a rank-two multiplicative monoid need not contain long additive progressions at all: many points in a short interval need not be equally spaced. A three-term progression of candidates would solve the S-unit equation

$$2^{n_1} M^{k_1} + 2^{n_3} M^{k_3} = 2^{n_2+1} M^{k_2},$$

and general S-unit theory yields finiteness statements, not the required nonexistence.

Failure point. Sparse exact structure and sharp local curvature are mutually consistent. The arbitrary-node determinant of Section 14 removes the equal-spacing hypothesis, but its packing exponent is still too weak to contradict the exact conditional count.

20.5 Radical fields and conjugates

The normalized radical $E = w^\vartheta$ is algebraic with exact binomial minimal polynomial $X^d - c/3^a$, and its conjugates are $\zeta_d^j E$; degree, norm, and denominator support are all kernel-verified (Section 6). What this does not supply is a way to transport the real analytic identity $\log E = \vartheta \log w$ to the other conjugates.

Failure point. The chosen positive real logarithm is not Galois-equivariant: the remaining roots differ by complex logarithmic periods, and taking norms merely reproduces $E^d = c/3^a$. Ramification is equally inconclusive, since the equation relates the constrained primes through a quotient of logarithms rather than through a field embedding; standard Kummer theory sees the radical equation but not the special value $\vartheta = \log 3/\log 2$.

Observation 20.2. A radical-field proof needs a genuinely mixed invariant: something simultaneously sensitive to the algebraic conjugates of E and to the logarithmic relation among $2, 3, w, E$. At present the algebraic and the real-analytic arguments meet only at the four-logarithm determinant.

20.6 The compact S-integral orbit

From the least generator define

$$U_k := \frac{M^k}{2^{\lfloor k\beta \rfloor}} = 2^{\{k\beta\}} \in \mathbb{Z}[1/2], \qquad V_k := \frac{A^k}{3^{\lfloor k\beta \rfloor}} = 3^{\{k\beta\}} \in \mathbb{Z}[1/3].$$

The points (U_k, V_k) lie on the compact analytic arc $\mathcal{A} = \{(2^t, 3^t) : 0 \le t < 1\}$ and are dense there because β is irrational. This is the compact-orbit reformulation of Section 17, and it invites o-minimal and S-integral point counting; Butler relates special cases of Wilkie-type counting to real three- and four-exponentials statements [5].

Failure point. The counting is not strong enough by an enormous margin. The logarithmic heights of the orbit points grow linearly in k, so the arc carries only about $\log H$ orbit points of height at most H, far below the subpolynomial count H^ε that Pila–Wilkie style theorems already tolerate outside algebraic parts. What would suffice is a denominator-sensitive refinement counting only points whose coordinate denominators are supported exactly on $\{2\}$ and on $\{3\}$ with synchronized exponents; no theorem of that strength appears in the inspected corpus.

20.7 Forcing an auxiliary base

If one could exhibit an integer $b > 1$ multiplicatively independent of 2 and 3 with b^x algebraic — or rational with suitably controlled denominator — then the six exponentials theorem, or the repository's rational-third-base theorem, would force $x \in \mathbb{Q}$ and hence $x \in \mathbb{Z}$. The five exponentials theorem [10, 8] similarly makes $e^{\gamma x}$ transcendental for every nonzero algebraic γ, given the four algebraic corner exponentials, and the kernel-verified 3-smooth classification is the formalized integer-level shadow of the six exponentials theorem.

Failure point. The hypotheses provide no such b. The natural auxiliary quantities $p^x = e^{x \log p}$ have transcendental coefficient $\log p$, so the five exponentials theorem does not convert a divisor of M or A into a contradiction; and the six exponentials theorem controls b^x, not the distinct question of whether b divides the integer 2^x. The obvious constructions $b = 2^u 3^v$, and any construction from M and A, collapse to the same rank-two logarithmic span. An auxiliary base must come from an arithmetic operation whose exponential behaviour stays controlled while its logarithm leaves that span — an algebraic independence problem in disguise.

20.8 Exploiting the primitive output pair

The primitive pair (M, A) is canonical and rigid: kernel-verified theorems show that it admits no matched depth, no common perfect power, and no nontrivial affine decomposition, and that the odd core and its coordinates are unique. Every structural handle one might hope to grip has therefore been shown to be absent by design rather than by accident.

Failure point. After all of that normalization, what remains is exactly the single relation $\log M\ \log 3 = \log A\ \log 2$, a product of logarithms lying outside the scope of linear-forms machinery.

20.9 The linear-form-in-log-products formulation of the wall

The obstruction can be given one further exact name. A counterexample is a pair of integers $m, A \geq 2$ with $\log m\ \log 3 = \log A\ \log 2$. Expanding both integers over their prime factorizations $m = \prod_p p^{m_p}$ and $A = \prod_p p^{a_p}$, failure of Conjecture 2.1 is equivalent to the nontrivial vanishing of

$$D \;=\; \sum_p \big(m_p \log 3 - a_p \log 2\big) \log p, \tag{69}$$

an *integer-coefficient linear form in the products of two prime logarithms* $\{\log p \cdot \log 2,\ \log p \cdot \log 3\}$. Baker's theory bounds linear forms in logarithms with *algebraic* coefficients [3]; here every coefficient of $\log p$ is itself a transcendental logarithm, and no lower-bound technology exists for such forms. Indeed even the $\mathbb{Q}$-linear independence of the family $\{\log p \cdot \log q : p < q \text{ prime}\}$ is open — it follows from, and is essentially a fragment of, the four exponentials circle of conjectures. Section 12 turns exactly that independence statement into a sufficient condition for the conjecture.

Failure point. Every route examined above — determinants, product formulas in the radical field, auxiliary exponentials, equidistribution against shrinking targets, orbit counting — terminates at a special case of this one independence statement. A genuine proof therefore requires either a fundamentally new lower bound for linear forms in log-products, or a mechanism exploiting the positivity and integrality of the coefficients (m_p, a_p) in (69) that has no analogue in the general four-exponentials setting.

21 Consolidated research program

The two source programs overlapped substantially; the exact structure theory of Section 5 settles some directions outright and re-prioritizes the rest. The list below is ordered by how directly each item attacks the quantitative deficit of Section 16 — the one missing logarithm between what the determinant hierarchy delivers and what a contradiction requires.

21.1 Priority A: p-adic amplification of semigroup determinants

The semigroup determinant $D = \det\big(m_i^{a_j}(m_i^{\vartheta})^{b_j}\big)$ is a positive integer, and under the exact monoid every entry has explicit valuations obtained from $m_i = 2^{n_i} M^{k_i}$ and $m_i^{\vartheta} = 3^{n_i} A^{k_i}$:

$$\begin{aligned} v_2\Big(\text{entry}_{ij}\Big) &= a_j\,(n_i + v_2(M)\,k_i) + b_j v_2(A)\,k_i, \\ v_3\Big(\text{entry}_{ij}\Big) &= a_j v_3(M)\,k_i + b_j\,(n_i + v_3(A)\,k_i). \end{aligned}$$

What this would buy is decisive: archimedean reasoning offers only the universal lower bound $|D| \geq 1$, whereas a systematic p-adic factor supplies $|D| \geq p^L$ and could erase the missing logarithm outright.

Question 21.1. *Can one choose $O(\log X)$ candidate rows and mixed-monomial columns so that the resulting nonzero determinant satisfies $\log |D|_p^{-1} \gg (\log X)^2$ for $p = 2$ or $p = 3$, while the archimedean estimate remains $\log |D|_\infty = o\big((\log X)^2\big)$?*

The difficulty is that a naive row or column factor is lost because the exponent set contains $(0, 0)$; only more subtle min-plus determinant estimates, applied after suitable row and column finite-difference operations, have a chance. The exact affine primitivity and the gcd condition on the normalized degrees should be fed into the estimate rather than discarded.

21.2 Priority B: exploit the full (n, k) grid, not sorted magnitudes

The arbitrary-node argument forgets the canonical coordinates and retains only the sorted integers m_i. Restoring them exposes rectangular, often Kronecker-like, bivariate exponential matrices, which is exactly the balanced 2×2 setting where ordinary six exponentials estimates stop; integrality, positivity, and the one-depth-zero normalization may create a margin absent from the general four-exponentials problem. The right experiment is not another large generic determinant but an optimized minor selected from triangular coordinate regions $n + k\beta \leq T$ and $a + b\vartheta \leq L$, with exact accounting of archimedean size, zero estimates, and 2- and 3-adic valuations. The hard part is that the optimization is genuinely two-parameter and the useful shapes are not predicted by any current heuristic, so a search must precede a theorem.

21.3 Priority C: denominator-sensitive counting on the compact arc

The compact-orbit reformulation asks for infinitely many points on $y = x^\vartheta$ in $[1, 2) \times [1, 3)$ whose first denominator is a power of 2, whose second denominator is a power of 3, and whose denominator exponents are synchronized by the same integer $\lfloor k\beta \rfloor$. A theorem asserting that a compact transcendental arc with nonzero curvature carries $o(H)$ points of logarithmic height at most H with denominators supported on two prescribed disjoint prime sets and a common exponent parameter would close the case, since the failed-conjecture orbit has $\asymp H$ such points. What makes it hard is that standard rational-point counting is sensitive to denominator *size* but not to denominator *support* or synchronization; Butler's connection between Wilkie-type statements and real exponential conjectures [5] suggests this direction sits on the transcendence frontier rather than being a routine application.

21.4 Priority D: mixed archimedean–radical invariants

The normalized radical $E = w^\vartheta$ has exact degree d and binomial minimal polynomial, so the Kummer field $\mathbb{Q}(E, \zeta_d)$ is completely explicit. The questions worth pursuing are whether the logarithmic relation $\log E \log 2 - \log w \log 3 = 0$ forces an impossible regulator or height relation there; whether simultaneous normalization at both outputs constrains the two normalized degrees through field intersections or discriminants beyond the known gcd condition; whether all complex branches $\log(\zeta_d^j E) = \log E + 2\pi i j/d$ can populate a multi-logarithm determinant with algebraic entries of controlled height; and whether a p-adic logarithm branch is compatible with the real identity after raising to the normalized degree. A warning is essential, and it is the same one that closed Section 20.5: merely taking norms or conjugates of $X^d - c/3^a$ reproduces known relations, so the new invariant must mix conjugation with the special logarithmic direction.

21.5 Priority E: formalize the new determinant results

The arbitrary-node repulsion theorem of Section 14 and the semigroup generalized-Vandermonde hierarchy of Section 15 are currently **[paper proof in this report]**. Formalizing them converts the strongest new tools in this report into reusable kernel-checked objects and, just as importantly, forces the constants in the gap theorem and the explicit dyadic bound to be stated uniformly. Lean drafts of `ArbitraryNodeDeterminant`, `MixedExponentSemigroup`, `SemigroupDeterminant`, and `FallingFactorialBound` already exist but are **[Lean draft, not yet accepted]**; the module layout is described in Section 22. The hard part is analytic bookkeeping rather than mathematics: divided differences at arbitrary nodes, falling-factorial growth, and generalized-Vandermonde positivity all need uniform statements before the main theorems can be assembled cheaply.

21.6 Priority F: force a third algebraic exponential

The denominator-controlled theorem in `RationalThirdBase` is a precise gateway: for an auxiliary integer q containing a prime outside $\{2, 3\}$ it suffices, in several forms, to prove that q^x is rational with denominator supported by the known outputs, after which monomial injectivity and the three-base determinant force $x \in \mathbb{Q}$ and hence $x \in \mathbb{Z}$. What this would buy is a complete proof. The obstacle is the one identified in Section 20.7: straight monomials cannot introduce a new prime direction, so candidates must come from norms, resultants, or algebraic combinations arising in the primitive radical field, whose xth powers would then have to simplify.

21.7 Priority G: five and sharp six exponentials

Formalizing the five exponentials theorem, or Roy's sharp six exponentials theorem [8], would add genuinely new restrictions — for instance transcendence of $e^{\gamma x}$ for algebraic $\gamma \neq 0$ — and would extend the rational-third-base result from a controlled denominator to an unrestricted rational output. The effort is comparable to the completed Gelfond–Schneider port, which is encouraging; the risk is that it be undertaken without a specific downstream lemma in mind, since as Section 20.7 shows these theorems constrain b^x and not the divisibility facts the conjecture actually needs.

21.8 Priority H: larger kernel-replayed regions and effective measures

The zero-free region $2^x < 4096$ is kernel-verified; the 2^{14} extension is in progress and 2^{20} is the natural next target, matching the region already covered by interval computation (**[interval computation]**) in Section 19. Every descent constant improves by bookkeeping once the region grows, so the payoff is broad but shallow. The bottleneck is a compact kernel-replayable proof object: adaptive convergent enclosures, chunked certificate modules, and tiers shared along ranges. In the same vein, effective irrationality measures for ϑ from Páde or hypergeometric methods would replace per-value certificate tiers with a uniform polynomial criterion and likely extend verified regions by orders of magnitude. Their limitation is sharp: they say nothing directly about the generator $\beta = \log_2 M$ for unknown M, which would require an effective measure derived from the log-product relation (69) and therefore lies beyond ordinary linear forms in logarithms.

21.9 Priority I: computational reconnaissance with exact certificates

Finite scanning is evidence, not a path to infinity, but computation remains valuable in three focused roles: searching candidate determinant shapes for large systematic 2- or 3-adic valuations, feeding Priority A and Priority B; enumerating small primitive radical patterns (w, d, a, c) and recording local obstructions with exact rational inequalities rather than floating-point rejection; and rejecting candidates early using the primitive-pair conditions — no matched depth, no common perfect power, no matched affine decomposition — while recording near misses keyed by odd core, denominator, and rational-power index. The goal must be pattern discovery and theorem generation; the difficulty is the discipline not to spend the budget on extending a decimal barrier that proves nothing new.

21.10 Priority J: the colossally abundant interface

Formalizing the original application — that the rational-power form of the conjecture implies that quotients of consecutive colossally abundant numbers are prime [1] — would connect the corpus to its historical source and give the whole development a visible consequence. It is logically downstream of everything above, and the work is combinatorial and definitional rather than transcendental, so it should follow the core structure modules rather than compete with them.

22 Lean formalization blueprint

The new arguments of Sections 14, 15, 16, 17 and 12 were written so that each analytic step is a reusable lemma rather than a step inside one monolithic proof. The intended module layout sits under `ExponentialIdentities/TwoBaseIntegerExponent/` alongside the existing kernel-verified corpus of Section 4.

Since the first draft of this blueprint was written, Lean sources have been committed for most of the modules below. They are drafts in the precise sense that matters here: the files exist, contain no `sorry`, and are written against the current corpus, but they have *not* yet been accepted by the Lean kernel in a completed build, and all but `FallingFactorialBound` are not yet on the aggregate import surface `ExponentialIdentities.lean`. Every such module is therefore marked **[Lean draft, not yet accepted]** below, and the underlying mathematics keeps whatever status Section 14 or Section 15 assigned it. Nothing in this section may be read as a claim of kernel verification.

22.1 `ArbitraryNodeDeterminant`

Status: **[Lean draft, not yet accepted]** — `ArbitraryNodeDeterminant.lean`; the underlying mathematics is **[paper proof in this report]** (Section 14).

This module formalizes the ordinary-power determinant and the arbitrary-node repulsion theorem of Section 14. Its dependencies are:

1. a determinant/divided-difference identity for the matrix whose columns are $1, x, \ldots, x^{r-1}, f(x)$ evaluated at the nodes, so that the determinant factors as the Vandermonde product times the rth divided difference of f;
2. a generalized mean value theorem for divided differences at distinct ordered nodes, which converts that divided difference into a derivative value at an interior point;
3. the existing derivative tower `rpowDerivativeFamily` and the falling-power coefficient `fallingRpowCoeff`, both already available in `HigherDifference`;
4. the integer-entry determinant lower bound: a matrix all of whose entries are integers has integer determinant, so a nonzero determinant has absolute value at least one.

The draft splits exactly along those four lines. It defines `nodeMatrix` and its integer shadow `nodeIntMatrix`, proves that candidate nodes make every entry an integer, derives `one_le_abs_det_nodeMatrix` from that, and then isolates the algebra of the final step in a hypothesis-carrying form: the analytic factorization $\det = V \cdot dd$ and the divided-difference bound $|dd| \leq B$ are inputs, and the conclusion is the repulsion inequality. A target statement for the fully analytic version is:

```
theorem vandermonde_lower_bound_of_twoBaseNaturalCandidates
    {r : N} (hr : 2 <= r)
    (m : Fin (r + 1) -> N)
    (hstrict : StrictMono m)
    (hcand : forall i, TwoBaseNaturalCandidate (m i)) :
    (r.factorial : R) / |fallingRpowCoeff logThreeDivLogTwo r| *
        (m 0 : R) ^ ((r : R) - logThreeDivLogTwo)
      <= vandermondeProduct r m
```

Listing 2: Proposed arbitrary-node theorem signature.

The intermediate form actually proved in the draft is the hypothesis-carrying one, which is worth stating separately because it is reusable for any f whose divided differences can be bounded:

```
theorem inv_le_vandermondeProduct_of_dividedDifference (r : N)
    (m : Fin (r + 1) -> N) (hstrict : StrictMono m)
    (hcand : forall i, TwoBaseNaturalCandidate (m i)) {dd B : R}
    (hB : 0 < B)
    (hdet : (nodeMatrix r m).det = vandermondeProduct r m * dd)
    (hdd_bound : |dd| <= B) (hne : (nodeMatrix r m).det <> 0) :
    1 / B <= vandermondeProduct r m
```

Listing 3: Packaged repulsion step (draft form).

Working with a `Fin`-indexed strictly monotone family and an explicit pair product is easier than working with a `Finset` of nodes; the `Finset` form is recovered at the end through the monotone enumeration `Finset.orderEmbOfFin`.

22.2 `GeneralizedVandermonde`

Status: Design note; the analytic content is **[paper proof in this report]** (Appendix D), and the nonvanishing half is drafted in `ExponentialPolynomialZeros`.

The positivity theorem for generalized Vandermonde determinants with arbitrary real exponents is useful well beyond this conjecture, and two formalization routes were available.

Wronskian route. Prove the Wronskian determinant formula of Appendix D directly, then invoke or develop the criterion that positive initial Wronskians produce an extended complete Chebyshev system and hence positive ordered evaluation determinants. This needs a small Chebyshev-system API that mathlib does not yet have.

Zero-count route. After the substitution $x = e^t$, prove by induction that a nonzero exponential polynomial with $r + 1$ distinct real exponents has at most r real zeros. Deduce nonvanishing of the evaluation determinant, and if the sign is wanted, obtain it by continuity and a coalescing-node limit.

The decisive observation for the present application is that only *nonvanishing* is needed downstream. The determinant in question has integer entries, and a nonzero integer already has absolute value at least one; the sign never enters the repulsion inequality. That removes the entire coalescing-limit argument from the critical path, and it is why the draft took the zero-count route and left strict positivity unproved.

22.3 `ExponentialPolynomialZeros`

Status: **[Lean draft, not yet accepted]** — `ExponentialPolynomialZeros.lean`.

This module carries the zero-count induction. Write $g(t) = \sum_{j \le n} c_j e^{\lambda_j t}$ with $\lambda_0 < \cdots < \lambda_n$. Normalizing by $e^{-\lambda_0 t}$ preserves the zero set; differentiating kills the constant term and lowers the number of exponents by one; Rolle's theorem between consecutive zeros supplies the zeros of the derivative. Induction on n then forces every coefficient to vanish.

Two Lean engineering difficulties dominate the effort, and neither is visible in the paper proof.

First, the Rolle step must produce n *distinct and ordered* interior points from $n + 1$ ordered zeros. Mathlib's `exists_hasDerivAt_eq_zero` yields one interior point per gap, but nothing about how the points from different gaps compare. The draft chooses them with `choose`, then proves strict monotonicity by chaining $y_i < x_{i+1} \le x_j < y_j$ through the monotonicity of the node family. When the hypothesis arrives as a `Finset` of at least $n+1$ reals rather than an indexed family, the monotone enumeration `Finset.orderEmbOfFin` converts it, after cutting the set down to exactly $n+1$ elements with `Finset.exists_subset_card_eq`.

Second, the induction reindexes. The derivative of an $(m + 1)$-term exponential polynomial is an m-term one whose exponents are $\lambda_{j+1} - \lambda_0$ and whose coefficients are $c_{j+1}(\lambda_{j+1} - \lambda_0)$, so every statement has to be transported along `Fin.succ`, and the induction has to be set up with the order quantified first so that `induction` may generalize over exponents, coefficients and nodes simultaneously. In the draft this is done through a private auxiliary theorem of the shape `forall (n : N) (lam c : Fin (n + 1) -> R), ...`, from which the user-facing statements are one-line consequences.

```
theorem card_le_of_expPoly_eq_zero {n : N} (lam c : Fin (n + 1) -> R)
    (hlam : StrictMono lam) (hc : exists j, c j <> 0) (S : Finset R)
    (hzero : forall t, t in S -> expPoly lam c t = 0) :
    S.card <= n

theorem det_rpowVandermonde_ne_zero {n : N} (x lam : Fin (n + 1) -> R)
    (hx : StrictMono x) (hx0 : 0 < x 0) (hlam : StrictMono lam) :
    (rpowVandermonde x lam).det <> 0
```

Listing 4: Zero-count theorem and its determinant form.

A vanishing determinant produces a nonzero kernel vector, via the lemma `Matrix.exists_mulVec_eq_zero_iff`, and it is a real combination of the functions $x \mapsto x^{\lambda_j}$ vanishing at $n+1$ distinct positive nodes. So the determinant statement follows from the zero-count statement.

22.4 `MixedExponentSemigroup` and `SemigroupDeterminant`

Status: **[Lean draft, not yet accepted]** — `MixedExponentSemigroup.lean` and `SemigroupDeterminant.lean`; the mathematics is **[paper proof in this report]** (Section 15).

The pair-indexed exponent is the whole of the definitional layer:

```
noncomputable def mixedExponent (ab : N x N) : R :=
  (ab.1 : R) + (ab.2 : R) * logThreeDivLogTwo
```

and the module proves, in this order:

- injectivity of `mixedExponent`, from irrationality of ϑ;
- integrality of $m^{a+b\vartheta} = m^a\,(m^\vartheta)^b$ for every candidate m, packaged as a natural-number witness `candidateRpowNat`;
- closure under addition, so that the image is an `AddSubmonoid R`;
- the elementary comparisons $a+b \le a+b\vartheta \le a+2b$ and $a+b\vartheta \le 3q$ for $a, b \le q$, which are what the counting arguments actually consume.

`SemigroupDeterminant` then builds the matrix $m_i^{\lambda_j}$ for candidate nodes and mixed exponents, shows every entry is a natural number, and concludes integrality and the lower bound $|\det| \ge 1$ whenever the determinant is nonzero — the nonvanishing being imported from `ExponentialPolynomialZeros`.

```
theorem one_le_abs_det_semigroupMatrix (r : N) (m : Fin (r + 1) -> N)
    (hcand : forall i, TwoBaseNaturalCandidate (m i))
    (ab : Fin (r + 1) -> N x N)
    (hne : (semigroupMatrix r m ab).det <> 0) :
    1 <= |(semigroupMatrix r m ab).det|
```

Listing 5: Semigroup determinant integrality.

It is deliberately unnecessary to define the globally sorted semigroup here. The determinant estimate accepts an arbitrary finite family of pairs together with a proof that their mixed exponents are strictly increasing; the sorted enumeration is a convenience of the paper proof, not a requirement of the formal one.

22.5 `FallingFactorialBound`

Status: [Lean draft, not yet accepted] — `FallingFactorialBound.lean`; already on the aggregate import surface.

This module carries the two elementary estimates behind every Newton-row bound of Section 15: the falling-coefficient bound $|(\lambda)_i|/i! \le 2^{\lambda+i}$ for $\lambda \ge 0$, and the strip bound $x^{\lambda-i} \le 2^{\lambda} N^{\lambda-i}$ for $N \le x \le 2N$, $N \ge 1$. The proof of the first compares each factor $|\lambda - k|$ with $m+1+k$ where $m = \lfloor \lambda \rfloor$, identifies $\prod_{k<i}(m+1+k)$ with $i!\binom{m+i}{i}$, and bounds the binomial coefficient by the row sum 2^{m+i}. Their product is the single inequality consumed downstream:

```
theorem abs_fallingRpowCoeff_div_factorial_mul_rpow_le
    {N x lam : R} (hN : 1 <= N) (h1 : N <= x) (h2 : x <= 2 * N)
    (hlam : 0 <= lam) (i : N) :
    |fallingRpowCoeff lam i| / (i.factorial : R) * x ^ (lam - (i : R))
      <= (2 : R) ^ (2 * lam + (i : R)) * N ^ (lam - (i : R))
```

Listing 6: Newton-row entry bound.

The determinant estimate then follows from `Matrix.det_apply` or, better, from a generic Leibniz-bound lemma worth adding once:

> if every entry in row i, column j is bounded by a separable quantity $A_i B_j$, then the determinant is at most $(r+1)! \prod_i A_i \prod_j B_j$.

22.6 `SemigroupWeightBound`

Status: [Lean draft, not yet accepted] — `SemigroupWeightBound.lean`.

For the explicit logarithmic-square corollary the full asymptotic count of the ordered semigroup is optional. What is needed is the square-box construction of Section 15 in *existence form*, which sidesteps real-valued order statistics entirely:

- the $(q+1)^2$ pairs in $[0,q]^2$ have pairwise distinct mixed exponents;
- every such exponent is at most $3q$, using $1 < \vartheta < 2$;
- taking $q = \lfloor \sqrt{r} \rfloor$ makes $r+1$ pairs fit, with total weight at most $6(r+1)\sqrt{r}$;
- dividing by $R_r = r(r+1)/2$ gives the deficit ratio $\Sigma_r / R_r \le 12/\sqrt{r}$.

The draft realizes the injective family explicitly by base-$(q+1)$ digits, $j \mapsto (j \bmod (q+1),\ j \div (q+1))$, so no sorting is performed and no canonical ordered semigroup is ever constructed. The statements are:

```
theorem exists_mixedExponent_family (r : N) :
    exists ab : Fin (r + 1) -> N x N, Function.Injective ab and
      (forall j, mixedExponent (ab j) <= 6 * Real.sqrt (r : R)) and
      (sum j, mixedExponent (ab j)) <= 6 * ((r : R) + 1) * Real.sqrt (r : R)
```

Listing 7: Square-box weight bound, existence form.

The module also contains the abstract counting bridge that turns any span lower bound into a cardinality bound, and which serves the ordinary-power corollary of Section 14 equally well: if $S \subseteq [N, N+H]$ and every $r+1$ elements of S contain two points at distance at least $L > 0$, then $\#S \le r + rH/L$. The proof partitions $[N, N+H]$ into $\lfloor H/L \rfloor + 1$ half-open blocks of length L, observes that each block meets S in at most r points, and applies pigeonhole.

22.7 `SemigroupDyadicBound`

Status: Planned; no draft yet. The mathematics is **[paper proof in this report]** (Section 15).

This is the module that combines the span theorem with the consecutive-window summation and produces the explicit $O((\log X)^2)$ dyadic bound. The final theorem could state:

```
theorem card_twoBaseNaturalCandidates_Icc_le_log_sq (X : N) (hX : 1 <= X) :
  ((Finset.Icc X (2 * X)).filter TwoBaseNaturalCandidate).card
    <= Nat.ceil (10000 * max 1 (Real.log X) ^ 2)
```

Listing 8: Proposed explicit dyadic bound.

A cleaner all-natural statement may replace the real logarithm by the binary ceiling logarithm, obtaining a larger but much easier constant times `Nat.clog 2 X ^ 2`. Since the repository already has logarithmic-square counting infrastructure in `MultiplicativeRank`, its finite-set definitions and cardinality lemmas can be reused while keeping the proof mechanism entirely independent — which is the point of proving the same bound twice.

22.8 `BlockGrowthReformulations`

Status: **[Lean draft, not yet accepted]** — `BlockGrowthReformulations.lean`; the mathematics is **[paper proof in this report]** (Section 17).

The exact conditional block formula proved in the corpus makes the dyadic block count $B(T) = \#\{\text{candidates in } [2^T, 2^{T+1})\}$ either identically 1 or bounded below by a positive multiple of T. That dichotomy collapses a whole family of apparently weaker growth hypotheses onto the conjecture itself, and the module states all four equivalences together so that a future argument may pick whichever is most convenient:

```
theorem alaogluErdosConjecture_block_growth_reformulations :
    (AlaogluErdosConjecture <-> exists C : N, forall T : N, dyadicBlockCount T <= C) and
    (AlaogluErdosConjecture <->
       (fun T : N => (dyadicBlockCount T : R)) =o[atTop] fun T : N => (T : R)) and
    (AlaogluErdosConjecture <->
       forall eps : R, 0 < eps ->
         frequently T in atTop, (dyadicBlockCount T : R) < eps * (T : R)) and
    (AlaogluErdosConjecture <-> forall T : N, dyadicBlockCount T = 1)
```

Listing 9: Block-growth reformulations.

The practical value is negative and worth stating plainly: proving mere boundedness, or even vanishing lower density along a subsequence, is exactly as hard as the conjecture. No softening of the counting target buys anything.

22.9 CompactOrbit

Status: [Lean draft, not yet accepted] — `CompactOrbit.lean`; the mathematics is **[paper proof in this report]** (Section 17).

This module states the conjecture as the nonexistence of a synchronized integral orbit on the compact arc $\{(2^t, 3^t) : 0 \le t < 1\}$. The orbit point is

$$(M^k/2^{\lfloor k\beta \rfloor},\ A^k/3^{\lfloor k\beta \rfloor}),$$

both coordinates divided by a power of their own base with the *same* integer exponent $\lfloor k\beta \rfloor$; that shared exponent is what makes the reformulation nontrivial.

```
theorem alaogluErdosConjecture_iff_no_compactOrbit :
    AlaogluErdosConjecture <->
      forall M A : N, forall beta : R,
        1 < M -> 1 < A -> Irrational beta -> 0 < beta ->
          exists k : N, 1 <= k and
            synchronizedOrbitPoint M A beta k not_in compactExponentialArc
```

Listing 10: Compact-orbit reformulation.

The forward direction routes through the canonical primitive generator, so the produced β is the least noninteger solution and M, A are its normalized outputs. The converse needs neither $M, A > 1$ nor $\beta > 0$: arc membership by itself already pins $M = 2^\beta$ and $A = 3^\beta$. Formalizing that asymmetry was the only delicate part.

22.10 LogProductIndependence

Status: [Lean draft, not yet accepted] — `LogProductIndependence.lean`; the mathematics is **[paper proof in this report]** (Section 12).

This module records the prime-log-product criterion of Section 12. The open hypothesis is never asserted; it is a named `Prop` that appears only as an explicit argument, so the exported theorem is unconditional Lean about a conditional mathematical statement.

```
def PrimeLogProductIndependence : Prop :=
  forall (S : Finset N) (c : N -> N -> Z),
    (forall p, p in S -> Nat.Prime p) ->
    (sum p in S, sum q in S, (c p q : R) * (Real.log p * Real.log q)) = 0 ->
    forall p, p in S -> forall q, q in S -> c p q + c q p = 0

theorem alaogluErdosConjecture_of_primeLogProductIndependence
    (hind : PrimeLogProductIndependence) : AlaogluErdosConjecture
```

Listing 11: Prime-log-product criterion.

The supporting lemma is the prime-factorization expansion $\log n = \sum_{p|n} v_p(n) \log p$, after which the relation $M^{\log 3} = A^{\log 2}$ becomes a vanishing integer combination of the products $\log p \log q$; symmetrized coefficientwise vanishing then forces M to be a power of 2 and the exponent to be an integer.

22.11 Proof-audit discipline

The discipline that governs every status label in this report is mechanical, and it is worth restating because several modules above are new. Each headline theorem must enter the aggregate import surface `ExponentialIdentities.lean` and be checked with `#print axioms`, so that the trust boundary of each result is visible rather than inferred. A result stays **[Lean draft, not yet accepted]** or **[paper proof in this report]** until a full build succeeds; the existence of a `sorry`-free source file is not verification, and neither is a successful build of an earlier revision. Numerical constants appearing in the tables of this report are explanatory only — the formal statements use exact expressions or safe rational constants such as 1/24 and 10000, chosen so that the arithmetic side conditions close by `norm_num` rather than by a tuned estimate.

23 Comparison of methods

23.1 Prime-valuation rank versus interpolation geometry

The module `MultiplicativeRank` proves that any three candidates have linearly dependent prime-exponent vectors, of rank at most two. It does so by a three-base, six-exponentials determinant: if three candidate bases were multiplicatively independent, integrality of all three ϑth powers would force ϑ rational, contradicting its irrationality. A finite candidate set therefore lies in a rational plane inside the free vector space on primes. Choosing two prime coordinates injectively encodes the set and yields an explicit logarithmic-square count.

The semigroup determinant of Section 15 proves a logarithmic-square dyadic bound by an entirely different mechanism.

Prime-valuation method	Semigroup determinant method
Uses prime factorization of candidate bases	Uses only integer values on $y = x^{\vartheta}$
Global multiplicative rank at most two	Local total positivity and interpolation
Invokes a six-exponentials consequence	Uses elementary determinant integrality once ϑ is fixed
Produces coordinates in two prime directions	Produces repulsion for arbitrary ordered nodes
Naturally global in magnitude	Naturally local in additive intervals
Kernel-verified **[kernel-verified Lean]**	Paper proof **[paper proof in this report]**, Lean draft **[Lean draft, not yet accepted]**

Their agreement at logarithmic-square scale is suggestive rather than coincidental. Both methods see a two-dimensional structure, but they see different two-dimensional structures: prime-exponent rank two on one side, mixed-exponent lattice dimension two on the other. Neither refinement alone has closed the gap, and a future proof may well need to combine the two dual lattices rather than push either one further in isolation.

23.2 Exact conditional rank one over two generators

Conditional on failure of the conjecture, the candidate monoid is not merely of rank at most two — it is exactly free on 2 and M. In prime-valuation space its points form the nonnegative lattice generated by the prime-exponent vectors of 2 and M. In mixed-exponent evaluation space, the values factor across a two-dimensional grid indexed by the pair (n, k) with value $2^n M^k$. This exact grid is far stronger than the unconditional rank theorem, and yet every presently known consequence remains compatible with it. That compatibility is the honest measure of how much room is left.

The strongest likely synthesis is a determinant whose rows use primitive-generator coordinates and whose columns use mixed exponents. Such a matrix would be simultaneously:

- integral, hence subject to the $|\det| \geq 1$ lower bound;
- totally positive after ordering by magnitude, hence nonvanishing for free;
- explicitly factored into powers of $2, 3, M, A$;
- constrained by affine primitivity and by simultaneous output normalization.

No current module combines all four properties. `MultiplicativeRank` has the first and third; the semigroup modules have the first and second; `PrimitiveGenerator` and `OutputNormalization` have the fourth. Building the module that has all four is the most concrete open formalization task suggested by this report.

23.3 Finite differences as confluent determinants

Forward differences are the equally spaced specialization of interpolation, and Wronskians are its confluent limit as nodes coalesce. The determinant hierarchy of Sections 14 and 15 therefore provides a single conceptual envelope for four things that the corpus currently treats separately:

- second-, third-, and arbitrary-order finite differences;
- arbitrary-node ordinary-power determinants;
- mixed-monomial generalized Vandermonde determinants;
- confluent determinants using derivatives or repeated coordinate directions.

The corresponding formalization strategy is to build a general interpolation-determinant layer once and recover `HigherDifference` as a specialization of it, rather than to keep the two developments parallel. That would remove duplicated real analysis and, more usefully, make experimental determinant choices cheap to certify: a new choice of nodes and exponents would need only a divided-difference bound, with integrality and nonvanishing supplied by the shared layer.

24 Conclusions

The Alaoglu–Erdős conjecture (Conjecture 2.1) remains open. Nothing in this report proves it, and no combination of the results collected here proves it. What has changed is that the reason can now be stated exactly rather than impressionistically.

The kernel-verified core is broad and its themes are complementary. Interpolation determinants settle three independent bases; Gelfond–Schneider excludes algebraic irrational exponents; exact certificates create a growing zero-free region, currently $2^x < 4096$; unique factorization yields a canonical primitive odd core and, conditionally, an exact solution monoid; arbitrary-order forward differences force sharp local additive sparsity, with the rational three-term criterion $h^{400} \leq n^{83}$ as its effective form; leastness gives exact affine primitivity; and unconditional equivalences with fractional-part gaps and localized radicals identify the entire remaining distance to the conjecture.

Conditional on a single counterexample, the corpus determines the whole picture. A counterexample would not produce a messy exceptional set that density or semigroup arguments might squeeze; it would produce the perfectly rigid free monoid

$$\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta, \qquad \mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\},$$

whose block counts, cumulative asymptotics, fractional parts, rotation statistics, rational-power index, radical degree, and simultaneous output normalization are all explicit and internally consistent with every verified gap and density bound. The surviving obstruction is correspondingly concentrated:

$$\text{exclude } w^{\log_2 3} \in \mathbb{Z}[1/3] \text{ for every primitive odd } w > 1.$$

That single exclusion is equivalent to, and exactly as open as, the original conjecture.

The new determinant hierarchy sharpens the picture quantitatively. The arbitrary-node theorem of Section 14 shows that candidate lattice points cannot cluster freely: every Vandermonde product over candidate nodes obeys a power-law lower bound. The mixed-exponent semigroup of Section 15 upgrades this to arbitrarily high-order positive integer determinants with a repulsion exponent tending to one, and yields a second, conceptually independent logarithmic-square counting theorem.

The most valuable outcome, however, is the quantified failure point of Section 16. Pure archimedean interpolation needs about $(\log X)^2$ nodes to operate at scale X, whereas the exact conditional monoid supplies only about $\log X$ of them. The method misses by exactly one factor of $\log X$ — not by a constant, and not by an epsilon. This makes future work sharper by exclusion: improving constants, tightening the falling-factorial bound, or extending finite searches cannot close a gap of that shape. The plausible routes are precisely those capable of supplying one additional logarithm of determinant strength: 2- or 3-adic divisibility used simultaneously with the archimedean estimate, a synchronized-denominator counting theorem on the compact exponential arc of Section 17, or a mechanism forcing a third algebraically independent exponential direction.

Section 12 isolates a different sufficient condition, and it is worth separating from the rest because it does not live at the four-exponentials boundary at all. If the products $\log p \log q$ of prime logarithms are linearly independent over $\mathbb{Q}$, the conjecture follows by an argument that is elementary once the hypothesis is granted. That independence is a consequence of Schanuel's conjecture and is not known unconditionally — indeed even the irrationality of $(\log 3)^2/(\log 2)^2$ is open — but it is a different classical conjecture, with a different literature and different partial results. Having two inequivalent sufficient conditions, one transcendence-theoretic and one arithmetic, is more informative than having one.

The near-term program is the one consolidated in Section 21: finish the exact conditional consequences, in particular the functional-analytic extension of blockwise equidistribution to arbitrary continuous tests and interval indicators; exploit the uniform higher-difference theorem combinatorially, allowing the order to grow with height; kernel-replay the 2^{20} region as a compact certificate; use the explicit radical field for a genuinely number-field argument; and pursue auxiliary-function methods tailored to integrality, where all corner values are positive integers and the primitive pair is canonical. Alongside those, the immediate formalization work is the module list of Section 22: bring the nine existing drafts through a full kernel build and onto the aggregate import surface, then attempt the combined determinant identified in Section 23.

A complete proof still requires a new idea. But the target is now narrow, the conditional structure is exact, the failure mode of the best available elementary method is measured rather than guessed, and the surrounding formal infrastructure is strong enough to test the next idea precisely.

A Status matrix for major statements

The status tags have the following meanings. **[kernel-verified Lean]** marks a statement that the Lean kernel accepts at the current repository state, with no `sorry`, no extra axioms, and no `native_decide`. **[paper proof in this report]** marks a paper proof given in this report that has not yet been formalized. **[Lean draft, not yet accepted]** marks a statement for which Lean source exists in the repository but has not yet been accepted by the kernel. **[interval computation]** marks a finite interval computation with a stated software trust boundary, which is not a kernel certificate. **[classical/open background]** marks classical or open background. No entry below asserts the full Alaoglu–Erdős conjecture; it remains open.

Statement	Status	Location
Classical frame		
$2^x, 3^x, 5^x \in \mathbb{Z} \Rightarrow x \in \mathbb{Z}$	**[kernel-verified Lean]**	`IntegerExponent`
Gelfond–Schneider theorem	**[kernel-verified Lean]**	`GelfondSchneider`
Four exponentials conjecture (hypothesis only)	**[classical/open background]**	[13, 14]

Statement	Status	Location
Full Alaoglu–Erdős conjecture	Open	Section 6, Section 21
Transcendence and reduction		
Nonintegral solution irrational and transcendental	[kernel-verified Lean]	`TwoBaseIntegerExponent`, `Transcendence`
$\vartheta = \log 3/\log 2$ transcendental	[kernel-verified Lean]	`Transcendence`
Four-exponentials conjecture $\Rightarrow$ Conjecture 2.1	[kernel-verified Lean]	`FourExponentialsReduction`
Prime-log-product independence $\Rightarrow$ Conjecture 2.1	**[Lean draft, not yet accepted]**	Section 12; `LogProductIndependence`
Zero-free regions and finite searches		
$2^x < 100$ and $2^x < 256 \Rightarrow x \in \mathbb{Z}$	[kernel-verified Lean]	`FiniteCheck`, `FiniteCheck256`
$2^x < 4096 \Rightarrow x \in \mathbb{Z}$ (so $x \geq 12$)	[kernel-verified Lean]	`FiniteCheck4096`
No nonintegral candidate with $2^x < 2^{20}$	**[interval computation]**	Section 19, Appendix F
Exact conditional structure		
Rank of candidate exponent vectors ≤ 2	[kernel-verified Lean]	`MultiplicativeRank`
Cross-valuation identity and conditional box growth	[kernel-verified Lean]	`CandidateStructure`
Common odd core $2^i w^j$ (existence)	[kernel-verified Lean]	`OddCore`
Counting $\mathrm{clog}_2 N \cdot \mathrm{clog}_3 N$; linear dyadic blocks	[kernel-verified Lean]	`OddCore`
Primitive (canonical) odd core; unique coordinates	[kernel-verified Lean]	`PrimitiveOddCore`
Unique least solution; affine primitive output pair	[kernel-verified Lean]	`LeastSolution`
Simultaneous primitive output normalization; gcd one; size dichotomy	[kernel-verified Lean]	`OutputNormalization`
Power-of-three denominator normalization	[kernel-verified Lean]	`ThreeDenominatorNormalization`
Rational-power index and exact solution monoid $\mathbb{N}_0 + \mathbb{N}_0\beta$	[kernel-verified Lean]	`RationalPowerIndex`, `PrimitiveGenerator`
Canonical primitive generator; exact candidate monoid $\{2^n M^k\}$	[kernel-verified Lean]	`PrimitiveGenerator`
Denominator-controlled rational third output and determinant margin	[kernel-verified Lean]	`RationalThirdBase`
Unrestricted rational-third-output determinant grid	Open	not yet formalized
3-smooth classification of further bases	[kernel-verified Lean]	`ThreeSmooth`
Counting, growth, and equidistribution		
Exact block counts and cumulative finite sum	[kernel-verified Lean]	`ExactCounting`
Cumulative $T^2/(2\beta) + O(T)$ asymptotic; subquadratic-growth equivalence	[kernel-verified Lean]	`QuadraticAsymptotic`
Vanishing nonzero block Fourier modes	[kernel-verified Lean]	`BlockWeyl`
Exact block averages for finite trigonometric polynomials	[kernel-verified Lean]	`BlockEquidistribution`
Full continuous-test/interval blockwise equidistribution	**[paper proof in this report]**	Section 5
Density dichotomy, gap equivalences, infinitude	[kernel-verified Lean]	`FractionalPartDensity`

Statement	Status	Location
Nonintegral candidate refinement; density zero	[kernel-verified Lean]	`NonintegerDensity`
Shrinking-target Diophantine bounds	[kernel-verified Lean]	`ShrinkingTarget`
Convergent-denominator growth $q_{n+1} < 2m^{q_n}$	[kernel-verified Lean]	`DenominatorGrowth`
Difference and progression obstructions		
Affine descent, $r + 8k \le x$	[kernel-verified Lean]	`ArithmeticRigidity`
Three-term AP obstruction under $h^5 \le n$	[kernel-verified Lean]	`ZeroDensity`
Sharp three-term and four-term AP obstructions ($r = 2, 3$)	[kernel-verified Lean]	`SharperProgressions`, `ThirdDifference`
Uniform arbitrary-order AP obstruction ($r \ge 2$)	[kernel-verified Lean]	`HigherDifference`
Effective rational three-term criterion $h^{400} \le n^{83}$	[kernel-verified Lean]	`SharpProgression`
Radical and localized reformulations		
Localized-radical and symmetric reformulations	[kernel-verified Lean]	`RadicalReformulation`, Appendix C
Primitive localized-radical reduction	[kernel-verified Lean]	`PrimitiveRadical`
Exact radical degree and binomial irreducibility	[kernel-verified Lean]	`RadicalDegree`
Candidate-form localization equivalence	[kernel-verified Lean]	`Localization`
New results of this report		
Generalized Vandermonde positivity (ordered chamber)	[paper proof in this report]	Section 14, Appendix D; drafted as `ExponentialPolynomialZeros`
Arbitrary-node Vandermonde repulsion and corollaries	[paper proof in this report]	Section 14; drafted as `ArbitraryNodeDeterminant`
Semigroup generalized-Vandermonde hierarchy (integrality half)	[paper proof in this report]	Section 15; drafted as `MixedExponentSemigroup`, `SemigroupDeterminant`
Semigroup gap theorem	[paper proof in this report]	Section 15; drafted as `FallingFactorialBound`
Explicit dyadic $O((\log X)^2)$ bound from that hierarchy	[paper proof in this report]	Section 15; drafted as `SemigroupWeightBound`
One-logarithm deficit analysis	[paper proof in this report]	Section 16
Compact synchronized-orbit reformulation	[paper proof in this report]	Section 17; drafted as `CompactOrbit`
Block-growth reformulations (bounded, $o(T)$, $\liminf$, $\equiv 1$)	[paper proof in this report]	Section 17; drafted as `BlockGrowthReformulations`

B Formalization inventory

All modules live under `Analysis/ExponentialIdentities/Lean/ExponentialIdentities/`, in namespace `LeanProofs`. The kernel-accepted modules listed first contain no `sorry`, no extra axioms, and no `native_decide`; certificate tables are replayed by the kernel.

B.1 Kernel-accepted modules

Module	Contents
`IntegerExponent` (+3 modules)	three-base theorem, interpolation determinants
`TwoBaseIntegerExponent`	conjecture statement, candidates, gaps, $2^x < 10$
`.../FiniteCheck`, `FiniteCheck256`	zero-free regions $2^x < 100$, $2^x < 256$
`.../FiniteCheck4096`	zero-free region $2^x < 4096$ (kernel `decide` table)
`.../ZeroDensity`	AP obstruction, Roth-number counting, $o(N)$
`.../NonintegerDensity`	nonintegral candidate refinement, density zero
`.../MultiplicativeRank`	rank ≤ 2, ϑ irrational, $(c\log_2 N)^2$
`.../CandidateStructure`	cross-valuation identity, conditional box growth
`.../OddCore`	common odd core, $c\log_2 N$ $c\log_3 N$, dyadic blocks
`.../PrimitiveOddCore`	gcd-normalized canonical core, unique coordinates
`.../LeastSolution`	unique positive least solution, exact affine primitivity
`.../ThreeDenominatorNormalization`	reduced denominator supported at 3
`.../RationalPowerIndex`	least rational-power index, exact solution monoid
`.../PrimitiveGenerator`	canonical primitive generator and unique coordinates
`.../ExactCounting`	exact unit/dyadic block counts and cumulative finite sum
`.../QuadraticAsymptotic`	sharp quadratic limit; subquadratic-growth equivalence
`.../BlockWeyl`	exact-block geometric sums and vanishing nonzero Fourier modes
`.../BlockEquidistribution`	finite trigonometric-polynomial block averages
`.../OutputNormalization`	simultaneous primitive outputs, gcd one, size dichotomy
`.../RationalThirdBase`	denominator divisibility and determinant margin
`.../RadicalReformulation`	localized-radical equivalences, base-swapped form
`.../PrimitiveRadical`	canonical primitive-power reduction of radical target
`.../RadicalDegree`	irreducible binomial, exact minimal polynomial and degree
`.../SharperProgressions`	sharp three-candidate curvature obstruction
`.../ThirdDifference`	exact third difference, four-candidate obstruction
`.../HigherDifference`	uniform arbitrary-order forward-difference obstruction
`.../SharpProgression`	effective rational three-term criterion $h^{400} \leq n^{83}$
`.../ArithmeticRigidity`	descent, $r + 8k \leq x$
`.../ThreeSmooth`	3-smooth classification of further bases
`.../ShrinkingTarget`	fractional-part shrinking targets, Diophantine lower bounds
`.../DenominatorGrowth`	convergent-denominator growth cap $q_{n+1} < 2m^{q_n}$
`.../Transcendence`	Gelfond–Schneider applications: solutions and $\log_2 3$
`.../FractionalPartDensity`	density dichotomy, gap equivalences, infinitude
`.../Localization`	candidate-form localization equivalence
`.../FourExponentialsReduction`	four exponentials $\Rightarrow$ the conjecture
`GelfondSchneider/...`	Gelfond–Schneider theorem (port of Karatarakis)

B.2 Drafted modules not yet accepted by the kernel

The following sources exist in the repository tree for the new results of this report but have *not* been accepted by the Lean kernel at the pinned state. Every statement they contain carries status **[Lean draft, not yet accepted]** and must not be cited as machine-checked.

Drafted module	Intended contents
`.../ArbitraryNodeDeterminant`	interpolation determinant at arbitrary ordered nodes; algebraic half of the repulsion theorem
`.../ExponentialPolynomialZeros`	zero-count theorem for real exponential polynomials; analytic engine for generalized Vandermonde positivity
`.../MixedExponentSemigroup`	pair-indexed exponent $a + b\vartheta$, injectivity, integrality of $m^{a+b\vartheta}$
`.../SemigroupDeterminant`	integrality half of the semigroup generalized-Vandermonde hierarchy

Drafted module	Intended contents
`.../FallingFactorialBound`	falling-factorial coefficient bound and dyadic-strip divided-difference bound
`.../SemigroupWeightBound`	crude mixed-exponent weight bound and window-summation counting bridge
`.../BlockGrowthReformulations`	boundedness, $o(T)$, $\liminf$, and $\equiv 1$ reformulations of the conjecture
`.../CompactOrbit`	compact synchronized-orbit reformulation with shared floor exponent
`.../LogProductIndependence`	prime-log-product independence as a sufficient condition distinct from four exponentials

Each headline theorem in these modules should be added to the aggregate import surface and checked with `#print axioms` before its status is promoted. Numerical constants in the tables of Appendix E are explanatory only; the formal statements should use exact expressions or safe rational constants such as 1/24 and 10000.

C A symmetric formulation

Everything above has a base-swapped version. Put $\eta = \log_3 2 = 1/\vartheta$. Failure of the conjecture is also equivalent to the existence of an integer $v > 1$ with $3 \nmid v$ and

$$v^\eta \in \mathbb{Z}[1/2].$$

Status: [kernel-verified Lean] — `not_alaogluErdosConjecture_iff_exists_threeFree_localized_radical` and `alaogluErdosConjecture_iff_threeFreeLocalizedRadicalExclusion` in `RadicalReformulation`.

For the primitive pair (M, A), factoring A into a power of 3 times a 3-coprime part produces a symmetric primitive core and rational-power index; the two normalizations give the same least exponent β, and their compatibility condition is exactly $\log M \log 3 = \log A \log 2$. The symmetry is useful in formalization — some divisibility arguments are easier on one side — and confirms that the localized-radical reduction of Section 6 is not an artifact of privileging the base 2.

D Detailed proof of the generalized Vandermonde sign

This appendix expands the generalized Vandermonde positivity lemma of Section 14 to make its analytic input fully transparent.

Let $0 \le \lambda_0 < \cdots < \lambda_r$ and define $f_j(x) = x^{\lambda_j}$ on $(0, \infty)$. For $k \le r$,

$$f_j^{(k)}(x) = (\lambda_j)_{\underline{k}}\, x^{\lambda_j - k}.$$

The kth initial Wronskian is

$$\begin{aligned} W_k(x) &= \det\big(f_j^{(i)}(x)\big)_{0\le i,j\le k} \\ &= x^{\sum_{j=0}^{k} \lambda_j - k(k+1)/2} \det\big((\lambda_j)_{\underline{i}}\big)_{0\le i,j\le k}. \end{aligned}$$

The polynomial $(z)_{\underline{i}}$ is monic of degree i. Replacing the monomial basis $1, z, \ldots, z^k$ by the falling-factorial basis is unit lower-triangular. Therefore

$$\det\big((\lambda_j)_{\underline{i}}\big)_{i,j=0}^{k} = \prod_{0\le i<j\le k} (\lambda_j - \lambda_i) > 0.$$

Hence every initial Wronskian is positive.

One standard characterization of extended complete Chebyshev systems [6] now implies that for $x_0 < \cdots < x_r$,

$$\det(f_j(x_i))_{i,j=0}^{r} > 0.$$

To avoid using that characterization as a black box, proceed inductively. Suppose a nontrivial combination

$$F(x) = \sum_{j=0}^{r} c_j x^{\lambda_j}$$

has $r+1$ distinct positive zeros. Put $x = e^t$ and factor the lowest exponential:

$$e^{-\lambda_0 t} F(e^t) = c_0 + \sum_{j=1}^{r} c_j e^{(\lambda_j - \lambda_0)t}.$$

Rolle's theorem gives at least r zeros of its derivative, an exponential polynomial with at most r terms. Induction gives a contradiction unless all coefficients vanish. Thus the evaluation determinant is nonzero. Its sign cannot change on the connected chamber $x_0 < \cdots < x_r$. Let the nodes coalesce as $x_i = x + i\varepsilon$ and divide by the positive ordinary Vandermonde. As $\varepsilon \downarrow 0$, the quotient tends to $W_r(x)/\prod_{k=0}^{r} k! > 0$. Hence the original determinant is positive everywhere in the ordered chamber.

Status: **[paper proof in this report]** — paper proof in this report. The zero-count engine is drafted in `ExponentialPolynomialZeros` but has not been accepted by the kernel (**[Lean draft, not yet accepted]**).

E Numerical constants

For reference,

$$\vartheta = 1.58496250072115618145\ldots, \qquad \vartheta(\vartheta-1) = 0.92714362797110482756\ldots.$$

The three-point span constant is

$$\left(\frac{8}{\vartheta(\vartheta-1)}\right)^{1/3} = 2.051072395661874\ldots.$$

For the ordinary-power order r, the raw span constant and exponent are

$$C_r = \left(\frac{r!}{|\,(\vartheta)_{\underline{r}}\,|}\right)^{2/(r(r+1))}, \qquad \alpha_r = \frac{2(r-\vartheta)}{r(r+1)}.$$

The continuous critical point for α_r solves

$$-r^2 + 2\vartheta r + \vartheta = 0,$$

so

$$r_* = \vartheta + \sqrt{\vartheta^2 + \vartheta} = 3.6090841940\ldots,$$

confirming that $r = 4$ is the optimal integer order.

For the semigroup hierarchy of Section 15, the first exponents of $\Lambda_\vartheta = \{a + b\vartheta : a, b \in \mathbb{N}_0\}$ and their pair labels are:

Index	Pair (a, b)	Exponent $a + b\vartheta$
0	$(0, 0)$	0
1	$(1, 0)$	1
2	$(0, 1)$	ϑ
3	$(2, 0)$	2
4	$(1, 1)$	$1 + \vartheta$
5	$(3, 0)$	3
6	$(0, 2)$	2ϑ
7	$(2, 1)$	$2 + \vartheta$
8	$(4, 0)$	4
9	$(1, 2)$	$1 + 2\vartheta$
10	$(3, 1)$	$3 + \vartheta$
11	$(0, 3)$	3ϑ
12	$(5, 0)$	5

These decimal values are explanatory. They are not part of any machine-checked statement, and formal versions of the corresponding theorems should carry exact expressions instead.

F Interval computation: script and outputs

The script below verifies one range. Set `START` to 1 or 2^{19} and `LIMIT` to 2^{19} or 2^{20} to obtain the two halves reported afterwards.

```
import math, time, hashlib
import mpmath as mp

START = 1
LIMIT = 1 << 19
DPS = 40

mp.iv.dps = DPS
ln2 = mp.iv.log(2)
ln3 = mp.iv.log(3)
theta_float = math.log(3.0) / math.log(2.0)

checked = 0
failures = []
min_lower = (float("inf"), None, None)
min_upper = (float("inf"), None, None)
manifest = hashlib.sha256()
start_time = time.time()

for m in range(START, LIMIT):
    if (m & (m - 1)) == 0:      # integral exponent: m is a power of two
        continue
```

```

    # Only proposes a neighboring integer.  The interval tests below certify it.
    n = math.floor(math.exp(theta_float * math.log(m)))

    lhs = mp.iv.log(m) * ln3
    lower_diff = lhs - mp.iv.log(n) * ln2
    upper_diff = mp.iv.log(n + 1) * ln2 - lhs

    lower_lb = float(lower_diff.a)
    upper_lb = float(upper_diff.a)
    if not (lower_lb > 0.0 and upper_lb > 0.0):
        failures.append((m, n, str(lower_diff), str(upper_diff)))
        break

    min_lower = min(min_lower, (lower_lb, m, n))
    min_upper = min(min_upper, (upper_lb, m, n))
    manifest.update(f"{m}:{n};".encode())
    checked += 1

print("checked_nonpowers=", checked)
print("failures=", len(failures))
print("min_lower_log_margin=", min_lower)
print("min_upper_log_margin=", min_upper)
print("sha256_m_n_manifest=", manifest.hexdigest())
print("elapsed_seconds=", time.time() - start_time)
assert not failures
```

Listing 12: Interval verification

The two 40-digit runs produced:

```
LIMIT=524288
DPS=40
checked_nonpowers=524268
failures=0
min_lower_log_margin=(1.4147089634746718e-15, 189427, 231498127)
min_upper_log_margin=(3.0841010256409534e-15, 306982, 497586055)
sha256_m_n_manifest=02b589461fa8c1721bc251575b98b1e3d58001f54cb7faa2cd6c59edb8c819a5
elapsed_seconds=18.505
```

Listing 13: Lower half output.

```
LIMIT=1048576
DPS=40
checked_nonpowers=524287
failures=0
min_lower_log_margin=(9.490232037370244e-16, 958460, 3023921230)
min_upper_log_margin=(2.3019781003173366e-15, 556061, 1275861767)
sha256_m_n_manifest=fccccf5de8673379d37ff0fc840ab091d9ff7a2352ff64a93091f01c21bf4cb4
elapsed_seconds=18.904
```

Listing 14: Upper half output.

The two SHA-256 manifests are

lower: `02b589461fa8c1721bc251575b98b1e3d58001f54cb7faa2cd6c59edb8c819a5`,

upper: `fccccf5de8673379d37ff0fc840ab091d9ff7a2352ff64a93091f01c21bf4cb4`.

Elapsed times are machine-dependent sanity checks; the counts, zero-failure results, extremal tuples, and hashes are the reproducibility data. The hashes cover the ASCII stream `m:n;` of proposed-and-certified pairs and make accidental changes to the candidate sequence detectable across reruns.

Status: [interval computation] — interval arithmetic in `mpmath` [15] at 40 digits. This is a software trust boundary, not a kernel certificate. The kernel-verified zero-free region remains $2^x < 4096$ in `FiniteCheck4096`.

G Suggested experimental determinant families

Before developing a full p-adic theory, several finite experiments can test whether the required amplification exists.

G.1 Rectangular row and column boxes

Choose rows (n, k) with $0 \le n < N_0$, $0 \le k < K_0$ and columns (a, b) with $0 \le a < A_0$, $0 \le b < B_0$. The entry is

$$(2^a 3^b)^n (M^a A^b)^k.$$

For complete rectangles, the matrix is a rearranged Kronecker product only when the row and column cardinalities match in a compatible way. Its determinant or maximal minors can be factored symbolically for small boxes. Record valuations at 2 and 3 as polynomials in $a_0 = v_2(M)$ and $b_0 = v_3(A)$.

G.2 Triangular boxes adapted to height

Use

$$n + k\beta \le T, \qquad a + b\vartheta \le L.$$

These are the natural row and column shapes from the exact counting theorem and the mixed-exponent semigroup. Search for square minors with minimal archimedean weight and maximal forced valuation. The linear boundary slopes are irrational, so floor-sum fluctuations may prevent cancellation patterns present in rectangles.

G.3 Finite-difference row operations

Order rows by n for fixed k, or by k for fixed n, and apply repeated difference operators. Differences introduce factors such as

$$2^a 3^b - 1, \qquad M^a A^b - 1.$$

These may have systematic 2- or 3-adic valuations for carefully selected column pairs. Lifting-the-exponent lemmas can then provide exact lower bounds. The risk is that the archimedean size of the same factors grows too quickly; experiments should track both sides.

G.4 Confluent columns

Instead of only distinct mixed monomials, consider derivatives with respect to an auxiliary continuous exponent parameter before specializing. Confluent generalized Vandermonde matrices introduce logarithmic factors, which are not integral, but common-denominator or linear-form constructions may restore arithmetic control. This is closer to Laurent's interpolation determinant method [10] and may reduce total exponent weight.

H Dependency map and Lean-oriented proof sketches

The conditional arithmetic criteria of this report were written with formalization in mind: each one is a short argument over an already kernel-verified base. This appendix records the shortest route from the existing corpus to each of them, and then gives skeleton statements for the four criteria whose formalization is least obvious.

Status: Design note. The snippets below are schematic ASCII transliterations and are not claimed to compile unchanged. The mathematics they target is **[paper proof in this report]**; none of it has been accepted by the kernel, and nothing here may be read as a claim of verification.

H.1 Dependency map

Each row names one new result, the principal existing input it is built on, and the extra mathematics a formalization would have to supply. All of the new results are conditional: they describe the structure forced on $\mathcal{S}$ and $\mathcal{C}$ by a hypothetical counterexample to Conjecture 2.1, and they are vacuous if the conjecture holds.

New result	Principal existing input	Additional mathematics
rational-function rigidity	canonical primitive generator; transcendence of β	injectivity of polynomial evaluation at a transcendental element
product and power criteria	unique (n, k) coordinates	quadratic coefficient comparison
quotient criterion	unique (n, k) coordinates	denominator clearing and coefficient comparison
three-smooth output test	transcendence of ϑ	unique factorization in $\mathbb{N}$
product transcendence	three-smooth-base theorem	classical six exponentials
common power degree	unique (n, k) coordinates	positive root uniqueness; divisibility
primitive statistics	total lattice count	Möbius inversion
gcd and lcm lattice	exact candidate monoid	prime-valuation min and max
ambient root saturation	primitive valuation gcd of w	Bézout and linear cone arithmetic
optimized differences	arbitrary-order difference theorem	gamma identity and Stirling estimate

Every entry in the middle column is [kernel-verified Lean] in the corpus catalogued in Section 4. In the right-hand column, only the six exponentials theorem is external background ([classical/open background]); the remaining items are ordinary mathlib-level algebra and analysis, which is what makes these criteria cheap to formalize once the base is in place.

Transliteration conventions. The listings write Unicode in plain ASCII: `R`, `Q`, `Z` and `N` are the real, rational, integer and natural types; arrows are `->` and `<->`; membership is `in` and divisibility is `dvd`; anonymous constructors use angle brackets; the two logical connectives use their two-character ASCII spellings; and the focusing dot is written as a period. Coercions are written as explicit casts such as `Int.cast` rather than as the coercion arrow.

H.2 Multiplication skeleton

A direct statement can use the repository predicate `TwoBaseIntegralSolution`, which is the defining conjunction $2^x \in \mathbb{Z}$ and $3^x \in \mathbb{Z}$.

```
theorem product_solution_iff_integer_factor
    {x y : R}
    (hx : TwoBaseIntegralSolution x)
    (hy : TwoBaseIntegralSolution y) :
    TwoBaseIntegralSolution (x * y) <->
      x in Set.range (Int.cast : Z -> R) \/
      y in Set.range (Int.cast : Z -> R) := by
  classical
  constructor
  . intro hxy
    by_cases hAE : AlaogluErdosConjecture
    . left
      exact hAE x hx
    . obtain <w, d, a, c, beta, ..., hcoord, ...> :=
        exists_canonical_primitiveGenerator_of_not_alaogluErdosConjecture hAE
      obtain <nx, kx, hxcoord> := (hcoord x).mp hx
      obtain <ny, ky, hycoord> := (hcoord y).mp hy
      obtain <np, kp, hpcoord> := (hcoord (x * y)).mp hxy
      -- If kx and ky are both positive, expanding the product gives a
      -- nonzero quadratic polynomial over Q vanishing at beta, which
      -- contradicts transcendence of beta.  Hence kx = 0 or ky = 0.
      ...
  . rintro (hxint | hyint)
    . exact solution_nsmul_left hxint hy
    . exact solution_nsmul_right hyint hx
```

Listing 15: Product criterion: skeleton.

In the finished theorem the integer range can be narrowed to $\mathbb{N}_0$, because every solution is nonnegative by `nonneg_of_two_rpow_integer`. The algebraic step should be isolated, since three of the four criteria reduce to it. Note that the coefficients must be rational — with arbitrary real a, b, c the statement is false — and that is exactly how the applications supply them, as differences of the integer coordinates:

```
lemma transcendental_quadratic_coeff_eq_zero
    {beta : R} {a b c : Q}
    (hbeta : Transcendental Q beta)
    (h : (a : R) * beta ^ 2 + (b : R) * beta + (c : R) = 0) :
```

```
    a = 0 /\ b = 0 /\ c = 0 := ...
```

Listing 16: The shared coefficient lemma.

Mathlib's polynomial evaluation API is probably the cleanest route: build the polynomial $aX^2 + bX + c$, rewrite `h` as an evaluation, and apply the definition of transcendence to conclude that the polynomial is zero.

H.3 Output smoothness skeleton

Define the predicate

$$\mathrm{ThreeSmooth}(m) :\Longleftrightarrow \exists u, v \in \mathbb{N}_0,\ m = 2^u 3^v.$$

A practical implementation should avoid a partial "natural output" function and instead carry explicit witnesses $M, A \in \mathbb{N}$ for the two integer powers:

```
theorem integer_of_outputs_threeSmooth
    {x : R} {M A : N}
    (hM : (2 : R) ^ x = (M : R))
    (hA : (3 : R) ^ x = (A : R))
    (hMs : ThreeSmooth M) (hAs : ThreeSmooth A) :
    x in Set.range (Int.cast : Z -> R) := by
  obtain <a, b, rfl> := hMs
  obtain <c, d, rfl> := hAs
  -- Taking logarithms base two: x = a + b * theta and x * theta = c + d * theta.
  have hpoly :
      (b : R) * logThreeDivLogTwo ^ 2
        + ((a : R) - (d : R)) * logThreeDivLogTwo - (c : R) = 0 := by
    ...
  have htheta := transcendental_logThreeDivLogTwo
  -- The coefficients are integers, so all three vanish:
  -- b = 0, a = d, c = 0, whence 2 ^ x = 2 ^ a and x = a.
  ...
```

Listing 17: Output smoothness: skeleton.

The converse direction is immediate: for integral $x = a$ the outputs are 2^a and 3^a. The interesting content is that three-smoothness of *both* outputs is already enough to force integrality, with no appeal to the conditional structure theory.

H.4 Common-power skeleton

The formal theorem is most reusable when it is stated for explicit output witnesses and for a fixed canonical generator, so that the coordinate comparison is available as a hypothesis rather than reconstructed inside the proof.

```
theorem simultaneous_perfectPower_iff_divides_coordinates
    (hfail : not AlaogluErdosConjecture)
    {x beta : R} {n k r : N}
    (hr : 0 < r)
    (hbeta : CanonicalPrimitiveGenerator beta)
    (hxcoord : x = (n : R) + (k : R) * beta) :
    (Exists fun U : N => (U : R) ^ r = (2 : R) ^ x) /\
    (Exists fun V : N => (V : R) ^ r = (3 : R) ^ x) <->
```

```
    r dvd n /\ r dvd k := by
  -- Convert the two perfect powers into a solution at x / r,
  -- apply uniqueness of coordinates, and compare.
  ...
```

Listing 18: Common powers: skeleton.

The positive-root step can be discharged either by strict monotonicity of $t \mapsto t^r$ on the nonnegative reals or by injectivity of the logarithm; the first is usually shorter in mathlib.

H.5 Ambient saturation skeleton

The only genuinely arithmetic lemma in the saturation argument is the reconstruction of a root of a power of a primitive number:

```
lemma primitive_core_root
    {w u q r : N}
    (hw : oddPrimeValuationGCD w = 1)
    (hr : 0 < r)
    (hpow : u ^ r = w ^ q) :
    Exists fun t : N => q = r * t /\ u = w ^ t := by
  -- Compare every coordinate of the prime factorizations.
  -- A Bezout combination of the positive valuations of w gives r | q;
  -- factorization extensionality then gives u = w ^ t.
  ...
```

Listing 19: Primitive-core root lemma.

The proof in this report only needs the case $q = dk$, but the general lemma is worth stating once: the same reconstruction is used implicitly in `PrimitiveOddCore` and in `OutputNormalization`, and a shared version would remove that duplication.

H.6 Suggested theorem order for a pull request

A low-conflict sequence, each step reviewable on its own, is:

1. add a small `ThreeSmooth` API together with the output-smoothness theorem;
2. add coordinate comparison lemmas for affine, quadratic and quotient expressions;
3. prove the product, power and quotient criteria;
4. prove the common-power and affine-descent criteria;
5. add the candidate divisibility and support results;
6. add primitive-core root reconstruction and ambient saturation;
7. only then add the asymptotic and six-exponentials extensions.

This ordering keeps the first several commits elementary, and it defers everything that depends on external transcendence input to the end, where it can be reviewed against the status discipline of Appendix A rather than mixed into routine algebra.

I Directed-rounding verifier

Appendix F gives the interval-arithmetic script behind the 2^{20} verification. This appendix gives the second, independent verifier: a C program that uses MPFR directed rounding rather than an interval library, and that extends the scanned range sixty-fourfold, to 2^{26}.

Status: [interval computation] — MPFR 4.2.2 at 192 bits with directed rounding modes. Like the 2^{20} scan this is a software trust boundary, not a kernel certificate. The kernel-verified zero-free region remains $2^x < 4096$ in `FiniteCheck4096`.

I.1 What the verifier certifies

For a non-power of two m, proving $m^\vartheta \notin \mathbb{N}$ amounts to exhibiting $n \in \mathbb{N}$ with $n < m^\vartheta < n+1$, and, as in Section 19, that pair of strict inequalities is certified in logarithmic form so that no exponentiation is ever performed. At precision $p = 192$ the program computes directed enclosures

$$L_m^- \le \log m \le L_m^+, \qquad L_2^- \le \log 2 \le L_2^+, \qquad L_3^- \le \log 3 \le L_3^+,$$

together with $L_n^+ \ge \log n$ and $L_{n+1}^- \le \log(n+1)$, and then rounds *downward* in

$$\delta_- = L_m^- L_3^- - L_n^+ L_2^+, \qquad \delta_+ = L_{n+1}^- L_2^- - L_m^+ L_3^+.$$

Positivity of δ_- proves the lower inequality and positivity of δ_+ proves the upper one. Every input on the n side is below 2^{42} and is therefore exactly representable at 192 bits.

An ordinary `long double` evaluation proposes $n \approx \lfloor m^\vartheta \rfloor$ and the offsets $0, -1, 1, -2, 2, -3, 3$ are tried in that order. The proposal has no logical role: a proposal is accepted only when both directed lower bounds are strictly positive, and a failure to locate any admissible n is reported as a failure rather than silently reclassified. Locator error can therefore produce a false alarm but not a false proof. Powers of two are skipped rather than tested, since $(2^q)^\vartheta = 3^q$ is the known integral candidate; an exact integral value at a non-power of two would satisfy neither strict inequality for any n and would be reported.

I.2 Source identity and environment

The recorded run used MPFR 4.2.2 linked as `libmpfr.so.6`, GCC 14.2.0 with `-O3 -fopenmp`, GNU OpenMP with five worker threads, Linux on x86-64, and 192-bit precision with the directed modes `MPFR_RNDD` and `MPFR_RNDU`. The SHA-256 digest of the exact executed source reproduced below is

`f888c0d89be9366f01e2d26f339dcbe446556d39d4c5dbcf8f579ec0d0052e1d`

The execution image contained the MPFR shared library but not its development header, so the source carries a guarded minimal declaration of the MPFR 4.x ABI; on a system with `mpfr.h` present the standard header branch is taken instead. That ABI detail is part of the software trust boundary and is disclosed here rather than hidden.

I.3 Source

The listing below is the executed source verbatim. Its language is C99; it is typeset in this report's generic code style, which has no C-specific highlighting.

```
#define _GNU_SOURCE
#include <stdio.h>
#include <stdlib.h>
#include <stdint.h>
#include <inttypes.h>
#include <math.h>
#include <omp.h>

#if defined(__has_include)
#  if __has_include(<mpfr.h>)
#    include <mpfr.h>
#    define HAVE_SYSTEM_MPFR_HEADER 1
#  endif
#endif

#ifndef HAVE_SYSTEM_MPFR_HEADER
/* Minimal declarations for the MPFR 4.x ABI.  The execution environment had
   libmpfr.so.6 but not the development header package.  A normal build with
   libmpfr-dev installed takes the branch above instead. */
typedef unsigned long mp_limb_t;
typedef long mpfr_prec_t;
typedef long mpfr_exp_t;
typedef int mpfr_sign_t;
typedef struct {
  mpfr_prec_t _mpfr_prec;
  mpfr_sign_t _mpfr_sign;
  mpfr_exp_t _mpfr_exp;
  mp_limb_t *_mpfr_d;
} __mpfr_struct;
typedef __mpfr_struct mpfr_t[1];
typedef __mpfr_struct *mpfr_ptr;
typedef const __mpfr_struct *mpfr_srcptr;
typedef enum {
  MPFR_RNDN = 0, MPFR_RNDZ = 1, MPFR_RNDU = 2,
  MPFR_RNDD = 3, MPFR_RNDA = 4, MPFR_RNDF = 5,
  MPFR_RNDNA = -1
} mpfr_rnd_t;
extern void mpfr_init2(mpfr_ptr, mpfr_prec_t);
extern void mpfr_clear(mpfr_ptr);
extern int mpfr_set_ui(mpfr_ptr, unsigned long, mpfr_rnd_t);
extern int mpfr_log(mpfr_ptr, mpfr_srcptr, mpfr_rnd_t);
extern int mpfr_mul(mpfr_ptr, mpfr_srcptr, mpfr_srcptr, mpfr_rnd_t);
extern int mpfr_sub(mpfr_ptr, mpfr_srcptr, mpfr_srcptr, mpfr_rnd_t);
extern int mpfr_cmp_ui(mpfr_srcptr, unsigned long);
extern double mpfr_get_d(mpfr_srcptr, mpfr_rnd_t);
extern const char *mpfr_get_version(void);
#endif

static inline int is_power_of_two_u64(uint64_t x) {
  return x && ((x & (x - 1)) == 0);
}

typedef struct {
  uint64_t checked;
```

```
  uint64_t failures;
  double min_lower;
  double min_upper;
  uint64_t min_lower_m, min_lower_n;
  uint64_t min_upper_m, min_upper_n;
  uint64_t first_fail_m, first_fail_guess;
} Stats;

int main(int argc, char **argv) {
  uint64_t start = 1;
  uint64_t limit = (1ULL << 20);
  int bits = 192;
  if (argc > 1) start = strtoull(argv[1], NULL, 0);
  if (argc > 2) limit = strtoull(argv[2], NULL, 0);
  if (argc > 3) bits = atoi(argv[3]);
  if (start < 1 || limit <= start) {
    fprintf(stderr, "bad range\n");
    return 2;
  }

  int nt = omp_get_max_threads();
  Stats *stats = calloc((size_t)nt, sizeof(Stats));
  if (!stats) return 3;
  for (int i = 0; i < nt; ++i) {
    stats[i].min_lower = INFINITY;
    stats[i].min_upper = INFINITY;
  }

  const long double theta = logl(3.0L) / logl(2.0L);
  double t0 = omp_get_wtime();

  mpfr_t two, three, ln2_lo, ln2_hi, ln3_lo, ln3_hi;
  mpfr_init2(two, bits); mpfr_init2(three, bits);
  mpfr_init2(ln2_lo, bits); mpfr_init2(ln2_hi, bits);
  mpfr_init2(ln3_lo, bits); mpfr_init2(ln3_hi, bits);
  mpfr_set_ui(two, 2, MPFR_RNDN);
  mpfr_set_ui(three, 3, MPFR_RNDN);
  mpfr_log(ln2_lo, two, MPFR_RNDD);
  mpfr_log(ln2_hi, two, MPFR_RNDU);
  mpfr_log(ln3_lo, three, MPFR_RNDD);
  mpfr_log(ln3_hi, three, MPFR_RNDU);

#pragma omp parallel
  {
    int tid = omp_get_thread_num();
    Stats *st = &stats[tid];
    mpfr_t xm, xn, xnp1;
    mpfr_t logm_lo, logm_hi, logn_hi, lognp1_lo;
    mpfr_t lhs_lo, lhs_hi, rhs_lo, rhs_hi, dl, du;
    mpfr_init2(xm, bits); mpfr_init2(xn, bits); mpfr_init2(xnp1, bits);
    mpfr_init2(logm_lo, bits); mpfr_init2(logm_hi, bits);
    mpfr_init2(logn_hi, bits); mpfr_init2(lognp1_lo, bits);
    mpfr_init2(lhs_lo, bits); mpfr_init2(lhs_hi, bits);
    mpfr_init2(rhs_lo, bits); mpfr_init2(rhs_hi, bits);
    mpfr_init2(dl, bits); mpfr_init2(du, bits);

#pragma omp for schedule(static)
    for (uint64_t m = start; m < limit; ++m) {
      if (is_power_of_two_u64(m)) continue;
```

```

    /* This is only a locator.  The MPFR inequalities below are the proof. */
    uint64_t guess = (uint64_t)floorl(expl(theta * logl((long double)m)));

    mpfr_set_ui(xm, (unsigned long)m, MPFR_RNDN);
    mpfr_log(logm_lo, xm, MPFR_RNDD);
    mpfr_log(logm_hi, xm, MPFR_RNDU);
    mpfr_mul(lhs_lo, logm_lo, ln3_lo, MPFR_RNDD);
    mpfr_mul(lhs_hi, logm_hi, ln3_hi, MPFR_RNDU);

    int ok = 0;
    uint64_t certn = 0;
    double dl_d = 0.0, du_d = 0.0;
    static const int offsets[7] = {0, -1, 1, -2, 2, -3, 3};
    for (int oi = 0; oi < 7; ++oi) {
      int off = offsets[oi];
      if (off < 0 && guess < (uint64_t)(-off) + 1) continue;
      uint64_t n = (uint64_t)((int64_t)guess + off);
      if (n < 1) continue;

      mpfr_set_ui(xn, (unsigned long)n, MPFR_RNDN);
      mpfr_set_ui(xnp1, (unsigned long)(n + 1), MPFR_RNDN);

      /* Directed lower bound for log(m)log(3)-log(n)log(2). */
      mpfr_log(logn_hi, xn, MPFR_RNDU);
      mpfr_mul(rhs_hi, logn_hi, ln2_hi, MPFR_RNDU);
      mpfr_sub(dl, lhs_lo, rhs_hi, MPFR_RNDD);
      if (mpfr_cmp_ui(dl, 0) <= 0) continue;

      /* Directed lower bound for log(n+1)log(2)-log(m)log(3). */
      mpfr_log(lognp1_lo, xnp1, MPFR_RNDD);
      mpfr_mul(rhs_lo, lognp1_lo, ln2_lo, MPFR_RNDD);
      mpfr_sub(du, rhs_lo, lhs_hi, MPFR_RNDD);
      if (mpfr_cmp_ui(du, 0) <= 0) continue;

      ok = 1;
      certn = n;
      dl_d = mpfr_get_d(dl, MPFR_RNDD);
      du_d = mpfr_get_d(du, MPFR_RNDD);
      break;
    }

    if (!ok) {
      ++st->failures;
      if (!st->first_fail_m || m < st->first_fail_m) {
        st->first_fail_m = m;
        st->first_fail_guess = guess;
      }
      continue;
    }

    ++st->checked;
    if (dl_d < st->min_lower) {
      st->min_lower = dl_d;
      st->min_lower_m = m;
      st->min_lower_n = certn;
    }
    if (du_d < st->min_upper) {
      st->min_upper = du_d;
```

```
        st->min_upper_m = m;
        st->min_upper_n = certn;
      }
    }

    mpfr_clear(xm); mpfr_clear(xn); mpfr_clear(xnp1);
    mpfr_clear(logm_lo); mpfr_clear(logm_hi);
    mpfr_clear(logn_hi); mpfr_clear(lognp1_lo);
    mpfr_clear(lhs_lo); mpfr_clear(lhs_hi);
    mpfr_clear(rhs_lo); mpfr_clear(rhs_hi);
    mpfr_clear(dl); mpfr_clear(du);
  }

  Stats total = {0};
  total.min_lower = INFINITY;
  total.min_upper = INFINITY;
  for (int i = 0; i < nt; ++i) {
    Stats *s = &stats[i];
    total.checked += s->checked;
    total.failures += s->failures;
    if (s->min_lower < total.min_lower) {
      total.min_lower = s->min_lower;
      total.min_lower_m = s->min_lower_m;
      total.min_lower_n = s->min_lower_n;
    }
    if (s->min_upper < total.min_upper) {
      total.min_upper = s->min_upper;
      total.min_upper_m = s->min_upper_m;
      total.min_upper_n = s->min_upper_n;
    }
    if (s->first_fail_m &&
        (!total.first_fail_m || s->first_fail_m < total.first_fail_m)) {
      total.first_fail_m = s->first_fail_m;
      total.first_fail_guess = s->first_fail_guess;
    }
  }

  double elapsed = omp_get_wtime() - t0;
  printf("mpfr_version=%s\n", mpfr_get_version());
  printf("range=[%" PRIu64 ",%" PRIu64 ")\n", start, limit);
  printf("precision_bits=%d\nthreads=%d\n", bits, nt);
  printf("checked_nonpowers=%" PRIu64 "\n", total.checked);
  printf("failures=%" PRIu64 "\n", total.failures);
  printf("min_lower_log_margin=%.17g at (m,n)=(%" PRIu64 ",%" PRIu64 ")\n",
         total.min_lower, total.min_lower_m, total.min_lower_n);
  printf("min_upper_log_margin=%.17g at (m,n)=(%" PRIu64 ",%" PRIu64 ")\n",
         total.min_upper, total.min_upper_m, total.min_upper_n);
  if (total.failures) {
    printf("first_failure=(m,guess)=(%" PRIu64 ",%" PRIu64 ")\n",
           total.first_fail_m, total.first_fail_guess);
  }
  printf("elapsed_seconds=%.6f\n", elapsed);

  mpfr_clear(two); mpfr_clear(three);
  mpfr_clear(ln2_lo); mpfr_clear(ln2_hi);
  mpfr_clear(ln3_lo); mpfr_clear(ln3_hi);
  free(stats);
  return total.failures ? 1 : 0;
}
```

Listing 20: Directed-rounding verifier (C99, MPFR, OpenMP).

I.4 Build and invocation

On a system with the MPFR development headers installed, the whole build and a single full-range run are:

```
gcc -O3 -fopenmp mpfr_scan.c -lmpfr -lm -o mpfr_scan
OMP_NUM_THREADS=5 ./mpfr_scan 1 67108864 192
```

Listing 21: Build and invocation (POSIX shell).

The recorded audit instead used sixteen shorter invocations over half-open subranges, so that each chunk produces an independently inspectable log. The exit status is nonzero exactly when some input failed to be certified, which makes the chunked run scriptable without parsing the output.

J Chunk audit and reproducibility

Status: [interval computation] — the numbers in this appendix are the output of the verifier of Appendix I. They are reproducible data at a software trust level, not kernel certificates; the kernel-verified exponent bound remains $x \geq 12$.

There are $2^{26} - 1 = 67{,}108{,}863$ positive integers below 2^{26}, of which exactly 26 are powers of two, namely 2^0 through 2^{25}. The verifier therefore tests $67{,}108{,}863 - 26 = 67{,}108{,}837$ inputs. The range was split at multiples of 2^{22} into one initial chunk $[1, 2^{22})$ and fifteen chunks of width 2^{22}. The canonical chunk log holds the full standard output of every invocation and has SHA-256 digest

`dcaa7f3b7e5be44acaee9015a2045da14ee42d8fa2d96b9619cdd44c072a6fd7`

Table 9 reports, for each chunk, the exact scanned integer interval, the number of non-powers certified, and the smallest directed lower and upper logarithmic margins recorded in that chunk. Every chunk returned a failure count of zero.

Table 9: Directed-rounding audit through 2^{26}.

Chunk	Integer range	Checked	Smallest lower gap	Smallest upper gap
0	1–4,194,303	4,194,281	1.1539×10^{-17}	3.0076×10^{-17}
1	4,194,304–8,388,607	4,194,303	4.6319×10^{-20}	1.3515×10^{-19}
2	8,388,608–12,582,911	4,194,303	1.2811×10^{-18}	1.3515×10^{-19}
3	12,582,912–16,777,215	4,194,304	4.6319×10^{-20}	2.9082×10^{-19}
4	16,777,216–20,971,519	4,194,303	1.1898×10^{-18}	1.3515×10^{-19}
5	20,971,520–25,165,823	4,194,304	5.4212×10^{-19}	1.5111×10^{-19}
6	25,165,824–29,360,127	4,194,304	4.6319×10^{-20}	1.7669×10^{-19}
7	29,360,128–33,554,431	4,194,304	1.4439×10^{-19}	5.4132×10^{-21}
8	33,554,432–37,748,735	4,194,303	4.2839×10^{-19}	1.3515×10^{-19}

Chunk	Integer range	Checked	Smallest lower gap	Smallest upper gap
9	37,748,736–41,943,039	4,194,304	3.9292×10^{-19}	1.0460×10^{-19}
10	41,943,040–46,137,343	4,194,304	1.9177×10^{-19}	1.5111×10^{-19}
11	46,137,344–50,331,647	4,194,304	1.0117×10^{-19}	3.4145×10^{-19}
12	50,331,648–54,525,951	4,194,304	5.6927×10^{-20}	1.2322×10^{-19}
13	54,525,952–58,720,255	4,194,304	3.6447×10^{-20}	1.1345×10^{-19}
14	58,720,256–62,914,559	4,194,304	6.7779×10^{-20}	5.4132×10^{-21}
15	62,914,560–67,108,863	4,194,304	5.0923×10^{-20}	3.9197×10^{-20}
	total	**67,108,837**		

J.1 Extremal records

The global lower-gap record is the chunk 13 value 3.6447×10^{-20}, attained at

$$(m, n) = (58{,}570{,}438,\ 2{,}048{,}703{,}434{,}093),$$

where a separate 100-digit evaluation gives $m^{\vartheta} - n = 1.07725158750919017548\ldots \times 10^{-7}$. The global upper-gap record is the chunk 7 value 5.4132×10^{-21}, attained at

$$(m, n) = (29{,}411{,}704,\ 687{,}581{,}893{,}443),$$

where $n + 1 - m^{\vartheta} = 5.36974926710988874953\ldots \times 10^{-9}$. The same upper margin reappears in chunk 14 at the scaled pair (58,823,408, 2,062,745,680,331); this is not a coincidence but the identity $(2m)^{\vartheta} = 3\, m^{\vartheta}$, which multiplies the input by two and its ϑ-power by three while leaving the logarithmic margin unchanged.

Both extremal inequalities were reevaluated independently at 100 decimal digits. That secondary check is not an independent proof system — it shares the same floating-point philosophy — but it does guard against transcription and aggregation mistakes, and the certified margins sit many orders of magnitude above the nominal 192-bit rounding scale of the logarithmic operations.

The 2^{20} run of Appendix F and the present run are independent implementations, one using interval arithmetic in `mpmath` and the other MPFR directed rounding, and they agree on the overlapping range $1 \leq m < 2^{20}$: no failures. Their recorded extremal metadata are not directly comparable, because the chunk boundaries differ.

J.2 Reproducibility checklist

To reproduce the finite claim independently:

1. obtain MPFR 4.2.x and GMP on a 64-bit platform;
2. save the source of Appendix I and verify its SHA-256 digest;
3. compile with optimization and OpenMP, or delete the OpenMP pragmas for a serial run;
4. run the sixteen half-open intervals of Table 9 at 192 bits;
5. verify that every invocation reports `failures=0`;
6. sum the reported `checked_nonpowers` and obtain 67,108,837;

7. compare the two global extremal records above;
8. optionally rerun at 256 or 320 bits, and on a second MPFR build;
9. for a theorem-grade result, translate the intervals into exact rational certificates and replay them in Lean rather than trusting the executable.

The finite statement is deterministic. Thread count changes execution time and the reduction order that produces the recorded minimum metadata, but it cannot change whether an individual directed inequality is certified. Step nine is the one that would change the status of the result rather than its range, and it is the reason the scan is presented here as data for certificate generation — in the sense of Section 21 — and not as an extension of the kernel-verified zero-free region.

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